PARITY FAMILIES OF LATTICE PATHS AND REAL-ROOTED SIGNED EULERIAN POLYNOMIALS

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Abstract. We classify the extremal families of lattice paths with pairwise even numbers of common edges for every length and endpoint, which gives Shankar's Catalan conjecture, compute the binary rank of grid graphs, and double the lower bound for odd families in the square. We prove that the signed Eulerian polynomials are real-rooted, as Fulman, Kim, Lee and Petersen conjectured, and settle conjectures on canon, caged and Dumont permutations.

1. Introduction

Two monotone lattice paths from $(0,0)$ to (n,n) share some of their unit edges. Shankar [35] asked how many such paths can be chosen so that any two of them share an even number of edges, and how many so that any two share an odd number. These are the analogues for lattice paths of the eventown and oddtown theorems on sets [5, 24, 3], which have recently been carried to subspaces of finite vector spaces by Dey [13]. Writing $\mathrm{M}_{\mathrm{ev}}(a,b)$ and $\mathrm{M}_{\mathrm{od}}(a,b)$ for the largest sizes of an even and of an odd family of paths from $(0,0)$ to (a,b) (precise definitions are in Section 2), Shankar proved

$$\mathrm{M}_{\mathrm{ev}}(n,n) = 2^n, \qquad \tfrac{1}{4}n^2 - O(1) \le \mathrm{M}_{\mathrm{od}}(n,n) \le n(n-1)+1,$$

exhibited C_n even families of 2^n paths, one for each noncrossing perfect matching of $[2n]$, and conjectured that there are no others [35, Conjecture 4.1]. His proof of $\mathrm{M}_{\mathrm{ev}}(n,n) \le 2^n$ is an induction that removes the last step of every path and controls a sum of squares of class sizes.

The first part of this paper (Sections 2–7) develops that induction into a structure theory. The second part (Sections 8–11) treats the conjectures and questions on permutations stated by Shankar and by his coauthors and colleagues: Fulman, Kim, Lee and Petersen [23], Dey and Sivasubramanian [16], Balachandran and Ramdas [4], Poddar and Sawant [31], and Jana, Mahato and Sivasubramanian [25]. Shankar's further work on descents, excedances and Eulerian-type polynomials [32, 36, 14, 15, 34] and on pattern avoidance [33, 10] supplies most of the tools used there.

1.1. Lattice paths.

The equality cases of Shankar's inequality are described by *skeletons*: a skeleton of $[m]$ is a noncrossing matching M of a subset of $[m]$ such that no unmatched point lies below an arc. To a skeleton we attach a system of sets of words, defined in Section 3.

Theorem A (Theorems 3.2 and 3.3)**.** *The extremal admissible systems of length m are exactly the skeleton systems, and there are $\binom{m}{\lfloor m/2 \rfloor}$ of them. In particular every even family of 2^n paths from $(0,0)$ to (n,n) is the family $\mathcal{F}_M$ of a unique noncrossing perfect matching M of $[2n]$, and there are exactly C_n such families.*

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The second statement is Shankar's conjecture; it was also obtained in an unrefereed manuscript of 27 September 2026, recorded in Section 3. Here it is a consequence of the first statement, which covers all lengths and all endpoints at once.

Theorem B (Theorems 4.3, 4.6 and 4.19)**.** *Let $a, b \geq 1$ and $m = a + b$.*

(i) *The binary rank of the adjacency matrix of the $a \times b$ grid graph is $ab - \gcd(a+1, b+1) + 1$.*

(ii) *A family of paths to (a, b) whose parity type matches that of m has at most $\sqrt{m^m/(a^a b^b)}$ members, and such a family with $2^{\min(a,b)}$ members exists; a family of the other parity type has at most $ab - \gcd(a+1, b+1) + 4$ members if m is even and at most $ab - \gcd(a+1, b+1) + 3$ if m is odd.*

(iii) *There are C_{n+1} odd families of 2^n paths from $(0, 0)$ to $(n+1, n)$, and $2^n \leq \mathrm{M}_{\mathrm{od}}(n+1, n) < 2^{n+1/2}$.*

For two and three rows of cells we construct families that are linear in the length and match the exact values computed for small rectangles (Theorems 4.16 and 4.18).

Theorem C (Theorems 5.2, 5.6, 5.7 and 5.10)**.** (i) *A region of the square between two lattice paths contains an even family of 2^n paths if and only if it contains both staircases $(NE)^n$ and $(EN)^n$; the extremal families in it correspond to Dyck paths below an explicit height profile.*

(ii) *An even family of Dyck paths of semilength n has at most 2^{n-1} members, and the extremal family is unique.*

(iii) *The intersection of the families $\mathcal{F}_{M_1}, \ldots, \mathcal{F}_{M_k}$ has 2^c members, where c is the number of connected components of the graph $M_1 \cup \cdots \cup M_k$ on $[2n]$; for $k = 2$ these are the loops of the meander formed by M_1 and M_2. The matrix $\left(|\mathcal{F}_M \cap \mathcal{F}_{M'}|\right)$ is the meander matrix of Di Francesco, Golinelli and Guitter at loop weight 2, and it is positive definite.*

Theorem D (Theorems 6.2 and 6.5)**.** *For every $n \geq 1$,*

$$\tfrac{1}{2}n^2 - 3n \;\leq\; \mathrm{M}_{\mathrm{od}}(n, n) \;\leq\; n(n-1) + 1,$$

and the upper bound drops to $n(n-1)$ or $n(n-1) - 1$ unless $n \equiv 0, 1 \pmod 8$.

The lower bound doubles the leading constant of [35]. It comes from windows of consecutive columns in which the odd condition reduces to an even condition one size smaller; the upper bound is the Arf invariant of a quadratic form on the cycle space of the grid.

Theorem E (Theorems 7.3, 7.5 and 7.7)**.** *For monotone paths in $\mathbb{Z}^d$ from the origin to $(n, \ldots, n)$, with n even if d is odd, the largest family with pairwise even numbers of common edges has size $d^{dn/2+o(n)}$ as $n \to \infty$. For $d \geq 3$ the weighted form of Shankar's inequality is an equality only for the trivial system of singletons.*

Thus the Catalan rigidity of Theorem A is a planar phenomenon. In the box $(1, 1, 1, 1, 1, 1)$ the maximum is 48 (Proposition 7.11).

1.2. **Permutations.** Let $A_n^{\pm}(t)$ be the generating polynomials of descents over the even and the odd permutations of $[n]$, and $E_n^{\pm}(t)$ those of excedances. Dey and Sivasubramanian [16] conjectured that $A_n^{\pm}$ is real-rooted for $n \equiv 0, 1 \pmod 4$, and Fulman, Kim, Lee and Petersen [23, Conjecture 1.3] extended the conjecture to all n. Shankar [32] proved that all four coefficient sequences are ultra-log-concave and conjectured that $E_n^{\pm}$ is real-rooted for $n \geq 5$.

Theorem F (Theorems 8.4 and 8.5)**.** *For every $n \geq 2$ the polynomials $A_n^+(t)$ and $A_n^-(t)$ have only real zeros. The polynomial $E_5^+(t) = 1 + 11t + 36t^2 + 11t^3 + t^4$ has no real zero, and for every $n \geq 6$ the polynomials $E_n^+(t)$ and $E_n^-(t)$ have only real zeros.*

The proof writes the four polynomials as $A_n + c(1-t)^{n-m}A_m$ with $|c| \le 1$ and $m \le \lceil n/2 \rceil$, and counts sign changes of this perturbation of the Eulerian polynomial at explicit test points obtained from the partial fraction expansion of the logistic function.

For Shankar's canon permutations [33], we prove that the descent polynomials of the 231-avoiding words are real-rooted, that the generating function of the 321-avoiding words is transcendental, with $Q_k \sim \frac{2}{\pi}\, 8^k/k$, and that the first of his two counting sequences is a Stieltjes moment sequence with an explicit density while the second is not (Theorems 9.10–9.16). For the caged subsequences of Balachandran, Ramdas and Shankar [4], we determine all extremal permutations of length $5k+4$ and show that restricting to n-cycles raises the minimum by at most one (Theorems 10.4 and 10.5). For the deformed Dumont permutations of Poddar, Sawant and Shankar [31], we give the cardinalities in closed form (Theorem 10.7). Finally, for the 2-Steiner distance matrix of a path we prove the coefficient conjecture of Jana, Mahato and Sivasubramanian [25] and show that their peak conjecture fails for every n with $16 \le n \le 60$ (Theorems 11.4 and 11.5). Table 10 in Section 12 lists every conjecture treated in the paper with its status.

1.3. **Methods and verification.** All proofs are given in the text. Computed values enter in three ways: the exact maxima for small boxes in Sections 4–7, which are reported in tables and guide the conjectures, except that Proposition 7.11 uses the value for the box $(1,1,1,1,1)$; the exact root counts of finitely many polynomials of small degree in Section 8; and the exact integer evaluations stated in Sections 9–11. These computations, a Lean 4 formalisation of the finite identities and the programs that check every numerical statement of the paper are archived in [8]. Earlier work of the author on group actions and on nerve complexes of coverings is in [6, 7, 9].

2. Words, common edges and admissible systems

The parity theorems for sets that give this subject its vocabulary are the following. A family of subsets of an n-element set whose members have odd size and meet pairwise in an even number of points has at most n members; if the sizes are even as well, the family has at most $2^{\lfloor n/2 \rfloor}$ members. The second statement is due to Berlekamp [5] and Graver [24], and both are standard applications of linear algebra over $\mathbb{F}_2$ [3, 22].[1] Shankar [35] replaced the ground set by the edge set of a grid and the subsets by monotone lattice paths. This section fixes the notation for that setting, introduces the collections of sets of words to which his method applies, and proves his inequality for them.

2.1. **Paths as words.** A *lattice path* of length m is a sequence of m unit steps in the plane that starts at $(0,0)$, each step going east (E) or north (N). We encode it by the word $x = x_1 \cdots x_m \in L(m) = \{0,1\}^m$ with $x_i = 1$ when the ith step is N and $x_i = 0$ when it is E. The *weight* $\mathrm{w}(x)$ is the number of letters 1, and $x_{<i} = x_1 \cdots x_{i-1}$ and $x_{\le i} = x_1 \cdots x_i$ are prefixes of x. The path of x ends at $(m - \mathrm{w}(x), \mathrm{w}(x))$; thus the paths from $(0,0)$ to (a,b) are the $\binom{a+b}{b}$ words of length $a+b$ and weight b. We write $E(x)$ for the set of the m edges of the path.

The *level* of a lattice point (u,v) is $u+v$. The ith step of every path leads from level $i-1$ to level i, and it starts at the point $(i-1-\mathrm{w}(x_{<i}), \mathrm{w}(x_{<i}))$. Two steps of two paths can therefore be the same edge only when they carry the same index i, and then they coincide exactly when

[1] For the first statement, the indicator vectors of the members have the identity as their Gram matrix over $\mathbb{F}_2$, so they are linearly independent. For the second, a family that cannot be enlarged is closed under symmetric difference, hence is a subspace $U \subseteq U^{\perp}$ of $\mathbb{F}_2^n$, and $\dim U + \dim U^{\perp} = n$.

they start at the same point and have the same direction. Hence

$$|E(x)\cap E(y)| = \#\{i\in[m]:\ x_i = y_i \text{ and } \mathrm{w}(x_{<i}) = \mathrm{w}(y_{<i})\} \qquad (x,y\in L(m)). \tag{2.1}$$

We say that position i *counts* for the pair (x,y) when it belongs to the set on the right. Measuring the intersection of two paths by their common edges goes back to Hiranya Dey, as recorded in [35]. Only the parity of (2.1) matters in this paper, and we put

$$B_m(x,y) = |E(x)\cap E(y)| \bmod 2\ \in \mathbb{F}_2.$$

For a statement P the bracket $[\,P\,]$ denotes the element 1 of $\mathbb{F}_2$ if P holds and 0 otherwise; an integer written where an element of $\mathbb{F}_2$ is expected stands for its residue. For $X\subseteq L(m-1)$ and a letter e we write $Xe = \{xe:\ x\in X\}\subseteq L(m)$.

Lemma 2.1. *Let $m\ge 1$.*

(a) *$B_m(x,x) = m$ and $B_m(x,y) = B_m(y,x)$ for all $x,y\in L(m)$.*

(b) *For $x,y\in L(m-1)$ and letters $e,e'\in\{0,1\}$,*

$$B_m(xe,ye') = B_{m-1}(x,y) + [\,e=e'\,]\,[\,\mathrm{w}(x)=\mathrm{w}(y)\,].$$

Proof. A path shares all its m edges with itself, and (2.1) is symmetric in x and y; this is (a). For (b), the positions $1,\dots,m-1$ count for (xe,ye') exactly when they count for (x,y), because only prefixes of x and y enter the condition. Position m counts when the last letters agree and the prefixes $x=(xe)_{<m}$ and $y=(ye')_{<m}$ have equal weight. □

Part (b) is the only tool needed for inductions on the length. Its simplest consequence will be used repeatedly: if x and y have the same weight, then appending the same letter to both adds 1 to B, while appending different letters leaves B unchanged.

Two symmetries preserve common edges. Exchanging the letters 0 and 1 in every word reflects all paths in the line $y=x$, so $B_m(\bar{x},\bar{y}) = B_m(x,y)$ for the complemented words $\bar{x},\bar{y}$. Reversing words of a common weight b and length $a+b$ rotates the rectangle $[0,a]\times[0,b]$ by a half turn about its centre, so B_m is also invariant under simultaneous reversal of two words of equal weight.[2]

Definition 2.2. A set $\mathcal{F}$ of lattice paths from $(0,0)$ to (a,b) is an *even family* if any two distinct members share an even number of edges, and an *odd family* if any two distinct members share an odd number of edges. The largest sizes of such families are denoted by $\mathrm{M}_{\mathrm{ev}}(a,b)$ and $\mathrm{M}_{\mathrm{od}}(a,b)$.

In terms of words, an even family is a set of words of length $a+b$ and weight b with $B_{a+b}(x,y)=0$ for $x\ne y$, and an odd family has $B_{a+b}(x,y)=1$ for $x\ne y$. The reflection gives $\mathrm{M}_{\mathrm{ev}}(a,b) = \mathrm{M}_{\mathrm{ev}}(b,a)$ and $\mathrm{M}_{\mathrm{od}}(a,b) = \mathrm{M}_{\mathrm{od}}(b,a)$.

2.2. **Admissible systems.** Deleting the last step of every path of a family produces paths with two different endpoints, and the parities between the two groups are no longer those inside a group. The induction therefore has to be carried out for collections of several sets of words. The following notion isolates what is preserved.

Definition 2.3. An *admissible system of length m* is a finite collection $\mathcal{A}$ of pairwise disjoint nonempty subsets of $L(m)$, called its *classes*, with the following properties.

(A1) All words of a class have the same weight.

(A2) $B_m(x,y) = m$ in $\mathbb{F}_2$ whenever x and y belong to the same class.

[2]For words of different weights reversal does not preserve B_m: the paths of 10 and 11 share their first edge, while the paths of the reversed words 01 and 11 share no edge.

(A3) For any two distinct classes A and A', the function B_m is constant on $A \times A'$.

The *norm* of $\mathcal{A}$ is $\|\mathcal{A}\| = \sum_{A\in\mathcal{A}} |A|^2$, and $\mathcal{A}$ is *extremal* if $\|\mathcal{A}\| = 2^m$. The same notation $\|\mathcal{X}\| = \sum_{X\in\mathcal{X}} |X|^2$ is used for any finite collection $\mathcal{X}$ of finite sets.

By Lemma 2.1(a), condition (A2) holds automatically for $x = y$; for $x \neq y$ it says that the two paths share a number of edges of the parity of m. The common weight of the words of a class A is written $\mathrm{w}(A)$. A subcollection of an admissible system is admissible, the empty collection is admissible with norm 0, and the collection of all 2^m singletons is admissible with norm 2^m, since (A2) and (A3) are empty conditions for singletons. So the bound proved below is attained in a trivial way; its content lies in what it says about systems with large classes.

Example 2.4. Let $\mathcal{F}$ be a set of paths from $(0,0)$ to (a,b) and $m = a+b$. Then $\{\mathcal{F}\}$ satisfies (A1) and (A3), and it satisfies (A2) exactly when $B_m(x,y) = m$ for all distinct $x, y \in \mathcal{F}$. So $\{\mathcal{F}\}$ is an admissible system if and only if $\mathcal{F}$ is an even family and m is even, or $\mathcal{F}$ is an odd family and m is odd. Its norm is $|\mathcal{F}|^2$.

2.3. **Shankar's inequality.** Let $\mathcal{A}$ be an admissible system of length $m \geq 1$. For a class $A \in \mathcal{A}$ and a letter $e \in \{0,1\}$ let

$$A^e = \{u \in L(m-1) :\ ue \in A\},$$

so that $A = A^0 0 \cup A^1 1$. We call A *split* if both A^0 and A^1 are nonempty. Following Shankar, we form the two collections

(2.2)

$$\mathcal{G}_0 = \{A^0 :\ A \in \mathcal{A},\ A^0 \neq \varnothing\} \cup \{A^1 :\ A \text{ split}\}, \qquad \mathcal{G}_1 = \{A^1 :\ A \in \mathcal{A},\ A^1 \neq \varnothing\} \cup \{A^0 :\ A \text{ split}\}.$$

Thus $\mathcal{G}_0$ contains the prefixes of all words that end in 0, grouped by class, and in addition the prefixes of the words ending in 1 from the split classes; $\mathcal{G}_1$ is obtained by exchanging the two letters. The sets A^0 and A^1 of a split class occur in both collections, and every other nonempty set A^e occurs in exactly one.

Lemma 2.5. *Let $A, A' \in \mathcal{A}$ and suppose that $A^0 \cap A'^1 \neq \varnothing$. Then $A \neq A'$, and neither A nor A' is split: $A^1 = \varnothing$ and $A'^0 = \varnothing$.*

Proof. Choose $u \in A^0 \cap A'^1$. The words $u0 \in A$ and $u1 \in A'$ have different weights, so $A \neq A'$ by (A1). Lemma 2.1 gives $B_m(u0, u1) = B_{m-1}(u,u) = m-1$, and by (A3) this is the value of B_m on all of $A \times A'$.

Suppose that A contains a word $v1$. Then $\mathrm{w}(v) = \mathrm{w}(u) - 1$, so by Lemma 2.1(b) both $B_m(u0, v1)$ and $B_m(u1, v1)$ equal $B_{m-1}(u,v)$. The first is m by (A2), because $u0$ and $v1$ lie in A; the second is $m-1$, because $v1 \in A$ and $u1 \in A'$. This is a contradiction, so $A^1 = \varnothing$.

Suppose that A' contains a word $v0$. Then $\mathrm{w}(v) = \mathrm{w}(u) + 1$, and both $B_m(u1, v0)$ and $B_m(u0, v0)$ equal $B_{m-1}(u,v)$. The first is m by (A2) for A', the second is $m-1$ by (A3). Hence $A'^0 = \varnothing$. □

Proposition 2.6. *Let $\mathcal{A}$ be an admissible system of length $m \geq 1$, and let $\mathcal{G}_0$, $\mathcal{G}_1$ be defined by* (2.2).

(i) *The sets listed in $\mathcal{G}_0$ are nonempty and pairwise disjoint, and $\mathcal{G}_0$ is an admissible system of length $m-1$. The same holds for $\mathcal{G}_1$.*

(ii) *$\|\mathcal{A}\| \leq \|\mathcal{G}_0\| + \|\mathcal{G}_1\|$, with equality if and only if $|A^0| = |A^1|$ for every split class A.*

Proof. (i) The two collections are exchanged by the reflection $x \mapsto \bar{x}$, so it suffices to treat $\mathcal{G}_0$. Two sets A^e and A'^e with $A \neq A'$ are disjoint because A and A' are. If a set A^0 of $\mathcal{G}_0$ met a

set A'^1 of $\mathcal{G}_0$, then A' would be split by the definition of $\mathcal{G}_0$ and not split by Lemma 2.5. So the sets of $\mathcal{G}_0$ are pairwise disjoint.[3]

All words of A^e have weight $\mathrm{w}(A)-e$, which is (A1). For $u, v \in A^e$, Lemma 2.1(b) and (A2) give $m = B_m(ue, ve) = B_{m-1}(u, v) + 1$, so $B_{m-1}(u, v) = m - 1$; this is (A2) for length $m-1$. Finally let A^e and $A'^{e'}$ be two different sets of $\mathcal{G}_0$ and $u \in A^e$, $v \in A'^{e'}$. Then

$$B_{m-1}(u, v) = B_m(ue, ve') + [\, e = e' \,]\,[\, \mathrm{w}(u) = \mathrm{w}(v) \,].$$

The second term depends only on the two sets, since weights are constant on each. If $A \neq A'$ the first term is constant by (A3). If $A = A'$, the two sets are A^0 and A^1, and the first term is m by (A2). So B_{m-1} is constant on $A^e \times A'^{e'}$, which is (A3).

(ii) A class that is not split equals $A^0 0$ or $A^1 1$, and contributes $|A|^2$ to $\|\mathcal{A}\|$ and the same amount to exactly one of $\|\mathcal{G}_0\|$, $\|\mathcal{G}_1\|$. A split class contributes $|A^0|^2 + |A^1|^2$ to each of $\|\mathcal{G}_0\|$ and $\|\mathcal{G}_1\|$, and by the Cauchy–Schwarz inequality

$$|A|^2 = \left(1 \cdot |A^0| + 1 \cdot |A^1|\right)^2 \leq 2(|A^0|^2 + |A^1|^2),$$

with equality exactly when $|A^0| = |A^1|$. Summing over the classes gives (ii). □

Theorem 2.7 (Shankar [35, Lemma 2.3])**.** *Every admissible system $\mathcal{A}$ of length $m \geq 0$ satisfies*

$$\|\mathcal{A}\| = \sum_{A \in \mathcal{A}} |A|^2 \leq 2^m.$$

If $m \geq 1$ and $\mathcal{A}$ is extremal, then $\mathcal{G}_0$ and $\mathcal{G}_1$ are extremal systems of length $m-1$, and $|A^0| = |A^1|$ for every split class A of $\mathcal{A}$.

Proof. For $m = 0$ the set $L(0)$ consists of the empty word, so $\mathcal{A}$ has at most one class, of size 1. For $m \geq 1$, Proposition 2.6 and the induction hypothesis give

$$\|\mathcal{A}\| \leq \|\mathcal{G}_0\| + \|\mathcal{G}_1\| \leq 2^{m-1} + 2^{m-1} = 2^m.$$

If $\|\mathcal{A}\| = 2^m$, both inequalities are equalities; the second forces $\|\mathcal{G}_0\| = \|\mathcal{G}_1\| = 2^{m-1}$, and the first gives the statement on split classes by Proposition 2.6(ii). □

Theorem 2.7 concerns words over two letters. Section 7 extends it to words over d letters, that is to monotone paths in $\mathbb{Z}^d$, where the bound becomes d^m, and a weighted form for two letters is proved in Section 4. The equality case of Theorem 2.7 is the subject of Section 3.[4]

2.4. **Shankar's theorems for the square.** For paths from $(0, 0)$ to (n, n) the two parity problems were treated in [35].

Theorem 2.8 (Shankar [35])**.** *Let $n \geq 1$.*

(i) $\mathrm{M}_{\mathrm{ev}}(n, n) = 2^n$.

(ii) $\mathrm{M}_{\mathrm{od}}(n, n) \leq n(n-1) + 1$.

Proof. (i) An even family $\mathcal{F}$ of paths from $(0, 0)$ to (n, n) is a one-class admissible system of length $2n$ by Example 2.4, so $|\mathcal{F}|^2 \leq 2^{2n}$ by Theorem 2.7. For the lower bound, let $\mathcal{F}$ consist of the 2^n words $w_1 \cdots w_n$ with $w_i \in \{01, 10\}$. Let $x, y \in \mathcal{F}$. Before each position $2i-1$ both prefixes have weight $i-1$, so position $2i-1$ counts exactly when $x_{2i-1} = y_{2i-1}$. In that case the two blocks w_i agree, the prefixes $x_{<2i}$ and $y_{<2i}$ have equal weight, and position $2i$ counts

[3]The lemma also explains why $\mathcal{G}_0$ is not simply the collection of all nonempty sets A^0 and A^1: a set A^0 may well meet a set A'^1 when neither class is split. This overlap is the starting point of Section 3.

[4]The inequality is far from sharp for a typical system. For $m = 2$ the system with the single class $\{01, 10\}$ and the system of four singletons are extremal, and Section 3 shows that there are no others. The collection $\{\{00\}, \{01, 10\}, \{11\}\}$, of norm 6, is not admissible, since $B_2(00, 01) = 1$ and $B_2(00, 10) = 0$.

as well; otherwise $x_{2i} \neq y_{2i}$ and position $2i$ does not count. Each block therefore contributes 0 or 2 to (2.1), and $\mathcal{F}$ is an even family.

(ii) This is the case $a = b = n$ of Corollary 4.12 in Section 4, where the bound is derived from the rank over $\mathbb{F}_2$ of the adjacency matrix of a grid graph. □

The two bounds have different orders of magnitude, exponential for even families and quadratic for odd ones. Shankar complemented (ii) by odd families of k^2 paths for $n = 2k$ and of $k(k+2)$ paths for $n = 2k+1$, so $\mathrm{M}_{\mathrm{od}}(n, n)$ is of order n^2; the odd problem is taken up in Section 6. The family used in the proof of (i) is one of many that attain 2^n. Shankar attached such a family to every noncrossing perfect matching of $\{1, \ldots, 2n\}$, proved that different matchings give different families, and conjectured that there are no others [35, Conjecture 4.1]. Figure 1 shows the two families for $n = 2$. The conjecture is proved in the next section.

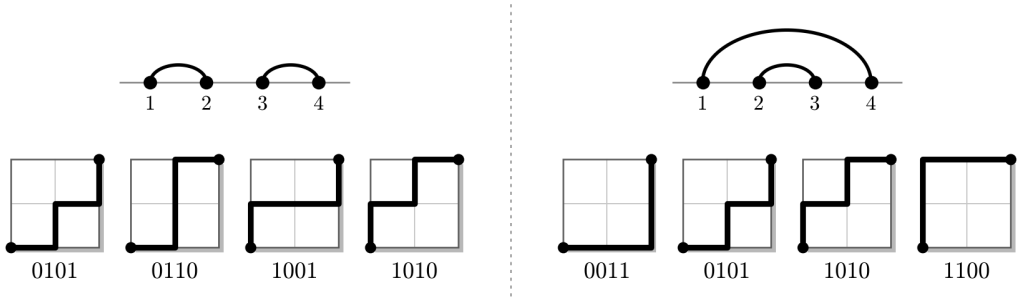


FIGURE 1. The two even families of four paths from $(0, 0)$ to $(2, 2)$. The first consists of the words $w_1 w_2$ with $w_1, w_2 \in \{01, 10\}$; the second consists of the words $x_1 x_2 \bar{x}_2 \bar{x}_1$, where $\bar{x} = 1 - x$. Any two paths of one family share 0 or 2 edges.

3. Skeleton systems and the Catalan classification

This section determines all admissible systems of norm 2^m. They are indexed by certain noncrossing partial matchings of $[m] = \{1, \ldots, m\}$, there are $\binom{m}{\lfloor m/2 \rfloor}$ of them, and those with a single class are the families that Shankar attached to noncrossing perfect matchings. This proves his conjecture on the number of even families of 2^n paths in the square.

3.1. **Skeletons and the statements.**

Definition 3.1. A *skeleton* of $[m]$ is a pair (M, S) in which M is a set of pairwise disjoint two-element subsets of $[m]$, the *arcs*, and $S = [m] \setminus \bigcup M$ is the set of *free points*, subject to two conditions:

(S1) no two arcs cross, that is, there are no arcs $\{p, q\}$ and $\{p', q'\}$ with $p < p' < q < q'$;

(S2) no free point lies under an arc, that is, there are no $s \in S$ and $\{p, q\} \in M$ with $p < s < q$.

Let $s_1 < \cdots < s_k$ be the free points. For $a \in L(k)$ let $C_a = C_a(M, S)$ be the set of words $x \in L(m)$ such that $x_p \neq x_q$ for every arc $\{p, q\}$ and $x_{s_j} = a_j$ for $1 \leq j \leq k$. The *skeleton system* of (M, S) is the collection $\Sigma(M, S) = \{C_a(M, S) : \ a \in L(k)\}$.

A word of C_a is obtained by writing a on the free points and filling each arc with 01 or 10; so $\Sigma(M, S)$ has $2^{|S|}$ classes of $2^{|M|}$ words each, and its norm is $2^{|S|} 4^{|M|} = 2^m$. A skeleton without

free points is a *noncrossing perfect matching* of $[m]$, which requires $m = 2n$ to be even; its system has the single class

$$\mathcal{F}_M = \{x \in L(2n) : \ x_p \neq x_q \text{ for every } \{p, q\} \in M\}.$$

The number of noncrossing perfect matchings of $[2n]$ is the Catalan number $C_n = \frac{1}{n+1}\binom{2n}{n}$ [38, 37]. Figure 2 shows a skeleton with four arcs and three free points.

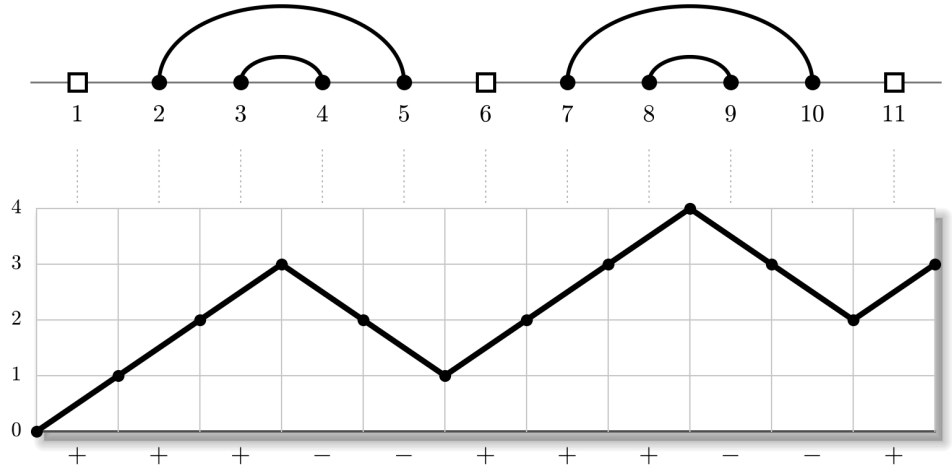


FIGURE 2. A skeleton of [11]: the arcs are $\{2,5\}$, $\{3,4\}$, $\{7,10\}$, $\{8,9\}$ and the free points are 1, 6, 11. Its system has eight classes of sixteen words; the class C_{101} consists of the words $1\,x_2x_3\bar{x}_3\bar{x}_2\,0\,x_7x_8\bar{x}_8\bar{x}_7\,1$ with $\bar{x} = 1 - x$. The lower part shows the sequence of signs used in Proposition 3.13: an up step at every free point and at the left end of every arc, a down step at every right end.

Theorem 3.2. *Let $m \geq 0$. An admissible system of length m is extremal if and only if it is the skeleton system $\Sigma(M, S)$ of a skeleton (M, S) of $[m]$. Distinct skeletons have distinct systems, and the number of extremal admissible systems of length m is $\binom{m}{\lfloor m/2 \rfloor}$.*

Theorem 3.3 (Catalan classification)**.** *Let $n \geq 1$ and let $\mathcal{F}$ be an even family of 2^n lattice paths from $(0,0)$ to (n,n). Then $\mathcal{F} = \mathcal{F}_M$ for exactly one noncrossing perfect matching M of $[2n]$. Consequently the number of even families of 2^n paths from $(0,0)$ to (n,n) is the Catalan number C_n.*[5]

Theorem 3.3 is [35, Conjecture 4.1]; Shankar proved that the families $\mathcal{F}_M$ are even and pairwise distinct, and verified the conjecture for $n \leq 5$. It follows from Theorem 3.2 in a few lines (Section 3.6). The proof of Theorem 3.2 is an induction on m along the last-letter splitting of Section 2. For an extremal system the two systems $\mathcal{G}_0$ and $\mathcal{G}_1$ of length $m-1$ turn out to be equal (Lemma 3.6), so by induction they are one skeleton system, and the classes of the system of length m are obtained by joining some classes of that skeleton system in pairs. A rigidity statement for two words of adjacent weights (Lemma 3.9) restricts the possible pairs, and in the end only two ways of joining remain: no pairs at all, which appends a free point, or all classes paired along the last free point, which closes an arc.

[5]An independent proof of this classification, dated 27 September 2026, appears in an unrefereed manuscript at `https://github.com/FDmd233/catalan-lattice-path-classification`.

3.2. **Skeleton systems are extremal.**

Proposition 3.4. *Let (M, S) be a skeleton of $[m]$ with k free points, let $a, a' \in L(k)$, and let $x \in C_a$ and $y \in C_{a'}$. Then*

$$\mathrm{w}(x) = |M| + \mathrm{w}(a) \qquad \text{and} \qquad B_m(x, y) = B_k(a, a'). \tag{3.1}$$

Hence $\Sigma(M, S)$ is an extremal admissible system of length m. Moreover (M, S) is determined by the union of the classes of $\Sigma(M, S)$.

Proof. Every arc carries one letter 1 of x, and the free points carry the letters of a; this gives the weight. For the second identity put $\Delta_i = \mathrm{w}(x_{<i}) - \mathrm{w}(y_{<i})$, so that position i counts for (x, y) exactly when $x_i = y_i$ and $\Delta_i = 0$. We compare the positions that count with those that count for (a, a'), treating free points and arcs separately.

Let $s = s_j$ be a free point. By (S2) every arc lies entirely to the left or entirely to the right of s, and an arc to the left contributes one letter 1 to $x_{<s}$ and one to $y_{<s}$. The free points to the left of s carry $a_1 \cdots a_{j-1}$ in x and $a'_1 \cdots a'_{j-1}$ in y. Therefore $\Delta_s = \mathrm{w}(a_{<j}) - \mathrm{w}(a'_{<j})$, and as $x_s = a_j$ and $y_s = a'_j$, position s counts for (x, y) exactly when position j counts for (a, a').

Let $\{p, q\}$ be an arc with $p < q$. A position strictly between p and q is not free by (S2), and the arc through it cannot cross $\{p, q\}$ by (S1), so it has both ends strictly between p and q. Each of these inner arcs adds one letter 1 to both words, whence $\Delta_q = \Delta_p + x_p - y_p$. If $x_p = y_p$, then also $x_q = y_q$ and $\Delta_q = \Delta_p$, so the positions p and q either both count or both do not. If $x_p \neq y_p$, then $x_q \neq y_q$ and neither position counts. Every arc thus contributes an even number to (2.1), and the free points contribute $|E(a) \cap E(a')|$. This proves (3.1).

The classes C_a are nonempty and pairwise disjoint. By (3.1) the weight is constant on C_a, the form B_m is constant on $C_a \times C_{a'}$, and on $C_a \times C_a$ it equals $B_k(a, a) = k = m - 2|M|$, which is m in $\mathbb{F}_2$. So $\Sigma(M, S)$ is admissible, and its norm was computed after Definition 3.1.

Let U be the union of the classes, the set of all words x with $x_p \neq x_q$ for every arc. For two positions $p \neq q$ that do not form an arc, the letters x_p and x_q of a word of U can be prescribed independently, in particular equal. So the arcs are exactly the pairs $\{p, q\}$ with $x_p \neq x_q$ for all $x \in U$, and U determines M and $S = [m] \setminus \bigcup M$. □

Example 3.5. The skeletons of [2] are the arc $\{1, 2\}$, with the system $\{\{01, 10\}\}$, and two free points, with the system of four singletons. The skeletons of [3] are the arc $\{1, 2\}$ with the free point 3, the arc $\{2, 3\}$ with the free point 1, and three free points; the pair $\{1, 3\}$ with the point 2 violates (S2). For the first of these the classes are $C_0 = \{010, 100\}$ and $C_1 = \{011, 101\}$, with $B_3 = 1$ inside each class and $B_3 = B_1(0, 1) = 0$ between them. The counts 2 and 3 agree with $\binom{2}{1}$ and $\binom{3}{1}$.

Condition (S2) cannot be dropped. For the arc $\{1, 3\}$ and the point 2 of [3] the analogous sets would be $\{001, 100\}$ and $\{011, 110\}$, and $B_3(001, 100) = 0$ instead of 1; the two paths EEN and NEE are edge-disjoint.[6]

3.3. **The two shorter systems coincide.** Throughout this subsection $\mathcal{A}$ is an extremal admissible system of length $m \geq 1$, and we use the sets A^0, A^1 and the collections $\mathcal{G}_0$, $\mathcal{G}_1$ of (2.2). By Theorem 2.7 both $\mathcal{G}_0$ and $\mathcal{G}_1$ are extremal systems of length $m - 1$. Put

$$\mathcal{K} = \{A^0,\ A^1 :\ A \text{ split}\}, \qquad \mathcal{P} = \{A^0 :\ A \in \mathcal{A},\ A^1 = \varnothing\}, \qquad \mathcal{Q} = \{A^1 :\ A \in \mathcal{A},\ A^0 = \varnothing\},$$

[6]The identity (3.1) says that the map $C_a \mapsto a$ carries the form B_m on a skeleton system to the form B_k on $L(k)$. In this sense every extremal system is a copy of the system of all singletons of $L(k)$, with each word replaced by a class of $2^{|M|}$ words.

so that $\mathcal{G}_0 = \mathcal{K} \cup \mathcal{P}$ and $\mathcal{G}_1 = \mathcal{K} \cup \mathcal{Q}$. Each of these two unions consists of pairwise disjoint sets, by Proposition 2.6, while a set of $\mathcal{P}$ and a set of $\mathcal{Q}$ may overlap: this happens when $u0$ and $u1$ both occur in classes of $\mathcal{A}$ that are not split. Figure 3 displays the three kinds of classes.

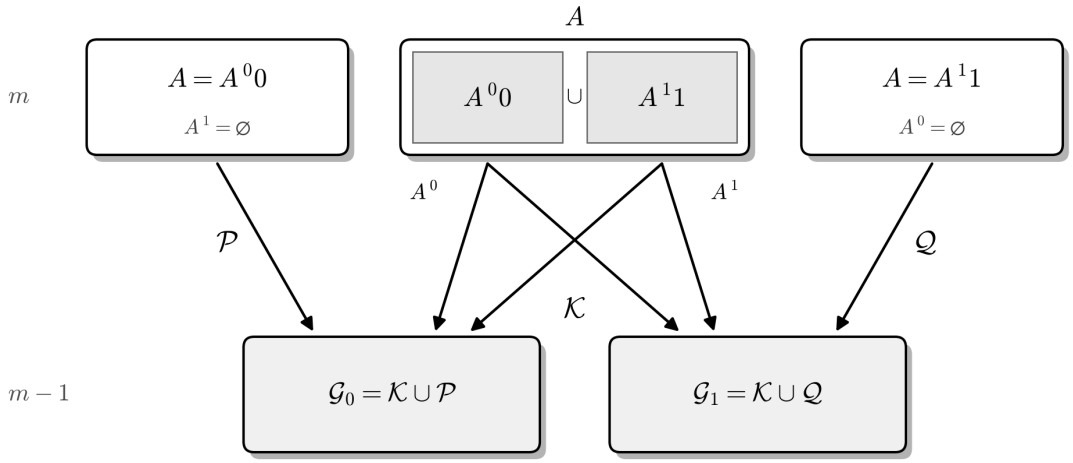


FIGURE 3. The last-letter splitting. A split class A sends both A^0 and A^1 to $\mathcal{G}_0$ and to $\mathcal{G}_1$ (the collection $\mathcal{K}$). A class whose words all end in 0 sends its set of prefixes to $\mathcal{G}_0$ only (the collection $\mathcal{P}$), and a class whose words all end in 1 sends it to $\mathcal{G}_1$ only (the collection $\mathcal{Q}$). For an extremal system $\mathcal{P} = \mathcal{Q}$.

Lemma 3.6. *If $\mathcal{A}$ is extremal, then $\mathcal{P} = \mathcal{Q}$ as collections of sets, and therefore $\mathcal{G}_0 = \mathcal{G}_1$.*

Proof. Let H be the bipartite graph whose vertices are the members of $\mathcal{P}$ on one side and the members of $\mathcal{Q}$ on the other, with $P \in \mathcal{P}$ adjacent to $Q \in \mathcal{Q}$ when $P \cap Q \neq \varnothing$. For a connected component Γ of H let $U_\Gamma \subseteq L(m-1)$ be the union of its vertices. We show first that

$$\mathcal{A}^* = \mathcal{K} \cup \{U_\Gamma : \ \Gamma \text{ a component of } H\}$$

is an admissible system of length $m-1$.

The sets U_Γ are nonempty. Sets in different components are disjoint, by the definition of H for a set of $\mathcal{P}$ and a set of $\mathcal{Q}$ and by Proposition 2.6 otherwise; so the sets U_Γ are pairwise disjoint. They are disjoint from the members of $\mathcal{K}$, since $\mathcal{K} \cup \mathcal{P}$ and $\mathcal{K} \cup \mathcal{Q}$ consist of pairwise disjoint sets. Adjacent vertices share a word, so all words of U_Γ have one weight. This gives (A1).

Next we claim that B_{m-1} is constant on $X \times Y$ for any two vertices or members of $\mathcal{K}$, X and Y, equal or not. If both lie in $\mathcal{G}_0$, or both in $\mathcal{G}_1$, this is the admissibility of that system, the constant being $m-1$ when $X = Y$. The remaining case is $X = A^0 \in \mathcal{P}$ and $Y = A'^1 \in \mathcal{Q}$. Then $A \neq A'$, and Lemma 2.1(b) gives $B_{m-1}(u, v) = B_m(u0, v1)$ for $u \in X$ and $v \in Y$, which is constant by (A3) for $\mathcal{A}$. Write $c(X, Y)$ for the constant. If X and X' have a word z in common, then $c(X, Y) = B_{m-1}(z, v) = c(X', Y)$ for every Y and any $v \in Y$. Hence $c(X, Y)$ does not change when X, or by symmetry Y, is replaced by a vertex of the same component.

It follows that B_{m-1} is constant on $X \times U_\Gamma$ for $X \in \mathcal{K}$ and on $U_\Gamma \times U_{\Gamma'}$ for two different components; this is (A3) for $\mathcal{A}^*$. For two vertices X, Y of the same component, $c(X, Y) = c(X, X) = m-1$; this is (A2) on U_Γ. On the members of $\mathcal{K}$, (A2) and (A3) hold because $\mathcal{K} \subseteq \mathcal{G}_0$. So $\mathcal{A}^*$ is admissible.

Now we compare norms. Theorem 2.7 applied to $\mathcal{A}^*$, and the extremality of $\mathcal{G}_0 = \mathcal{K} \cup \mathcal{P}$, give

$$\|\mathcal{K}\| + \sum_{\Gamma} |U_\Gamma|^2 \;\le\; 2^{m-1} \;=\; \|\mathcal{K}\| + \sum_{\Gamma} \sum_{P \in \Gamma \cap \mathcal{P}} |P|^2.$$

The members of $\mathcal{P}$ in a component are pairwise disjoint subsets of U_Γ, so

$$|U_\Gamma|^2 \;\ge\; \Big(\sum_{P \in \Gamma \cap \mathcal{P}} |P| \Big)^2 \;\ge\; \sum_{P \in \Gamma \cap \mathcal{P}} |P|^2$$

for every Γ. Summed over Γ these inequalities go in the direction opposite to the previous display, so each of them is an equality. Since $U_\Gamma \neq \varnothing$, equality on the left shows that Γ contains a member of $\mathcal{P}$ and that U_Γ is the union of the members of $\mathcal{P}$ in Γ; equality on the right, for nonempty sets, shows that there is only one such member. So $U_\Gamma = P$ for the unique $P \in \Gamma \cap \mathcal{P}$. The same argument with $\mathcal{G}_1 = \mathcal{K} \cup \mathcal{Q}$ gives $U_\Gamma = Q$ for the unique $Q \in \Gamma \cap \mathcal{Q}$. Every component therefore consists of one member of $\mathcal{P}$ and one member of $\mathcal{Q}$, and these are the same set. Hence $\mathcal{P} = \mathcal{Q}$. □

By Lemma 3.6, each class of $\mathcal{G}_0$ either is a member of $\mathcal{K}$, and then it is joined in $\mathcal{A}$ with exactly one other class of $\mathcal{G}_0$ to form a split class, or it is a member P of $\mathcal{P} = \mathcal{Q}$, and then both $P0$ and $P1$ are classes of $\mathcal{A}$. When $\mathcal{G}_0$ is a skeleton system, this is recorded by the index words of its classes.

Definition 3.7. Let $k \ge 0$. A *grouping* of $L(k)$ is a partition of $L(k)$ into blocks that are singletons $\{c\}$ or ordered pairs (a, b) with $a \neq b$. Its *items* are $(a, 0)$ and $(b, 1)$ for every pair (a, b), and $(c, 0)$ and $(c, 1)$ for every singleton $\{c\}$. For $a, b, c \in L(k)$ put

$$f_{a,b}(c) = B_k(a, c) + B_k(b, c) \in \mathbb{F}_2.$$

The grouping is *admissible* if the following two conditions hold.

(G1) Every pair (a, b) satisfies $\mathrm{w}(a) = \mathrm{w}(b) + 1$ and $B_k(a, b) = k + 1$.

(G2) For every pair (a, b) and every item (c, e) with $c \notin \{a, b\}$,

$$f_{a,b}(c) = [\, e = 0 \,]\, [\, \mathrm{w}(c) = \mathrm{w}(a) \,] + [\, e = 1 \,]\, [\, \mathrm{w}(c) = \mathrm{w}(b) \,].$$

An item (c, e) stands for the set $C_c e$ of words of length m, and the items of a block together make up one or two classes. Two consequences of (G2) will be used. Let (a, b) be a pair of an admissible grouping.

(G3) If $\{c\}$ is a singleton, then $\mathrm{w}(c) \notin \{\mathrm{w}(a), \mathrm{w}(b)\}$ and $f_{a,b}(c) = 0$.

(G4) If $c \in L(k)$ and $\mathrm{w}(c) \notin \{\mathrm{w}(a), \mathrm{w}(b)\}$, then $f_{a,b}(c) = 0$.

For (G3), apply (G2) to the items $(c, 0)$ and $(c, 1)$: this gives $[\, \mathrm{w}(c) = \mathrm{w}(a) \,] = f_{a,b}(c) = [\, \mathrm{w}(c) = \mathrm{w}(b) \,]$, and since $\mathrm{w}(a) \neq \mathrm{w}(b)$ both brackets vanish. For (G4), the word c is different from a and b and occurs in at least one item (c, e), for which the right-hand side of (G2) is 0.

Lemma 3.8. *Let $\mathcal{A}$ be an extremal admissible system of length $m \ge 1$, and suppose that $\mathcal{G}_0 = \Sigma(M', S')$ for a skeleton (M', S') of $[m-1]$ with k free points. Write $C_a = C_a(M', S')$. Then there is an admissible grouping of $L(k)$ such that the classes of $\mathcal{A}$ are the sets $C_a 0 \cup C_b 1$ for its pairs (a, b) and the sets $C_c 0$ and $C_c 1$ for its singletons $\{c\}$.*

Proof. By Lemma 3.6 the classes C_a, $a \in L(k)$, are the members of $\mathcal{K}$ and of $\mathcal{P} = \mathcal{Q}$. For a split class A write $A^0 = C_a$ and $A^1 = C_b$ and let (a, b) be a pair; then $A = C_a 0 \cup C_b 1$. For a member C_c of $\mathcal{P} = \mathcal{Q}$ there are a class A with $A = C_c 0$ and a class A' with $A' = C_c 1$; let $\{c\}$ be a singleton. The blocks so defined partition $L(k)$, and every class of $\mathcal{A}$ arises from exactly one block.

By Proposition 3.4, the words of C_a have weight $|M'| + \mathrm{w}(a)$ and $B_{m-1}(x, y) = B_k(a, a')$ for $x \in C_a$, $y \in C_{a'}$; also $m - 1 = k + 2|M'|$. Let (a, b) be a pair. Condition (A1) for the class $C_a0 \cup C_b1$ gives $\mathrm{w}(a) = \mathrm{w}(b) + 1$. For $x \in C_a$ and $y \in C_b$, condition (A2) and Lemma 2.1(b) give $m = B_m(x0, y1) = B_{m-1}(x, y) = B_k(a, b)$, and $m = k + 1$ in $\mathbb{F}_2$. This is (G1).

Let (c, e) be an item with $c \notin \{a, b\}$, and let A' be the class of $\mathcal{A}$ that contains C_ce. It differs from $A = C_a0 \cup C_b1$, so by (A3) the values $B_m(x0, ze)$ and $B_m(y1, ze)$ agree for $x \in C_a$, $y \in C_b$, $z \in C_c$. By Lemma 2.1(b) and (3.1) these values are

$$B_k(a, c) + [\, e = 0\,]\,[\,\mathrm{w}(a) = \mathrm{w}(c)\,] \qquad \text{and} \qquad B_k(b, c) + [\, e = 1\,]\,[\,\mathrm{w}(b) = \mathrm{w}(c)\,],$$

and their equality is (G2). □

3.4. **A rigidity lemma for two words of adjacent weights.** Condition (G4) says that no word outside the two weight levels of a pair (a, b) distinguishes a from b through the form B_k. The next lemma shows that this is very restrictive. For $j \geq 1$ let $(10)^j$ and $(01)^j$ denote the words of length $2j$ obtained by repeating 10 and 01.

Lemma 3.9 (Rigidity)**.** *Let $k \geq 1$ and let $a, b \in L(k)$ satisfy*

(R1) $\mathrm{w}(a) = \mathrm{w}(b) + 1$ *and* $B_k(a, b) = k + 1$*;*

(R2) $B_k(a, c) = B_k(b, c)$ *for every $c \in L(k)$ with $\mathrm{w}(c) \notin \{\mathrm{w}(a), \mathrm{w}(b)\}$.*

Then either $a = u1$ and $b = u0$ for some $u \in L(k-1)$, or $k = 2j+1 \geq 3$ is odd and $a = 1(10)^j$, $b = 0(01)^j$.

The second alternative is genuine: for $k = 3$ the pair $a = 110$, $b = 001$ satisfies (R1) and (R2), as the case $k = 3$ of the proof shows. It will be excluded in Proposition 3.12 by using (G2) in full. The paths of $1(10)^j$ and $0(01)^j$ run on the two sides of the staircase of unit squares along the diagonal and have no edge in common (Figure 4).

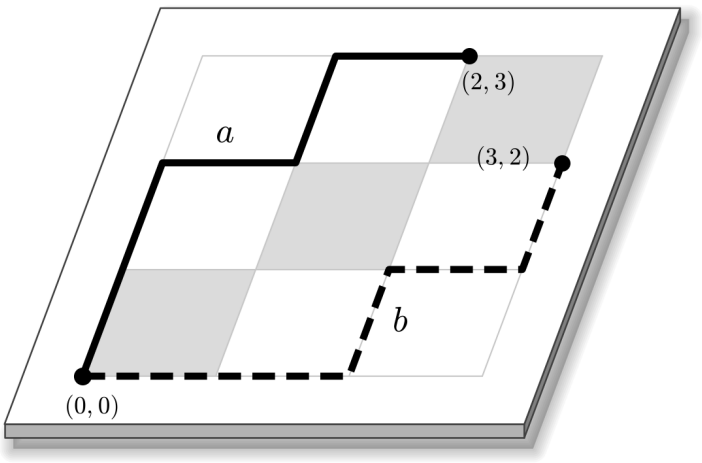


FIGURE 4. The exceptional pair of Lemma 3.9 for $k = 5$: the paths of $a = 11010$ (above) and $b = 00101$ (below). They occupy different points at every level from 1 to 5, so they share no edge, and $B_5(a, b) = 0 = k + 1$. The unit squares with lower left corners $(0, 0)$, $(1, 1)$ and $(2, 2)$ lie between them.

The proof compares a word c with the word obtained from it by exchanging two consecutive letters. For $p \in L(k)$ and $0 \leq \ell \leq k$ the path of p has exactly one point at level ℓ, namely $(\ell - \mathrm{w}(p_{\leq \ell}), \mathrm{w}(p_{\leq \ell}))$. Let $V_p(v) \in \mathbb{F}_2$ be 1 if the path of p passes through the lattice point v

and 0 otherwise. For a lattice point v let Q_v be the unit square with lower left corner v and ∂Q_v the set of its four sides.

Lemma 3.10. *Let $p \in L(k)$ and let v be a lattice point of level $\ell \le k-2$ with nonnegative coordinates.*

(a) $|E(p) \cap \partial Q_v| = V_p(v) + V_p(v+(1,1))$ *as integers.*

(b) *Let $c \in L(k)$ pass through v and through $v+(1,1)$, and let c' be obtained from c by exchanging the letters in positions $\ell+1$ and $\ell+2$. Then*

$$B_k(p,c) + B_k(p,c') = V_p(v) + V_p(v+(1,1)) \qquad \text{in } \mathbb{F}_2.$$

Proof. (a) The bottom and the left side of Q_v are the two edges that start at v. As $\ell < k$, a path through v continues along exactly one of them, and a path that avoids v contains neither. The top and the right side are the two edges that end at $v+(1,1)$, a point of level $\ell+2 \le k$; a path through that point arrives along exactly one of them, and a path avoiding it contains neither.

(b) The letters of c in positions $\ell+1$ and $\ell+2$ are different, since they lead from v to $v+(1,1)$. So c and c' coincide outside these positions, where one reads 01 and the other 10: the two paths pass around Q_v on opposite sides, and the symmetric difference of $E(c)$ and $E(c')$ is ∂Q_v. Hence $|E(p) \cap E(c)| + |E(p) \cap E(c')| \equiv |E(p) \cap \partial Q_v| \pmod 2$, and (a) finishes the proof. □

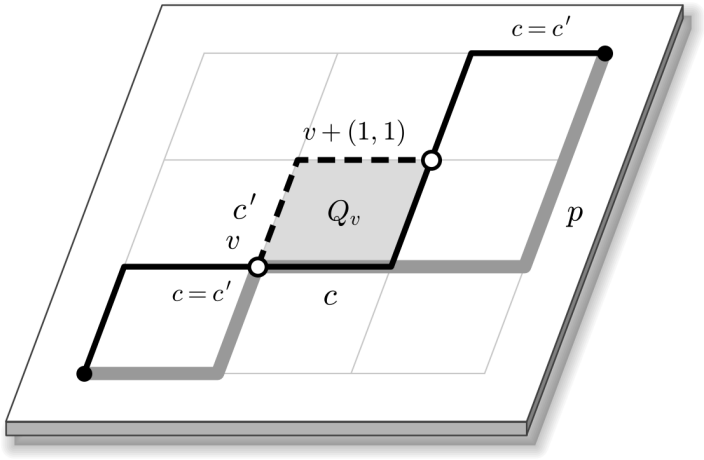


FIGURE 5. Lemma 3.10. Exchanging two consecutive different letters of c moves its path across one unit square Q_v, so the edge sets of c and c' differ by the four sides of Q_v. A path p contains one of the two sides at v if it passes through v, and one of the two sides at $v+(1,1)$ if it passes through that point.

Proof of Lemma 3.9. Put $f = f_{a,b}$, so that (R2) reads $f(c) = 0$ for $\mathrm{w}(c) \notin \{\mathrm{w}(a), \mathrm{w}(b)\}$. For $p \in L(k)$ and a letter e let $\lambda_e(p)$ be the length of the initial run of letters e in p (zero if $p_1 \ne e$). The path of the constant word 0^k follows the horizontal axis, and the path of p shares with it exactly its first $\lambda_0(p)$ edges; likewise for 1^k and the vertical axis. Thus

$$B_k(p, 0^k) = \lambda_0(p), \qquad B_k(p, 1^k) = \lambda_1(p) \qquad \text{in } \mathbb{F}_2. \tag{3.2}$$

The cases $k \le 3$. For $k = 1$ the only pair with $\mathrm{w}(a) = \mathrm{w}(b) + 1$ is $(1,0)$, of the first kind with u empty. For $k = 2$, (R1) requires $B_2(a,b) = 1$. If $b = 00$, then $a \in \{01, 10\}$ and by (3.2)

$B_2(a, 00) = \lambda_0(a)$, which is 1 only for $a = 01$. If $\mathrm{w}(b) = 1$, then $a = 11$ and $B_2(11, b) = \lambda_1(b)$, which is 1 only for $b = 10$. Both pairs, $(01, 00)$ and $(11, 10)$, are of the first kind. For $k = 3$, (R1) requires $B_3(a, b) = 0$. If $b = 000$, then $\lambda_0(a)$ is even, so $a \in \{100, 001\}$; the word $c = 111$ has weight $3 \notin \{1, 0\}$, and $f(111) = \lambda_1(a) + \lambda_1(000) = \lambda_1(a)$ must vanish, which excludes $a = 100$. So $(a, b) = (001, 000)$. If $\mathrm{w}(b) = 2$, the reflection $x \mapsto \bar{x}$, which turns the pair (a, b) into the pair $(\bar{b}, \bar{a})$ and preserves (R1) and (R2), reduces to the previous case and gives $(a, b) = (111, 110)$. If $\mathrm{w}(b) = 1$, both 000 and 111 are available in (R2), so $\lambda_0(a) \equiv \lambda_0(b)$ and $\lambda_1(a) \equiv \lambda_1(b)$ modulo 2. The residues of (λ_0, λ_1) are $(0, 0), (0, 1), (1, 0)$ for $a = 110, 101, 011$ and $(0, 1), (1, 0), (0, 0)$ for $b = 100, 010, 001$, so (a, b) is one of $(101, 100)$, $(011, 010)$, $(110, 001)$. The first two are of the first kind and the third is the exceptional pair with $j = 1$.[7]

The function J. From now on $k \geq 4$. Let $J = V_a + V_b$, a function on lattice points with values in $\mathbb{F}_2$. At a given level each of the two paths has one point, so J equals 1 at exactly two points of the level if the paths are apart there, and vanishes on the whole level if they meet. Call a lattice point $v = (x, y)$ with $x, y \geq 0$ and level $\ell \leq k - 2$ *testable* if some word c with $\mathrm{w}(c) \notin \{\mathrm{w}(a), \mathrm{w}(b)\}$ passes through v and $v + (1, 1)$. For such c, the word c' of Lemma 3.10(b) has the same weight, so $f(c) = f(c') = 0$; adding the identities of Lemma 3.10(b) for $p = a$ and $p = b$ gives

$$J(v) = J\big(v + (1, 1)\big) \qquad \text{for every testable point } v. \tag{3.3}$$

A word through $v = (x, y)$ and $v + (1, 1)$ has $y + 1$ letters 1 among its first $\ell + 2$ letters, and its remaining $k - \ell - 2$ letters are arbitrary. Its weight can therefore be any integer in the interval $[\, y + 1,\ y + k - \ell - 1 \,]$, which contains $k - \ell - 1$ integers, whereas the excluded weights $\mathrm{w}(b)$ and $\mathrm{w}(b) + 1$ are two consecutive integers. Consequently:

- (T1) every point of level $\ell \leq k - 4$ is testable, since the interval has at least three elements;
- (T2) a point (x, y) of level $k - 3$ is testable unless $y = \mathrm{w}(b) - 1$, since the interval is $\{y + 1, y + 2\}$.

Propagation along diagonals. For $0 \leq \ell \leq k$ let D_ℓ be the set of the numbers $x - y$ for the points (x, y) of level ℓ with $J(x, y) = 1$. It has no element or two elements, they are congruent to ℓ modulo 2 and lie in $[-\ell, \ell]$, and $D_0 = \varnothing$. Let $\ell \leq k - 4$ and let $d \equiv \ell$ be an integer with $|d| \leq \ell$. The points with $x - y = d$ at the levels ℓ and $\ell + 2$ are v and $v + (1, 1)$ for a point v with nonnegative coordinates, which is testable by (T1). So (3.3) yields

$$D_{\ell+2} \cap [-\ell, \ell] = D_\ell \qquad (0 \leq \ell \leq k - 4). \tag{3.4}$$

In particular $D_\ell \subseteq D_{\ell+2}$, and since neither set has more than two elements, $D_{\ell+2} = D_\ell$ whenever $D_\ell \neq \varnothing$.

Suppose that for one of the two parities there is a level $\ell \leq k-2$ of that parity with $D_\ell \neq \varnothing$, let ℓ_0 be the smallest one, and suppose $\ell_0 \geq 2$. Then $D_{\ell_0-2} = \varnothing$, and (3.4) shows that D_{ℓ_0} has no element in $[-(\ell_0 - 2), \ell_0 - 2]$. Hence $D_{\ell_0} = \{\ell_0, -\ell_0\}$: at level ℓ_0 the two paths are at $(\ell_0, 0)$ and $(0, \ell_0)$, so one word begins with 0^{ℓ_0} and the other with 1^{ℓ_0}, and the paths are apart at all levels $1, \ldots, \ell_0$. But $D_{\ell_0-2} = \varnothing$ means that they meet at level $\ell_0 - 2$. This forces $\ell_0 = 2$, and then $D_1 \neq \varnothing$. In summary:

(3.5)
the smallest level $\ell_0 \leq k-2$ of a given parity with $D_{\ell_0} \neq \varnothing$ is 1 or 2, and $\ell_0 = 2$ forces $D_1 \neq \varnothing$.

We distinguish three cases.

Case 1: $a_1 = b_1$. Then $D_1 = \varnothing$. By (3.5) no odd level $\ell \leq k - 2$ has $D_\ell \neq \varnothing$, since the smallest would be 1, and no even level either, since the smallest would be 2 and would force

[7] The pair $(110, 001)$ does satisfy (R1) and (R2): the two paths are edge-disjoint, so $B_3 = 0$, and the only words of weight outside $\{1, 2\}$ are 000 and 111, for which the residues were just compared.

$D_1 \neq \varnothing$. So the paths meet at every level up to $k-2$, and the words a and b agree in their first $k-2$ letters. They are different words; let $t \in \{k-1, k\}$ be the first position where they differ. If $t = k-1$, the positions $1, \dots, k-2$ count for (a, b), position $k-1$ does not, and position k does not either because the prefixes $a_{<k}$ and $b_{<k}$ differ in exactly one letter. Then $B_k(a, b) = k-2$, contrary to (R1). So $t = k$, and $\mathrm{w}(a) = \mathrm{w}(b) + 1$ gives $a = u1$, $b = u0$ with $u = a_{<k}$.

Case 2: $a_1 \neq b_1$ and $D_\ell = \varnothing$ for every even $\ell \leq k-2$. As $k \geq 4$, the paths meet at level 2, necessarily at $(1, 1)$; one word begins with 10 and the other with 01. Both weights are then at least 1, so $c = 0^k$ is allowed in (R2). By (3.2), $f(0^k) = \lambda_0(a) + \lambda_0(b) = 0 + 1$, a contradiction.

Case 3: $a_1 \neq b_1$ and $D_\ell \neq \varnothing$ for some even $\ell \leq k-2$. By (3.5) the smallest such level is 2, and the argument that led to (3.5) gives $D_2 = \{2, -2\}$; also $D_1 = \{1, -1\}$. By (3.4) and the remark following it,

$$D_\ell = \{1, -1\} \text{ for odd } \ell \leq k-2, \qquad D_\ell = \{2, -2\} \text{ for even } 2 \leq \ell \leq k-2.$$

In particular the paths are apart at every level from 1 to $k-2$.

Let k be even. At level $k-2$ the values of $x - y$ on the two paths are 2 and -2, so at level $k-1$ they lie in $\{1, 3\}$ and in $\{-3, -1\}$, and the paths are apart there as well. A position $i \geq 2$ counts only if the paths meet at level $i-1$, and position 1 does not count because $a_1 \neq b_1$. So $B_k(a, b) = 0$, whereas $k+1$ is odd; this contradicts (R1).

Let $k = 2j+1$ be odd; then $j \geq 2$. Let p^+ be the one of the two words whose path has $x - y = 2$ at level 2, and p^- the other. The value of $x - y$ changes by ± 1 at each step, and on the path of p^+ it must lie in D_ℓ at every level $\ell \leq k-2$; starting from 2 at level 2 it is therefore 1 at odd levels and 2 at even levels. So p^+ begins with $0(01)^{j-1}$, and its path passes through $v^+ = (j, j-2)$ at level $k-3$ and through $(j, j-1)$ at level $k-2$. Symmetrically, p^- begins with $1(10)^{j-1}$ and passes through $v^- = (j-2, j)$ and $(j-1, j)$. We have $J(v^+) = J(v^-) = 1$.

Suppose v^+ is testable. By (3.3), $J(j+1, j-1) = 1$, so one of the paths passes through $(j+1, j-1)$, a point of level $k-1$. It is not the path of p^-, which is at $(j-1, j)$ one level earlier. So the letter of p^+ in position $k-1$ is 0, and $\mathrm{w}(p^+) \leq (j-1) + 0 + 1 = j$. Suppose v^- is testable. In the same way the path of p^- passes through $(j-1, j+1)$, its letter in position $k-1$ is 1, and $\mathrm{w}(p^-) \geq j+1$.

The points v^+ and v^- have level $k-3$ and different second coordinates, so by (T2) at least one of them is testable. If v^+ is not testable, then $j-2 = \mathrm{w}(b)-1$, so both weights are at most j; but v^- is testable and gives a weight at least $j+1$. If v^- is not testable, then $j = \mathrm{w}(b) - 1$, so both weights are at least $j+1$; but v^+ is testable and gives a weight at most j. Hence both points are testable, $\mathrm{w}(p^+) \leq j$ and $\mathrm{w}(p^-) \geq j+1$. As the two weights differ by 1, we get $\mathrm{w}(p^+) = j$ and $\mathrm{w}(p^-) = j+1$, so $p^+ = b$ and $p^- = a$. The word b begins with $0(01)^{j-1}$, has 0 in position $k-1$ and weight j, so $b = 0(01)^{j-1}01 = 0(01)^j$; and a begins with $1(10)^{j-1}$, has 1 in position $k-1$ and weight $j+1$, so $a = 1(10)^{j-1}10 = 1(10)^j$. □

3.5. **Admissible groupings.** A pair of the form $(u1, u0)$ with $u \in L(k-1)$ is called *canonical*. For canonical pairs the function f of Definition 3.7 is explicit.

Lemma 3.11. *For $u \in L(k-1)$ and $c \in L(k)$,*

$$f_{u1,u0}(c) = [\,\mathrm{w}(c_{<k}) = \mathrm{w}(u)\,].$$

Proof. By Lemma 2.1(b), $B_k(ue, c) = B_{k-1}(u, c_{<k}) + [\,e = c_k\,]\,[\,\mathrm{w}(u) = \mathrm{w}(c_{<k})\,]$ for $e \in \{0, 1\}$. Exactly one of the two letters e equals c_k, so the sum over e is the right-hand side. □

Proposition 3.12. *Let $k \geq 0$. An admissible grouping of $L(k)$ either has no pair, or it has no singleton and its pairs are the canonical pairs $(u1, u0)$, $u \in L(k-1)$.*

Proof. Suppose that the grouping has a pair; then $k \geq 1$. By (G1) and (G4) every pair satisfies (R1) and (R2), so by Lemma 3.9 every pair is canonical or is the exceptional pair $(a^*, b^*) = (1(10)^j, 0(01)^j)$ with $k = 2j + 1 \geq 3$.

The exceptional pair does not occur. Suppose (a^*, b^*) is a pair of the grouping. We have $\mathrm{w}(a^*) = j + 1$ and $\mathrm{w}(b^*) = j$; the word a^* ends in 0 and b^* ends in 1, so $\mathrm{w}(a^*_{<k}) = j + 1$ and $\mathrm{w}(b^*_{<k}) = j - 1$. There are $\binom{2j+1}{j} \geq 3$ words of weight j; let $c \neq b^*$ be one of them. By (G3) the word c does not form a singleton, so it belongs to a pair other than (a^*, b^*), which is canonical, say $(u1, u0)$. Here $\mathrm{w}(u) = j - 1$ if $c = u1$ and $\mathrm{w}(u) = j$ if $c = u0$. Since the blocks are disjoint, a^* and b^* are different from $u1$ and $u0$, and (G2) applies to the pair $(u1, u0)$ and the items $(a^*, 0)$ and $(b^*, 1)$. Together with Lemma 3.11 it gives

$$[\,\mathrm{w}(u) = j + 1\,] = f_{u1,u0}(a^*) = [\,\mathrm{w}(a^*) = \mathrm{w}(u1)\,] = [\,\mathrm{w}(u) = j\,],$$
$$[\,\mathrm{w}(u) = j - 1\,] = f_{u1,u0}(b^*) = [\,\mathrm{w}(b^*) = \mathrm{w}(u0)\,] = [\,\mathrm{w}(u) = j\,].$$

The first line shows $\mathrm{w}(u) \neq j$ and the second shows $\mathrm{w}(u) \neq j - 1$, which contradicts the choice of u.

Every canonical pair occurs. All pairs are now canonical. Let $\Lambda \subseteq \{0, \ldots, k\}$ be the set of weights of the words that belong to pairs; it is not empty. By (G3) a word whose weight lies in Λ is not a singleton, so it belongs to a canonical pair. Let $t \in \Lambda$. If $t \geq 1$, choose a word $v1$ of weight t; the canonical pair containing it is $(v1, v0)$, and $\mathrm{w}(v0) = t - 1$, so $t - 1 \in \Lambda$. If $t \leq k - 1$, choose a word $v0$ of weight t; its pair is $(v1, v0)$, so $t + 1 \in \Lambda$. Hence $\Lambda = \{0, \ldots, k\}$. There is then no singleton, every word $v0$ lies in the pair $(v1, v0)$, and the pairs are exactly the canonical ones. □

Proof of Theorem 3.2, first two assertions. By Proposition 3.4, skeleton systems are extremal and distinct skeletons have distinct systems. We show by induction on m that an extremal admissible system $\mathcal{A}$ of length m is a skeleton system. For $m = 0$ the norm 1 forces $\mathcal{A}$ to consist of the class formed by the empty word, which is $\Sigma(\varnothing, \varnothing)$.

Let $m \geq 1$. By Theorem 2.7, $\mathcal{G}_0$ is an extremal system of length $m - 1$, so by the induction hypothesis $\mathcal{G}_0 = \Sigma(M', S')$ for a skeleton (M', S') of $[m - 1]$, with free points $s_1 < \cdots < s_k$. Lemma 3.8 provides an admissible grouping of $L(k)$ that describes the classes of $\mathcal{A}$, and Proposition 3.12 leaves two possibilities.

If the grouping has no pair, the classes of $\mathcal{A}$ are the sets C_c0 and C_c1 with $c \in L(k)$. These are the classes of $(M', S' \cup \{m\})$, which is a skeleton of $[m]$ because the point m lies under no arc. So $\mathcal{A} = \Sigma(M', S' \cup \{m\})$.

Otherwise $k \geq 1$ and the classes of $\mathcal{A}$ are the sets $C_{u1}0 \cup C_{u0}1$ with $u \in L(k - 1)$. Such a set consists of the words $x \in L(m)$ with $x_p \neq x_q$ for the arcs $\{p, q\}$ of M', with $x_{s_i} = u_i$ for $i < k$, and with $x_{s_k} \neq x_m$. Put $M = M' \cup \{\{s_k, m\}\}$ and $S = S' \setminus \{s_k\}$. Every position strictly between s_k and m lies on an arc of M', because s_k is the last free point, and such an arc has both ends to the right of s_k, because s_k lies under no arc of M'. So the new arc crosses no arc of M' and covers no free point, (M, S) is a skeleton of $[m]$, and $\mathcal{A} = \Sigma(M, S)$. □

The induction shows how extremal systems grow with the length: either a free point is appended, which doubles the number of classes and keeps their size, or the last free point is joined to the new point by an arc, which halves the number of classes and doubles their size. All classes of an extremal system have the same size $2^{|M|}$, and their number $2^{|S|}$ is a power of 2 with $|S| \equiv m \pmod 2$.[8]

[8]In particular an admissible system with exactly two classes, or with classes of two different sizes, has norm strictly less than 2^m. This is the source of the strict inequalities for rectangles in Section 4.

3.6. **The Catalan classification and the number of skeletons.**

Proof of Theorem 3.3. By Example 2.4, $\{\mathcal{F}\}$ is an admissible system of length $2n$, and its norm is $|\mathcal{F}|^2 = 2^{2n}$. By Theorem 3.2 it is the system of a skeleton (M, S) of $[2n]$. A skeleton system has $2^{|S|}$ classes, so $S = \varnothing$; thus M is a noncrossing perfect matching and $\mathcal{F} = \mathcal{F}_M$. The matching is unique by Proposition 3.4. Conversely, for every noncrossing perfect matching M the set $\mathcal{F}_M$ consists of 2^n words of weight n, and by (3.1) with $k = 0$ any two of them satisfy $B_{2n} = 0$; so $\mathcal{F}_M$ is an even family of 2^n paths from $(0, 0)$ to (n, n). The even families of size 2^n are therefore in bijection with the noncrossing perfect matchings of $[2n]$. □

Proposition 3.13. *Let $m \geq 0$ and $0 \leq j \leq m/2$. The number of skeletons of $[m]$ with j arcs, and hence with $k = m - 2j$ free points, is*

$$s(m, j) = \binom{m}{j} - \binom{m}{j-1} = \frac{m - 2j + 1}{m + 1}\binom{m+1}{j}, \tag{3.6}$$

and the total number of skeletons of $[m]$ is $\binom{m}{\lfloor m/2 \rfloor}$.

Proof. To a skeleton (M, S) attach the sequence $\varepsilon \in \{1, -1\}^m$ with $\varepsilon_i = -1$ if i is the larger end of an arc and $\varepsilon_i = 1$ otherwise. Each entry -1 is preceded by the entry 1 of its own arc, so all partial sums of ε are nonnegative; the total sum is k.

Conversely let $\varepsilon \in \{1, -1\}^m$ have nonnegative partial sums. Read it from left to right and match every -1 with the nearest unmatched 1 to its left, which exists because the partial sum before the -1 is positive. Let M be the set of matched pairs and S the set of positions of the entries 1 that remain unmatched. If two arcs $\{p, q\}$, $\{p', q'\}$ had $p < p' < q < q'$, then p' would be unmatched when q is read and nearer to q than p; if a point $s \in S$ had $p < s < q$ for an arc $\{p, q\}$, the same would hold for s. So (M, S) is a skeleton. The two constructions are inverse to each other: for a skeleton, the positions strictly between the ends of an arc $\{p, q\}$ are covered by arcs lying strictly between p and q, so by induction on $q - p$ they are matched among themselves and p is the nearest unmatched 1 when q is read.

It remains to count the sequences with nonnegative partial sums and total sum $k = m - 2j$, which have j entries -1. There are $\binom{m}{j}$ sequences with j entries -1. Those with some negative partial sum are mapped, by reversing the signs of all entries after the first partial sum equal to -1, bijectively onto the sequences with total sum $-k-2$, which have $m-j+1$ entries -1.[9] Their number is $\binom{m}{m-j+1} = \binom{m}{j-1}$. This gives the first expression in (3.6), and the second follows from $\binom{m}{j} = \frac{m+1-j}{m+1}\binom{m+1}{j}$ and $\binom{m}{j-1} = \frac{j}{m+1}\binom{m+1}{j}$. The sum of $\binom{m}{j} - \binom{m}{j-1}$ over $0 \leq j \leq \lfloor m/2 \rfloor$ telescopes to $\binom{m}{\lfloor m/2 \rfloor}$. □

Proposition 3.13 completes the proof of Theorem 3.2. For $m = 2n$ and $j = n$ it gives $s(2n, n) = \binom{2n}{n} - \binom{2n}{n-1} = C_n$ noncrossing perfect matchings, in agreement with Theorem 3.3.

3.7. **Refined enumeration of the extremal systems.** An extremal system $\Sigma(M, S)$ of length m has 2^k classes of size 2^j, where $k = |S|$ and $j = |M|$. Proposition 3.13 therefore counts the extremal systems by their number of classes: there are $s(m, j)$ extremal systems of length m with 2^{m-2j} classes. The numbers $s(m, j)$ are the ballot numbers; the rows for $m \leq 8$ are

$$(1),\ (1),\ (1,1),\ (1,2),\ (1,3,2),\ (1,4,5),\ (1,5,9,5),\ (1,6,14,14),\ (1,7,20,28,14).$$

[9] This is André's reflection. The image has total sum $-1-(k+1) = -k-2$; the first partial sum equal to -1 is unchanged, so applying the same rule to the image recovers the sequence, and every sequence with negative total sum has a partial sum equal to -1.

This subsection records further statistics. Two structural remarks are used. First, by (S2) the free points $s_1 < \cdots < s_k$ of a skeleton cut $[m]$ into $k+1$ intervals, some of them possibly empty, and the arcs inside each interval form a noncrossing perfect matching of it; conversely any choice of $k+1$ noncrossing perfect matchings, placed between consecutive free points, is a skeleton. Counting skeletons with $r-1$ free points in this way, (3.6) shows that the number of r-tuples of noncrossing perfect matchings with n arcs in total is

$$c(r,n) = s(2n+r-1,n) = \frac{r}{2n+r}\binom{2n+r}{n} \qquad (r \geq 1,\ n \geq 0). \tag{3.7}$$

Second, the proof of Theorem 3.2 describes a map from the skeletons of $[m]$ to those of $[m-1]$: delete the point m if it is free, and if it is the right end of an arc $\{s,m\}$, delete the arc and let s be free. A skeleton of $[m-1]$ with k free points has one preimage with $k+1$ free points and, if $k \geq 1$, one further preimage with $k-1$ free points.[10]

An arc of a skeleton is *outermost* if it lies under no other arc, and the *depth* of a skeleton is the largest number of arcs $\{p_1,q_1\},\dots,\{p_h,q_h\}$ with $p_1 < \cdots < p_h < q_h < \cdots < q_1$.

Proposition 3.14 (Outermost arcs)**.** *Let $k \geq 0$, $j \geq 1$ and $1 \leq r \leq j$. The number of skeletons with k free points, j arcs and exactly r outermost arcs is*

$$\binom{k+r}{k}\frac{r}{2j-r}\binom{2j-r}{j}.$$

Proof. Reading from left to right, the free points and the outermost arcs form a sequence of $k+r$ objects, and every arrangement of k points and r arcs occurs, since a free point is never under an arc; this accounts for the factor $\binom{k+r}{k}$. Inside the r outermost arcs one places an r-tuple of noncrossing perfect matchings with $j-r$ arcs in total, and by (3.7) there are $c(r,j-r) = \frac{r}{2j-r}\binom{2j-r}{j-r} = \frac{r}{2j-r}\binom{2j-r}{j}$ such tuples. □

Proposition 3.15 (Totals)**.** *Let $N_m = \binom{m}{\lfloor m/2 \rfloor}$. Summed over all skeletons of $[m]$, the number of free points is $2^m - N_m$ and the number of arcs is $\frac{1}{2}\big((m+1)N_m - 2^m\big)$. Hence a skeleton of $[m]$ chosen uniformly at random has on average $2^m/N_m - 1$ free points, which is asymptotic to $\sqrt{\pi m/2}$.*

Proof. Let F_m be the total number of free points and Z_m the number of skeletons of $[m]$ without free points. By the description of the preimages given above, a skeleton of $[m-1]$ with k free points contributes 1 to N_m and $k+1$ to F_m, and if $k \geq 1$ it contributes 1 and $k-1$ once more. Therefore

$$N_m = 2N_{m-1} - Z_{m-1}, \qquad F_m = (F_{m-1}+N_{m-1}) + \big(F_{m-1} - (N_{m-1} - Z_{m-1})\big) = 2F_{m-1} + Z_{m-1}.$$

Adding the two relations gives $F_m + N_m = 2(F_{m-1}+N_{m-1})$, and $F_0 + N_0 = 1$, so $F_m + N_m = 2^m$. A skeleton with k free points has $(m-k)/2$ arcs, so the total number of arcs is $\frac{1}{2}(mN_m - F_m)$. The asymptotic statement follows from $N_m \sim 2^m\sqrt{2/(\pi m)}$. □

The identity $F_m + N_m = 2^m$ says that the numbers $k+1$ of intervals between free points add up to 2^m over all skeletons of $[m]$.[11]

[10] In the second case s becomes the last free point of the image; it lies under no arc, because an arc over s would cross $\{s,m\}$. So the image is a skeleton, and the two operations of the proof of Theorem 3.2 are the only ways back.

[11] The same recursion handles other weights. If a_m denotes the total number of classes in all extremal systems of length m, that is the sum of 2^k over all skeletons, then $a_m = 2a_{m-1} + \frac{1}{2}(a_{m-1} - Z_{m-1})$, where Z_{m-1} is $C_{(m-1)/2}$ for odd m and 0 for even m. This gives $a_m = 1, 2, 5, 12, 30, 74, 185, 460$ for $m = 0,\dots,7$.

Proposition 3.16 (Bounded depth)**.** *For $h \geq 0$ let $S_h(m)$ be the number of skeletons of $[m]$ of depth at most h. Then*

$$S_h(m) = \frac{4}{2h+3} \sum_{\substack{1 \leq i \leq 2h+1 \\ i \text{ odd}}} \cos^2\Big(\frac{i\pi}{2(2h+3)}\Big) \Big(2\cos\frac{i\pi}{2h+3}\Big)^m.$$

In particular $S_0(m) = 1$, and $S_1(m)$ is the Fibonacci number F_{m+1}, where $F_1 = F_2 = 1$.

Proof. Read a skeleton from left to right and record after each position the number of arcs that have been opened and not yet closed. A left end raises this number by 1, a right end lowers it by 1, and a free point leaves it unchanged and can occur only when it is 0, by (S2). The largest value reached is the depth. Skeletons of depth at most h thus correspond to closed walks of length m from the vertex 0 in the graph Γ_h which is the path with vertices $0, 1, \ldots, h$ together with a loop at 0; the inverse map matches each down step with the nearest unmatched up step to its left, as in the proof of Proposition 3.13. So $S_h(m) = (T^m)_{00}$ for the adjacency matrix T of Γ_h.

For odd i with $1 \leq i \leq 2h+1$ put $\theta_i = i\pi/(2h+3)$ and $v_i(t) = \cos\big((2t+1)\theta_i/2\big)$ for $0 \leq t \leq h+1$. The identity $\cos(\alpha-\theta) + \cos(\alpha+\theta) = 2\cos\theta\cos\alpha$ gives $v_i(t-1) + v_i(t+1) = 2\cos\theta_i\, v_i(t)$ for $1 \leq t \leq h$, and with $\alpha = \theta_i/2$ it gives $v_i(0) + v_i(1) = 2\cos\theta_i\, v_i(0)$. Moreover $v_i(h+1) = \cos(i\pi/2) = 0$. So $(v_i(0), \ldots, v_i(h))$ is an eigenvector of T with eigenvalue $2\cos\theta_i$; these $h+1$ eigenvalues are distinct, and the eigenvectors form an orthogonal basis because T is symmetric. Their squared norms are

$$\sum_{t=0}^{h} \cos^2\frac{(2t+1)\theta_i}{2} = \frac{h+1}{2} + \frac{1}{2}\sum_{t=0}^{h}\cos\big((2t+1)\theta_i\big) = \frac{h+1}{2} + \frac{\sin\big((2h+2)\theta_i\big)}{4\sin\theta_i} = \frac{2h+3}{4},$$

since $(2h+2)\theta_i = i\pi - \theta_i$ and i is odd. The spectral decomposition

$$(T^m)_{00} = \sum_i \frac{v_i(0)^2}{\|v_i\|^2}\,(2\cos\theta_i)^m$$

is the stated formula. For $h = 0$ the graph is a single loop. For $h = 1$ a skeleton of depth at most 1 is a sequence of free points and arcs $\{i, i+1\}$ covering $[m]$, and the number of such sequences satisfies the Fibonacci recursion with the values 1 and 2 for $m = 1, 2$. □

For $h = 2$ the characteristic polynomial of T is $x^3 - x^2 - 2x + 1$, so

$$S_2(m) = S_2(m-1) + 2S_2(m-2) - S_2(m-3) \qquad (m \geq 3),$$

and the values for $m = 0, \ldots, 10$ are $1, 1, 2, 3, 6, 10, 19, 33, 61, 108, 197$.

Proposition 3.17 (Log-concavity)**.** *For every $m \geq 0$ the sequence of the numbers $s(m, j)$, $0 \leq j \leq \lfloor m/2 \rfloor$, is strictly log-concave, and in particular unimodal. The polynomial $p_m(t) = \sum_j s(m, j)\, t^j$ has only real roots if and only if $m \leq 4$.*

Proof. By (3.6), $(m+1)\, s(m, j) = (m+1-2j)\binom{m+1}{j}$. On the range $0 \leq j \leq m/2$ the first factor is positive and linear in j, hence log-concave, and the binomial coefficients $\binom{m+1}{j}$ are strictly log-concave. A product of a positive log-concave sequence and a positive strictly log-concave sequence is strictly log-concave.

For $m \leq 4$ the polynomials are 1, 1, $1 + t$, $1 + 2t$ and $1 + 3t + 2t^2 = (1 + t)(1 + 2t)$. Let $m \geq 5$ and $n = \lfloor m/2 \rfloor \geq 2$, and suppose that p_m, which has degree n and constant term 1,

has only real roots. Then $p_m(t) = \prod_{i=1}^{n}(1+\rho_i t)$ with real numbers ρ_i, and

$$\sum_i \rho_i = s(m,1) = m-1, \qquad \sum_{i<i'} \rho_i\rho_{i'} = s(m,2) = \frac{m(m-3)}{2},$$

so that $\sum_i \rho_i^2 = (m-1)^2 - m(m-3) = m+1$. The Cauchy–Schwarz inequality $(\sum_i \rho_i)^2 \le n\sum_i \rho_i^2$ gives $(m-1)^2 \le n(m+1) \le m(m+1)/2$, that is $m^2 - 5m + 2 \le 0$, which fails for $m \ge 5$. □

For $m = 5$ the polynomial is $1 + 4t + 5t^2$, with discriminant -4. So the distribution of the number of classes over the extremal systems of a given length is log-concave, but from $m = 5$ on its generating polynomial is not real-rooted, in contrast with the polynomials studied in Section 8.

4. Rectangles

We now consider paths from $(0,0)$ to (a,b) with $a,b \ge 1$, of length $m = a+b$, and both parities. The two problems behave differently according to whether the parity required of the pairwise intersections agrees with the parity of m. If it agrees, a family is a one-class admissible system and Shankar's inequality applies; a weighted form of that inequality gives a bound that is exponential in $\min(a,b)$ and polynomial in the longer side. If it disagrees, the Gram matrix of the family over $\mathbb{F}_2$ is nonsingular or nearly so, and the family is bounded by the rank of the adjacency matrix of a grid graph, which is computed exactly. Explicit families for one, two and three rows of cells show how far these bounds are from the truth.

4.1. **The two types of families.**

Definition 4.1. Let $\mathcal{F}$ be a set of lattice paths from $(0,0)$ to (a,b) and $m = a+b$. We call $\mathcal{F}$ a *matching family* if $B_m(x,y) = m$ for all distinct $x, y \in \mathcal{F}$, and a *mismatched family* if $B_m(x,y) = m+1$ for all distinct $x, y \in \mathcal{F}$. The largest sizes of such families are denoted by $M(a,b)$ and $M^*(a,b)$.

Thus $M(a,b) = \mathrm{M}_{\mathrm{ev}}(a,b)$ and $M^*(a,b) = \mathrm{M}_{\mathrm{od}}(a,b)$ when m is even, and $M(a,b) = \mathrm{M}_{\mathrm{od}}(a,b)$ and $M^*(a,b) = \mathrm{M}_{\mathrm{ev}}(a,b)$ when m is odd. By Example 2.4 the matching families are exactly the one-class admissible systems. Both functions are symmetric in a and b, by the reflection in the diagonal.

Lemma 4.2. *If $\mathcal{F}$ is a matching family of paths from $(0,0)$ to (a,b) and $e \in \{0,1\}$, then $\mathcal{F}e$ is a matching family of paths of length $m+1$, and the same holds for mismatched families. Consequently M and M^* are nondecreasing in each argument, and*

$$\mathrm{M}_{\mathrm{od}}(a+1,b) \ge \mathrm{M}_{\mathrm{ev}}(a,b), \qquad \mathrm{M}_{\mathrm{ev}}(a+1,b) \ge \mathrm{M}_{\mathrm{od}}(a,b).$$

Proof. All words of $\mathcal{F}$ have weight b, so Lemma 2.1(b) gives $B_{m+1}(xe, ye) = B_m(x,y) + 1$ for $x, y \in \mathcal{F}$. The paths of $\mathcal{F}0$ end at $(a+1,b)$ and those of $\mathcal{F}1$ at $(a,b+1)$. □

Geometrically, all paths receive the same additional last edge, so every pairwise intersection grows by one edge while the length grows by one step.[12]

The main results of this section on general rectangles are collected in the following statement; the proofs are given in the subsections indicated.

Theorem 4.3. *Let $a,b \ge 1$, $m = a+b$ and $r = ab - \gcd(a+1, b+1) + 1$.*

[12]The same holds for a common first step: the words ex and ey share the first edge, and the remaining parts are translates of the paths of x and y. So $e\mathcal{F}$ is of the same type as $\mathcal{F}$ as well, even when the words of $\mathcal{F}$ have different weights.

(i) $2^{\min(a,b)} \le M(a,b) \le \sqrt{\dfrac{m^m}{a^a\, b^b}} \le 2^{m/2}$, *and the last inequality is strict unless* $a = b$. *In particular* $M(a,b) = 2^{m/2}$ *if and only if* $a = b$.
(ii) $M^*(a,b) \le r+3$ *if* m *is even, and* $M^*(a,b) \le r+2$ *if* m *is odd.*
(iii) *The number* r *is the rank over* $\mathbb{F}_2$ *of the adjacency matrix of the* $a \times b$ *grid graph.*

Part (i) is proved in Section 4.2, part (iii) in Section 4.3 and part (ii) in Section 4.4.

4.2. **Matching families: a weighted inequality and the box bound.** For an admissible system $\mathcal{A}$ of length m and positive real numbers ω_0, ω_1 put

$$\|\mathcal{A}\|_\omega = \sum_{A \in \mathcal{A}} |A|^2\, \omega_0^{\,m-\mathrm{w}(A)}\, \omega_1^{\,\mathrm{w}(A)}.$$

The weight attached to a class is the monomial $\omega_0^u \omega_1^v$ of the common endpoint (u,v) of its paths, and $\|\mathcal{A}\|_{(1,1)} = \|\mathcal{A}\|$.

Theorem 4.4 (Weighted inequality)**.** *Let* $\mathcal{A}$ *be an admissible system of length* $m \ge 0$ *and* $\omega_0, \omega_1 > 0$. *Then*

$$\|\mathcal{A}\|_\omega \le (\omega_0 + \omega_1)^m.$$

Proof. For $m = 0$ there is at most one class, consisting of the empty word. Let $m \ge 1$ and let $\mathcal{G}_0$, $\mathcal{G}_1$ be the systems of length $m-1$ defined in (2.2); they are admissible by Proposition 2.6. Write $\mu(A) = \omega_0^{\,m-\mathrm{w}(A)} \omega_1^{\,\mathrm{w}(A)}$ for a class of $\mathcal{A}$, and $\mu'(X) = \omega_0^{\,m-1-\mathrm{w}(X)} \omega_1^{\,\mathrm{w}(X)}$ for a class of $\mathcal{G}_0$ or $\mathcal{G}_1$. Since $\mathrm{w}(A^e) = \mathrm{w}(A) - e$, we have $\mu(A) = \omega_e\, \mu'(A^e)$ whenever $A^e \ne \varnothing$.

A class A with $A^1 = \varnothing$ satisfies $|A|^2 \mu(A) = \omega_0\, |A^0|^2 \mu'(A^0)$, and A^0 is a class of $\mathcal{G}_0$. A class with $A^0 = \varnothing$ satisfies $|A|^2 \mu(A) = \omega_1\, |A^1|^2 \mu'(A^1)$, and A^1 is a class of $\mathcal{G}_1$. For a split class the Cauchy–Schwarz inequality gives

$$|A|^2 = \Big(\sqrt{\omega_0}\,\frac{|A^0|}{\sqrt{\omega_0}} + \sqrt{\omega_1}\,\frac{|A^1|}{\sqrt{\omega_1}}\Big)^2 \le (\omega_0+\omega_1)\Big(\frac{|A^0|^2}{\omega_0} + \frac{|A^1|^2}{\omega_1}\Big),$$

and multiplying by $\mu(A)$,

$$|A|^2 \mu(A) \le (\omega_0+\omega_1)\big(|A^0|^2 \mu'(A^0) + |A^1|^2 \mu'(A^1)\big).$$

Here A^0 and A^1 are classes of both $\mathcal{G}_0$ and $\mathcal{G}_1$, so the right-hand side is ω_0 times their contribution to $\|\mathcal{G}_0\|_\omega$ plus ω_1 times their contribution to $\|\mathcal{G}_1\|_\omega$. Summing over all classes, each class of $\mathcal{G}_0$ and of $\mathcal{G}_1$ being used once, we obtain

$$\|\mathcal{A}\|_\omega \le \omega_0 \|\mathcal{G}_0\|_\omega + \omega_1 \|\mathcal{G}_1\|_\omega \le (\omega_0+\omega_1)(\omega_0+\omega_1)^{m-1}$$

by induction. □

For $\omega_0 = \omega_1 = 1$ this is Theorem 2.7. The system of all singletons gives equality for every ω, by the binomial theorem; the gain lies in the freedom to choose ω when the classes sit at few endpoints. The version for words over d letters is Theorem 7.3 of Section 7.

Theorem 4.5 (Box bound)**.** *For all* $a, b \ge 1$ *with* $m = a + b$,

$$M(a,b) \le \sqrt{\frac{m^m}{a^a\, b^b}} < \Big(\frac{e\, m}{b}\Big)^{b/2}.$$

More generally, if a set of words of length m *with* $B_m(x,y) = m$ *for all pairs contains* N_v *words of weight* v *for* $0 \le v \le m$, *then* $\sum_v N_v^2\, \omega_0^{m-v} \omega_1^v \le (\omega_0+\omega_1)^m$ *for all* $\omega_0, \omega_1 > 0$.

Proof. Let $\mathcal{F}$ be a set of words of length m with $B_m = m$ on all pairs. Its nonempty weight classes form an admissible system: (A1) and (A2) hold by assumption, and (A3) holds with the constant m. Theorem 4.4 gives the second statement. For a matching family of paths from $(0,0)$ to (a,b) there is one class, of weight b, and the choice $\omega_0 = a$, $\omega_1 = b$ gives $|\mathcal{F}|^2 a^a b^b \le m^m$.[13] Finally

$$\frac{m^m}{a^a b^b} = \Big(\frac{m}{b}\Big)^b \Big(1 + \frac{b}{a}\Big)^a < \Big(\frac{m}{b}\Big)^b e^b. \qquad \square$$

Proof of Theorem 4.3(i). By symmetry let $a \ge b$. Lemma 4.2 and Theorem 2.8(i) give

$$M(a,b) \ge M(b,b) = 2^b.$$

The upper bound is Theorem 4.5. The function $t \mapsto t \log t$ is strictly convex for $t > 0$, so

$$a \log a + b \log b \ge m \log(m/2),$$

with equality only for $a = b$. Thus $m^m/(a^a b^b) \le 2^m$, with equality only for $a = b$. If $a \ne b$ then $M(a,b) < 2^{m/2}$, and $M(n,n) = 2^n$ by Theorem 2.8(i). $\square$

For fixed b the box bound is polynomial in a, of degree $b/2$, whereas Theorem 2.7 alone gives $2^{(a+b)/2}$. Two special cases are used later:

$$M(a,2) < \frac{e\,(a+2)}{2}, \qquad M(n+1,n) \le \sqrt{\frac{(2n+1)^{2n+1}}{(n+1)^{n+1}\,n^n}} < 2^{\,n+1/2}. \tag{4.1}$$

The lower bound $2^{\min(a,b)}$ is not sharp in general either; already $M(4,2) = 5$ (Table 1), between 4 and the box bound $\sqrt{6^6/(4^4\,2^2)} \approx 6.75$.

4.3. **The binary rank of grid graphs.** Let $A_{a,b}$ be the adjacency matrix over $\mathbb{F}_2$ of the grid graph whose vertices are the pairs (i,j) with $1 \le i \le a$ and $1 \le j \le b$, two vertices being adjacent when they agree in one coordinate and differ by 1 in the other. We think of the vertices as the ab unit cells of the rectangle $[0,a] \times [0,b]$, two cells being adjacent when they share a side.

Theorem 4.6. *For all $a, b \ge 1$,*

$$\operatorname{rank}_{\mathbb{F}_2} A_{a,b} = ab - \gcd(a+1, b+1) + 1.$$

In particular $A_{a,b}$ is invertible over $\mathbb{F}_2$ if and only if $a+1$ and $b+1$ are coprime, and the kernel of $A_{n,n}$ has dimension n.

For $a = b$ the value $n(n-1)$ was found by Shankar [35]. Kernels of such matrices over $\mathbb{F}_2$ are the configurations of the σ-game of Sutner [39], where they are studied through polynomials of Chebyshev type;[14] the proof below is self-contained. Define $f_s \in \mathbb{F}_2[x]$ by

$$f_0 = 0, \qquad f_1 = 1, \qquad f_{s+1} = x f_s + f_{s-1} \quad (s \ge 1),$$

so that $f_2 = x$, $f_3 = x^2 + 1$, $f_4 = x^3$, and f_s is monic of degree $s-1$ for $s \ge 1$.

Lemma 4.7. *For all $i, j \ge 0$, $\gcd(f_i, f_j) = f_{\gcd(i,j)}$, with the convention $\gcd(i,0) = i$.*

[13] This choice is optimal: the function $(\omega_0 + \omega_1)^m \omega_0^{-a} \omega_1^{-b}$ is homogeneous of degree 0, and on the line $\omega_0 + \omega_1 = m$ the product $\omega_0^a \omega_1^b$ is largest at (a,b).

[14] In the σ-game every vertex of a graph carries a lamp, and pressing a vertex switches the lamps of its neighbours. The patterns that can be produced from darkness form the column space of the adjacency matrix over $\mathbb{F}_2$, so Theorem 4.6 says that $2^{ab - \gcd(a+1,b+1)+1}$ patterns are reachable on the $a \times b$ grid.

Proof. Let $R = \left(\begin{smallmatrix} x & 1 \\ 1 & 0 \end{smallmatrix}\right)$. By induction on $s \geq 1$, using the recursion,

$$R^s = \begin{pmatrix} f_{s+1} & f_s \\ f_s & f_{s-1} \end{pmatrix}.$$

Taking determinants gives $f_{s+1}f_{s-1} + f_s^2 = 1$, since $\det R = 1$ in $\mathbb{F}_2$; hence $\gcd(f_s, f_{s+1}) = 1$. Comparing the upper right entries of $R^{p+q} = R^p R^q$ gives $f_{p+q} = f_{p+1}f_q + f_p f_{q-1}$ for $p, q \geq 1$. Now let $i > j \geq 1$. With $p = i - j$ and $q = j$,

$$\gcd(f_i, f_j) = \gcd(f_{i-j+1}f_j + f_{i-j}f_{j-1},\, f_j) = \gcd(f_{i-j}f_{j-1},\, f_j) = \gcd(f_{i-j}, f_j),$$

the last step because f_{j-1} is prime to f_j. So the gcd is unchanged when the larger index is reduced by the smaller, and the Euclidean algorithm on the indices ends with $\gcd(f_g, f_0) = f_g$ for $g = \gcd(i, j)$. The cases $i = j$ and $j = 0$ are trivial. □

Let P_a be the adjacency matrix of the path with a vertices, the $a \times a$ matrix over $\mathbb{F}_2$ with entries 1 exactly in the positions $(i, i \pm 1)$.

Lemma 4.8. *For every polynomial $g \in \mathbb{F}_2[x]$,*

$$\dim \ker g(P_a) = \deg \gcd(g, f_{a+1}).$$

Proof. Let $\chi_a = \det(xI + P_a)$ be the characteristic polynomial. Expanding along the last row gives $\chi_a = x\chi_{a-1} + \chi_{a-2}$ with $\chi_0 = 1$ and $\chi_1 = x$, so $\chi_a = f_{a+1}$. Let e_1 be the first unit vector. The vector $P_a^s e_1$ has the entry 1 in coordinate $s + 1$ and vanishes in all later coordinates, for $0 \leq s \leq a - 1$; so $e_1, P_a e_1, \ldots, P_a^{a-1} e_1$ form a basis of $\mathbb{F}_2^a$. The map $u \mapsto u(P_a)e_1$ from $\mathbb{F}_2[x]$ to $\mathbb{F}_2^a$ is therefore surjective, and its kernel is an ideal (h) with $\deg h = a$. By the Cayley–Hamilton theorem f_{a+1} lies in this ideal, and both polynomials are monic of degree a, so $h = f_{a+1}$. Under the resulting isomorphism $\mathbb{F}_2[x]/(f_{a+1}) \cong \mathbb{F}_2^a$ the matrix $g(P_a)$ acts as multiplication by g. With $d = \gcd(g, f_{a+1})$, the residue of u is killed by g exactly when f_{a+1}/d divides u, and the multiples of f_{a+1}/d modulo f_{a+1} form a space of dimension $\deg d$. □

Proof of Theorem 4.6. Write a vector $z \in \mathbb{F}_2^{ab}$ as the sequence of its rows $z_1, \ldots, z_b \in \mathbb{F}_2^a$, where z_j collects the coordinates (i, j), and put $z_0 = z_{b+1} = 0$. The neighbours of the cell (i, j) are $(i \pm 1, j)$ and $(i, j \pm 1)$, so $A_{a,b}z = 0$ means

$$P_a z_j + z_{j-1} + z_{j+1} = 0 \qquad (1 \leq j \leq b).$$

These equations say $z_{j+1} = P_a z_j + z_{j-1}$, hence $z_j = f_j(P_a)z_1$ for $0 \leq j \leq b+1$ by the recursion for the f_j, and the condition $z_{b+1} = 0$ becomes $f_{b+1}(P_a)z_1 = 0$. Conversely every z_1 in the kernel of $f_{b+1}(P_a)$ defines a vector of the kernel of $A_{a,b}$. So $z \mapsto z_1$ is an isomorphism from $\ker A_{a,b}$ onto $\ker f_{b+1}(P_a)$, and by Lemmas 4.8 and 4.7

$$\dim \ker A_{a,b} = \deg \gcd(f_{b+1}, f_{a+1}) = \deg f_{\gcd(a+1,b+1)} = \gcd(a + 1, b + 1) - 1. \qquad \square$$

The proof shows that a vector of $\ker A_{a,b}$ is determined by its first row. The rank is always even, as it must be for the matrix of an alternating form. Figure 6 displays the nullity for small grids.

4.4. **Mismatched families: the Gram matrix of the path space.** Let Γ be the graph with vertex set $\{0, \ldots, a\} \times \{0, \ldots, b\}$ whose edges are the unit segments between these points, and let $\langle\, , \rangle$ be the standard bilinear form on $\mathbb{F}_2^{E(\Gamma)}$. A path is identified with the indicator vector of its edge set, so that $\langle x, y \rangle = B_m(x, y)$ for two paths from $(0, 0)$ to (a, b). Let W be the span of all these paths. For $1 \leq i \leq a$ and $1 \leq j \leq b$ let $c_{i,j}$ be the sum of the four sides of the cell (i, j), the unit square with upper right corner (i, j), and let $H = 1^b 0^a$ be the path

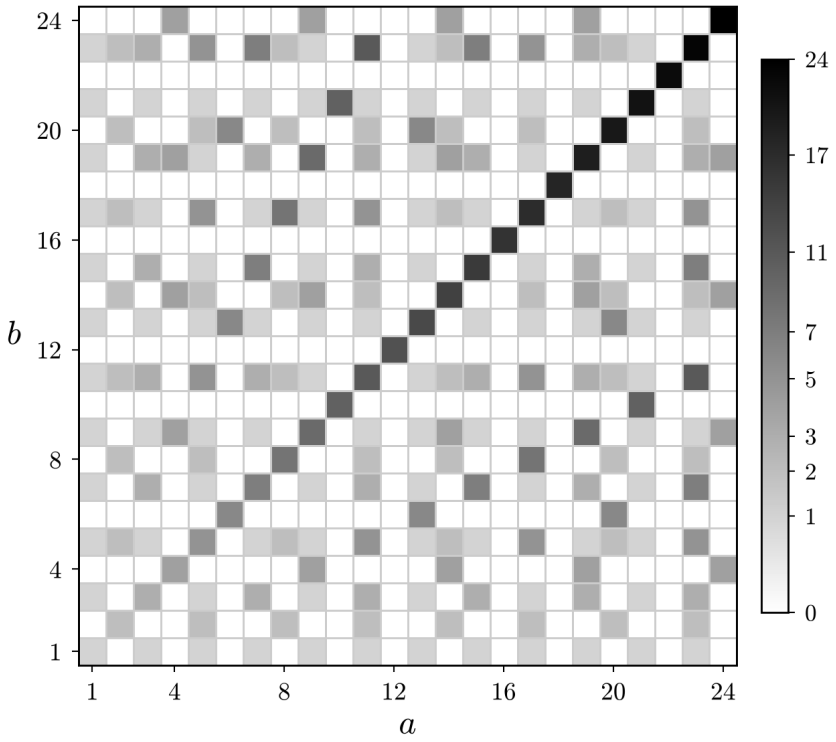


FIGURE 6. The nullity $\gcd(a+1, b+1) - 1$ of the adjacency matrix of the $a \times b$ grid graph over $\mathbb{F}_2$ for $1 \le a, b \le 24$. On the diagonal it equals n; it vanishes exactly when $a+1$ and $b+1$ are coprime.

along the left and the top side of the rectangle. For $a = b$ the following basis is the one used in [35].

Lemma 4.9. *The vectors H and $c_{i,j}$, $1 \le i \le a$, $1 \le j \le b$, form a basis of W; thus* $\dim W = ab + 1$. *In this basis the Gram matrix of $\langle\,,\rangle$ on W is*

$$G_W = \begin{pmatrix} m & \tau^T \\ \tau & A_{a,b} \end{pmatrix}, \qquad \tau_{i,j} = [\,i = 1\,] + [\,j = b\,] \in \mathbb{F}_2.$$

Proof. Let Z be the cycle space of Γ, the space of edge sets that meet every vertex in an even number of edges. As Γ is connected, $\dim Z = |E| - |V| + 1 = a(b+1) + b(a+1) - (a+1)(b+1) + 1 = ab$.[15] The ab cell boundaries lie in Z and are linearly independent: in a nonempty set of cells choose a cell with the largest j and, among these, with the largest i; its top side belongs to no other cell of the set, so the sum of the boundaries is not zero. Hence the $c_{i,j}$ form a basis of Z.

The sum of two paths from $(0,0)$ to (a,b) lies in Z, since both paths have degree 1 at the two corners and degree 0 or 2 elsewhere. So $W = \mathbb{F}_2 H + (W \cap Z)$. Every cell boundary is the sum of two paths, namely of a path that passes the lower left corner of the cell and continues with 01 and the path that continues with 10 instead; thus $Z \subseteq W$. As $H \notin Z$, the first claim follows.

Two different cells share one side if they are adjacent and none otherwise, and $\langle c_{i,j}, c_{i,j} \rangle = 4 = 0$; this gives the block $A_{a,b}$. The path H consists of the left and the top side of the

[15] For a spanning tree, each of the $|E| - |V| + 1$ remaining edges closes one cycle with the tree; these cycles are independent, and an element of Z that contains no edge outside the tree is a forest with all degrees even, hence empty.

rectangle. The boundary of the cell (i,j) has one edge on the left side if $i=1$ and none otherwise, and one edge on the top side if $j=b$ and none otherwise; this gives τ. Finally $\langle H,H\rangle = m$. □

Proposition 4.10. *Let* $r = \operatorname{rank} A_{a,b}$.

(a) $\operatorname{rank} G_W \le r+2$.

(b) *If* m *is even, then* $\operatorname{rank} G_W = r$ *when* τ *lies in the column space of* $A_{a,b}$, *and* $\operatorname{rank} G_W = r+2$ *otherwise.*

(c) *If* $a=b=n$, *then* $\tau = A_{n,n}u$ *for the vector* $u_{i,j} = [\,i<j\,]$, *and* $\operatorname{rank} G_W = n(n-1)$.

Proof. (a) Adding one row and one column to a matrix raises its rank by at most 2.

(b) For even m all basis vectors of Lemma 4.9 have an even number of edges. Since $x \mapsto \langle x,x\rangle$ is additive over $\mathbb{F}_2$, the form is alternating on W and its rank is even. If $\tau = A_{a,b}u$, replace H by $H' = H + \sum_{i,j} u_{i,j}c_{i,j}$. Then $\langle H', c_{i,j}\rangle = \tau_{i,j} + (A_{a,b}u)_{i,j} = 0$ and $\langle H', H'\rangle = 0$, so in the new basis the Gram matrix is $A_{a,b}$ bordered by zeros, of rank r. If τ is not in the column space, the rows of G_W other than the first already have rank $r+1$, and an even rank between $r+1$ and $r+2$ is $r+2$, as r is even.

(c) We compute $(A_{n,n}u)_{i,j}$, the number modulo 2 of the neighbours (i',j') of (i,j) with $i'<j'$. If $i \ge j+1$, no neighbour has this property, and $\tau_{i,j}=0$ because $i\ge 2$ and $j \le n-1$. If $i \le j$, the neighbours $(i-1,j)$ and $(i,j+1)$ have the property whenever they exist, that is for $i\ge 2$ and for $j\le n-1$ respectively; the neighbours $(i+1,j)$ and $(i,j-1)$ have it both if $i \le j-2$ and neither if $i \ge j-1$. So $(A_{n,n}u)_{i,j} = [\,i\ge 2\,]+[\,j\le n-1\,] = [\,i=1\,]+[\,j=n\,] = \tau_{i,j}$. The rank follows from (b) and Theorem 4.6. □

Theorem 4.11. *Let* $a,b \ge 1$ *and* $r = ab - \gcd(a+1,b+1)+1$. *A mismatched family of paths from* $(0,0)$ *to* (a,b) *has at most* $\operatorname{rank} G_W + 1 \le r+3$ *members if* m *is even, and at most* $\operatorname{rank} G_W \le r+2$ *members if* m *is odd.*

Proof. Let $\mathcal{F} = \{x_1,\dots,x_N\}$ be a mismatched family and $G_{\mathcal{F}} = (\langle x_s, x_t\rangle)_{s,t}$ its Gram matrix. Expressing the x_s in the basis of Lemma 4.9 by a matrix X gives $G_{\mathcal{F}} = X^T G_W X$, so $\operatorname{rank} G_{\mathcal{F}} \le \operatorname{rank} G_W$.

If m is odd, the diagonal entries of $G_{\mathcal{F}}$ are $m=1$ and the others are $m+1=0$, so $G_{\mathcal{F}}$ is the identity and $N \le \operatorname{rank} G_W$.

If m is even, $G_{\mathcal{F}} = J+I$ with the all-ones matrix J of size N. Over $\mathbb{F}_2$ we have $J^2 = NJ$ and $(J+I)^2 = NJ+I$. For even N this is the identity, so $J+I$ is invertible; for odd N the same applies to its principal submatrix of size $N-1$. In both cases $\operatorname{rank} G_{\mathcal{F}} \ge N-1$, and $N \le \operatorname{rank} G_W + 1$. The bounds in terms of r follow from Proposition 4.10(a). □

This proves Theorem 4.3(ii); part (iii) is Theorem 4.6.

Corollary 4.12 (Shankar [35])**.** $\mathrm{M}_{\mathrm{od}}(n,n) \le n(n-1)+1$ *for every* $n \ge 1$.

Proof. For $a=b=n$ the odd families are the mismatched ones, and $\operatorname{rank} G_W = n(n-1)$ by Proposition 4.10(c). □

For even m the bound of Theorem 4.11 is $r+1$ or $r+3$ according to whether τ lies in the column space of $A_{a,b}$. A computation by Gaussian elimination for all $a,b \le 10$ with $a+b$ even, recorded in the verification archive [8], shows that in this range τ lies in the column space exactly in the following cases: a and b both even; $a=b$; and a,b odd with $a \equiv b \equiv 1 \pmod 4$.

4.5. **One row of cells.** For $b=1$ both problems are solved by a direct count.

Proposition 4.13. *For every* $a \ge 1$,

$$M(a,1) = 2 \qquad \textit{and} \qquad M^*(a,1) = \lfloor a/2 \rfloor + 1.$$

Proof. The paths from $(0,0)$ to $(a,1)$ are $x^{(i)} = 0^i 1 0^{a-i}$ for $0 \le i \le a$. For $i < j$ the paths $x^{(i)}$ and $x^{(j)}$ share the first i edges of the bottom side and the last $a-j$ edges of the top side and nothing else, so $B_{a+1}(x^{(i)}, x^{(j)}) = i+j+a$. A set of indices gives a matching family when $i+j$ is odd for all its pairs, and a mismatched family when $i+j$ is even for all its pairs. Among three integers two have the same parity, so a matching family has at most two members, and $\{x^{(0)}, x^{(1)}\}$ is one. A mismatched family consists of indices of one parity, and the $\lfloor a/2 \rfloor + 1$ even indices form the largest one. □

So for one row the matching problem has a bounded answer, far below the box bound $\sqrt{(a+1)^{a+1}/a^a}$, which grows like $\sqrt{e\,a}$, and the mismatched problem has a linear answer, about half of the bound $r+2$ or $r+3$ of Theorem 4.11, where $r \ge a-1$.

4.6. **A formula for several rows.** For fixed b a path from $(0,0)$ to (a,b) is determined by the abscissas $0 \le i_1 \le \cdots \le i_b \le a$ of its north steps: it is the word $0^{i_1} 1\, 0^{i_2 - i_1} 1 \cdots 1\, 0^{a - i_b}$, and its tth north step leads from $(i_t, t-1)$ to (i_t, t). We denote this path by $(i_1, \ldots, i_b)$ and put $i_0 = 0$ and $i_{b+1} = a$. For two real intervals I and I' let $\ell(I \cap I')$ be the length of their intersection, which is 0 when they are disjoint or meet in a point.

Lemma 4.14. *For two paths $p = (i_1, \ldots, i_b)$ and $q = (k_1, \ldots, k_b)$ from $(0,0)$ to (a,b),*

$$|E(p) \cap E(q)| = \sum_{t=0}^{b} \ell\big([i_t, i_{t+1}] \cap [k_t, k_{t+1}]\big) + \#\{t \in [b] :\ i_t = k_t\}.$$

Proof. The horizontal edges of p at height t are the unit segments of $[i_t, i_{t+1}] \times \{t\}$, for $0 \le t \le b$, so the two paths share $\ell([i_t, i_{t+1}] \cap [k_t, k_{t+1}])$ horizontal edges at height t. The only vertical edge of p between the heights $t-1$ and t is $\{i_t\} \times [t-1, t]$, and it is an edge of q exactly when $i_t = k_t$. □

4.7. **Two rows of cells.** For $b = 2$ we write the paths as (i, i') with $0 \le i \le i' \le a$. The length is $a+2$, so a family is matching when $|E(p) \cap E(q)| \equiv a$ for all its pairs and mismatched when $|E(p) \cap E(q)| \equiv a+1$. Put $d(p,q) = |E(p) \cap E(q)| - a$.

Lemma 4.15. *Let $p = (i, i')$ and $q = (k, k')$ be two paths from $(0,0)$ to $(a,2)$, named so that $i < k$, or $i = k$ and $i' \le k'$. Then, modulo 2,*

(A) $d(p,q) \equiv i + k'$ *if* $i' < k$;
(B) $d(p,q) \equiv i + k' + [\,i = i'\,] + [\,k = k'\,]$ *if* $i' = k$;
(C) $d(p,q) \equiv i + i' + k + k' + [\,i = k\,] + [\,i' = k'\,]$ *if* $i' > k$.

Proof. By Lemma 4.14 with $i \le k$,

$$|E(p) \cap E(q)| = i + \max\big(0, \min(i', k') - k\big) + a - \max(i', k') + \#\{i = k\} + \#\{i' = k'\},$$

where $\#\{\cdot\}$ is 1 or 0. In case (A) we have $i \le i' < k \le k'$, the middle term vanishes, $\max(i', k') = k'$ and there is no coincidence. In case (B) the middle term vanishes again, $\max(i', k') = k'$, and since $i' = k$ the coincidence $i = k$ means $i = i'$ and the coincidence $i' = k'$ means $k = k'$. In case (C) the middle term is $\min(i', k') - k$, and $\min(i', k') - \max(i', k') \equiv i' + k'$. □

Theorem 4.16. *For every $a \ge 2$,*

$$M(a,2) \ge \Big\lfloor \frac{a}{2} \Big\rfloor + 3, \qquad M^*(a,2) \ge \begin{cases} \lfloor (5a-1)/4 \rfloor, & a \text{ odd}, \\ \lfloor (5a-2)/4 \rfloor, & a \text{ even}. \end{cases}$$

Proof. Matching families. Let $e = 2\lfloor a/2 \rfloor \geq 2$ and take the paths $D_k = (k, k)$ for even k with $0 \leq k \leq a$, together with $U = (0, 1)$ and $V = (e-1, e)$. These are $\lfloor a/2 \rfloor + 3$ different paths. We show $d \equiv 0$ for all pairs. For D_k, D_l with $k < l$ rule (A) gives $k + l$. For D_0 and U, in this order, rule (B) gives $0 + 1 + 1 + 0$. For U and D_k with $k \geq 2$ rule (A) gives $0 + k$. For V and D_e rule (B) gives $(e-1) + e + 0 + 1$. For D_k and V with $k \leq e - 2$ rule (A) gives $k + e$. For U and V rule (A) gives $0 + e$ if $e \geq 4$, and rule (B) gives $0 + 2 + 0 + 0$ if $e = 2$.

Mismatched families, a odd. Take the paths

$$P_j = (2j, 2j+1) \ \ (0 \leq 2j \leq a-3), \qquad Q_t = (4t, 4t+3) \ \ (0 \leq 4t \leq a-5),$$
$$R_o = (o, a) \ \ (o \text{ odd}, \ 1 \leq o \leq a).$$

The second coordinate is less than a for the paths P_j and Q_t, and the difference of the coordinates is 1 for P_j and 3 for Q_t; so the listed paths are different, and their number is $\frac{a-1}{2} + \lfloor \frac{a-1}{4} \rfloor + \frac{a+1}{2} = \lfloor \frac{5a-1}{4} \rfloor$. We show $d \equiv 1$ for all pairs.

Two paths P_j, P_l with $j < l$ fall under (A), with $d \equiv 2j + 2l + 1$; two paths Q_t, Q_u with $t < u$ fall under (A), with $d \equiv 4t + 4u + 3$. For $R_o, R_{o'}$ with $o < o'$: if $o' < a$, rule (C) gives $o + a + o' + a + 0 + 1$; if $o' = a$, rule (B) gives $o + a + 0 + 1$. In both cases $d \equiv 1$, as o, o' and a are odd.

For P_j and R_o: if $o \geq 2j + 3$, rule (A) gives $2j + a$; if $o = 2j + 1$, rule (B) gives $2j + a + 0 + 0$, because $2j + 1 \leq a - 2$; if $o \leq 2j - 1$, rule (C) with R_o first gives $o + a + 2j + (2j + 1) + 0 + 0$. Each of these is odd.

For P_j and Q_t: if $2j + 1 < 4t$, rule (A) gives $2j + 4t + 3$, and if $2j \geq 4t + 4$, rule (A) with Q_t first gives $4t + 2j + 1$. Otherwise $2j$ is $4t$ or $4t + 2$. If $2j = 4t$, rule (C) with P_j first gives $4t + (4t + 1) + 4t + (4t + 3) + 1 + 0$. If $2j = 4t + 2$, rule (C) with Q_t first gives $4t + (4t + 3) + (4t + 2) + (4t + 3) + 0 + 1$. Each of these is odd.

For Q_t and R_o, where $4t + 3 \leq a - 2$: if $o \geq 4t + 5$, rule (A) gives $4t + a$; if $o = 4t + 3$, rule (B) gives $4t + a + 0 + 0$; if $o = 4t + 1$, rule (C) gives $4t + (4t + 3) + (4t + 1) + a + 0 + 0$; if $o \leq 4t - 1$, rule (C) with R_o first gives $o + a + 4t + (4t + 3) + 0 + 0$. Each of these is odd.

Mismatched families, a even. The rules of Lemma 4.15, and the case into which a pair falls, do not change when 1 is added to all four coordinates, because each rule is a sum of an even number of coordinates and of indicators of coincidences. Hence the family constructed above for the odd number $a - 1$, shifted by one unit to the right, consists of paths from $(0, 0)$ to $(a, 2)$ with $d \equiv 1$ on all pairs; its members are $(2j + 1, 2j + 2)$, $(4t + 1, 4t + 4)$ and $(o + 1, a)$. Add the path $Y = (0, a)$, which differs from all of them. With a shifted path (k, k'), $1 \leq k < a$, the path Y comes first and falls under (C), which gives $a + k + k' + [\, k' = a \,]$; this is odd for $(2j + 1, 2j + 2)$ and $(4t + 1, 4t + 4)$, where $k + k'$ is odd and $k' \leq a - 2$, and for $(o + 1, a)$ with $o + 1 < a$, where $k + k'$ is even and $k' = a$. With the shifted path (a, a) rule (B) gives $0 + a + 0 + 1$. The family has $\lfloor (5(a-1) - 1)/4 \rfloor + 1 = \lfloor (5a - 2)/4 \rfloor$ members. □

Figure 7 shows the three constructions. Together with (4.1), Theorem 4.16 determines the order of the matching problem for two rows.

Corollary 4.17. *For every $a \geq 2$,*

$$\left\lfloor \frac{a}{2} \right\rfloor + 3 \leq M(a, 2) < \frac{e\,(a+2)}{2} \qquad \text{and} \qquad \left\lfloor \frac{5a-2}{4} \right\rfloor \leq M^*(a, 2) \leq 2a + 4 - \gcd(a+1, 3).$$

In particular $M(a, 2)$ and $M^*(a, 2)$ grow linearly in a.

Proof. The bounds for M are Theorem 4.16 and Theorem 4.5. For M^*, the rank of Theorem 4.6 is $r = 2a - \gcd(a + 1, 3) + 1$, and Theorem 4.11 gives $M^*(a, 2) \leq r + 3$. □

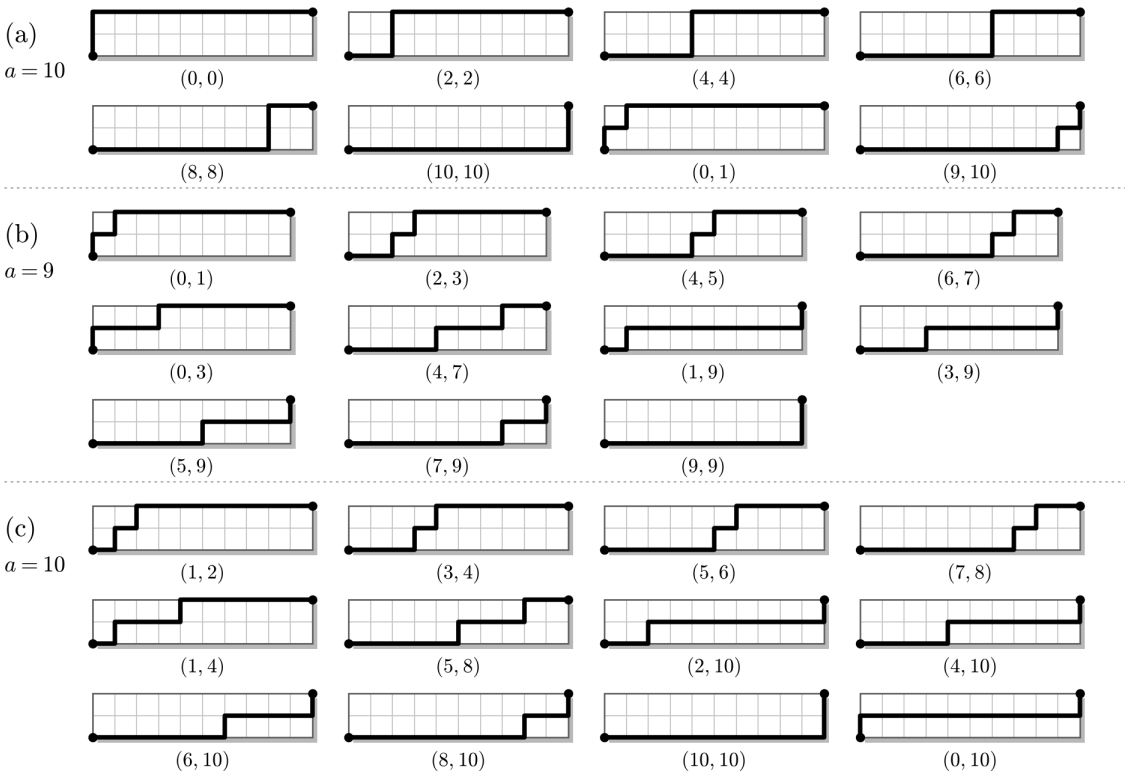


FIGURE 7. The families of Theorem 4.16, each path labelled by the abscissas (i, i') of its two north steps: the matching family for $a = 10$, formed by the double steps on the even lines together with $(0, 1)$ and $(9, 10)$; the mismatched family for $a = 9$, formed by the paths $(2j, 2j+1)$, $(4t, 4t+3)$ and $(o, 9)$; and its shift for $a = 10$ completed by $(0, 10)$.

An exhaustive search over all families of paths from $(0, 0)$ to $(a, 2)$, recorded in the verification archive [8], shows that the two lower bounds of Theorem 4.16 are the exact values of $M(a, 2)$ and $M^*(a, 2)$ for $2 \le a \le 22$. In this range the constant $e/2 \approx 1.36$ of the box bound is therefore to be compared with the true slope $\frac{1}{2}$, and the slope 2 of the rank bound with the true slope $\frac{5}{4}$.

4.8. **Three rows of cells.** For $b = 3$ and even a the length $a + 3$ is odd, so the mismatched families are the even families. Theorem 4.11 bounds them by $r + 2 \le 3a + 2$. The following construction has about two thirds of this size.

Theorem 4.18. *Let $a \ge 4$ be even, and let $\mathcal{T}_a$ consist of the paths*

$$A_z = (0, 1, z) \quad (z \text{ odd},\ 1 \le z \le a-1), \qquad B_z = (2, 3, z) \quad (z \text{ even},\ 4 \le z \le a-2),$$
$$C_z = (z, a-1, a) \quad (z \text{ odd},\ 1 \le z \le a-1), \quad D_z = (z, a-3, a-2) \quad (z \text{ even},\ 2 \le z \le a-4),$$

together with $O = (0, 0, 0)$ and $O' = (a, a, a)$. Then $\mathcal{T}_a$ is an even family of paths from $(0, 0)$ to $(a, 3)$. It has $2a - 2$ members if $a \ne 6$ and 9 members if $a = 6$. Hence

$$\mathrm{M}_{\mathrm{ev}}(a, 3) \ge 2a - 2 \quad (a \text{ even},\ a \ge 4,\ a \ne 6), \qquad \mathrm{M}_{\mathrm{ev}}(6, 3) \ge 9.$$

Proof. The number of paths. There are $a/2$ paths A_z, as many paths C_z, and $(a-4)/2$ paths of each of the kinds B_z and D_z. The first two coordinates are $(0, 1)$ for A_z and $(2, 3)$ for B_z;

the first coordinate of C_z is odd, and that of D_z is even and at least 2. So a path can be of two kinds only if $(2,3,z) = (z', a-3, a-2)$, which means $a = 6$, $z = 4$, $z' = 2$. The paths O and O' are of none of the four kinds. This gives $a + (a-4) + 2 = 2a - 2$ paths for $a \neq 6$, and 9 paths for $a = 6$, where $B_4 = D_2$.

A symmetry. The half turn $(x, y) \mapsto (a - x, 3 - y)$ of the rectangle preserves common edges and maps the path (i_1, i_2, i_3) to $(a - i_3, a - i_2, a - i_1)$. It exchanges A_z with C_{a-z}, B_z with D_{a-z}, and O with O'. It is therefore enough to treat the pairs of the types AA, BB, AB, AC, AD, BD, OA, OB, OC, OD and OO': the types CC, DD, CD, BC, $O'C$, $O'D$, $O'A$ and $O'B$ are the images of AA, BB, AB, AD, OA, OB, OC and OD. In each case we evaluate the formula of Lemma 4.14: the four lengths for $t = 0, 1, 2, 3$, and then the coincidences.

A_z and $A_{z'}$ with $z < z'$. The lengths are 0, 1, $z - 1$ and $a - z'$, and the first two coordinates coincide. The total $a + 2 + z - z'$ is even.

B_z and $B_{z'}$ with $z < z'$. The lengths are 2, 1, $z - 3$ and $a - z'$, and the first two coordinates coincide. The total is again $a + 2 + z - z'$.

A_z and B_y. The lengths are $\ell([0,0] \cap [0,2]) = 0$, $\ell([0,1] \cap [2,3]) = 0$, $\ell([1,z] \cap [3,y]) = \max(0, \min(z,y) - 3)$ and $\ell([z,a] \cap [y,a]) = a - \max(z,y)$, and no coordinates coincide, z being odd and y even. For $z = 1$ the total is $a - y$. For $3 \leq z < y$ it is $(z-3) + (a-y)$, a sum of two even numbers. For $z > y$ it is $(y-3) + (a-z)$, a sum of two odd numbers.

A_z and C_y. The lengths are $\ell([0,0] \cap [0,y]) = 0$, $\ell([0,1] \cap [y, a-1]) = 0$ since $y \geq 1$, $\ell([1,z] \cap [a-1,a]) = 0$ since $z \leq a - 1$, and $\ell([z,a] \cap [a,a]) = 0$. No coordinates coincide: $0 \neq y$, $1 \neq a - 1$ and $z \neq a$. The two paths are edge-disjoint.

A_z and D_y, where $a \geq 6$. The lengths are 0; $\ell([0,1] \cap [y, a-3]) = 0$ since $y \geq 2$; $\ell([1,z] \cap [a-3, a-2])$, which is 1 for $z = a - 1$ and 0 for $z \leq a - 3$; and $\ell([z,a] \cap [a-2,a])$, which is 1 for $z = a - 1$ and 2 for $z \leq a - 3$. No coordinates coincide: $0 \neq y$, $1 \neq a - 3$, and $z \neq a - 2$ by parity. The total is 2.

B_z and D_y. For $a = 6$ these are the same path, so let $a \geq 8$. The lengths are $\ell([0,2] \cap [0,y]) = 2$, $\ell([2,3] \cap [y, a-3]) = [\, y = 2 \,]$, $\ell([3,z] \cap [a-3, a-2]) = [\, z = a-2 \,]$ and $\ell([z,a] \cap [a-2,a]) = 2$, read as integers. The first coordinates coincide when $y = 2$, the second never because $3 \neq a - 3$, the third when $z = a - 2$. The total is 4, 6 or 8.

Pairs containing O. The path $O = (0,0,0)$ has the intervals $[0,0]$, $[0,0]$, $[0,0]$ and $[0,a]$, so only the length for $t = 3$ can be positive. With A_z this length is $a - z$ and the first coordinates coincide; the total $a - z + 1$ is even. With B_z the length is $a - z$, without coincidence, and z is even. With C_z the length is $\ell([0,a] \cap [a,a]) = 0$, without coincidence. With D_y it is $\ell([0,a] \cap [a-2,a]) = 2$, without coincidence. With $O' = (a,a,a)$, whose intervals are $[0,a]$, $[a,a]$, $[a,a]$, $[a,a]$, all lengths vanish and nothing coincides. □

By an exhaustive search recorded in [8], $\mathrm{M}_{\mathrm{ev}}(a,3) = 6, 10, 14, 18, 22$ for $a = 4, 6, 8, 10, 12$. So $\mathcal{T}_a$ is a largest even family for $a = 4, 8, 10, 12$, and for $a = 6$ it misses the maximum by one. The same search gives $M^*(a,3) = 3, 6, 7, 10, 11, 14, 15, 18, 18, 22, 23$ for $a = 3, \ldots, 13$. By Lemma 4.2, Theorem 4.18 also gives $M^*(a,3) \geq 2a - 4$ for odd $a \geq 5$, $a \neq 7$, and $M^*(a,b) \geq 2a - 2$ for all $b \geq 3$ and even $a \geq 4$, $a \neq 6$.[16] Figure 8 shows $\mathcal{T}_8$.

4.9. **Odd lengths and the Catalan numbers.** For paths from $(0,0)$ to $(n+1, n)$ the matching families are the odd families, and they inherit the Catalan structure of the square.

[16]Large parts of the mismatched families for one, two and three rows are sets of the following kind. For fixed words α, β and $L \geq 1$, two paths $\alpha\, 0^z 1\, 0^{L-z} \beta$ and $\alpha\, 0^{z'} 1\, 0^{L-z'} \beta$ share $|\alpha| + |\beta| + L - |z - z'|$ edges. If z runs over the integers of one parity in $[0, L]$, this number has the parity of the length plus one, so the set is a mismatched family. The paths A_z, for instance, are obtained with $\alpha = 101$ and β empty.

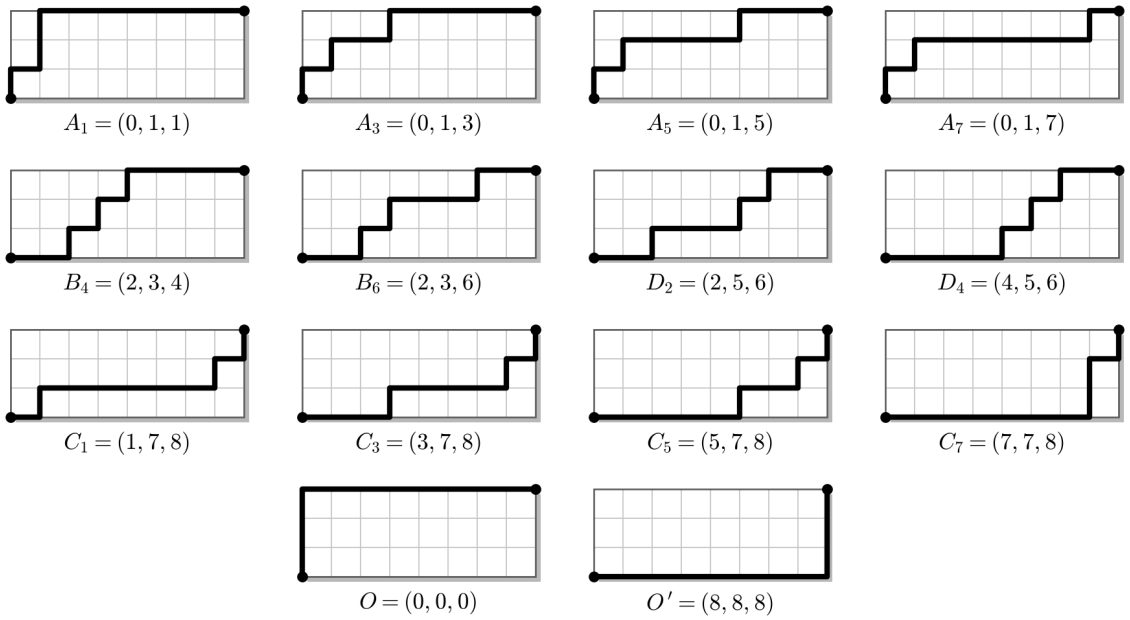


FIGURE 8. The even family $\mathcal{T}_8$ of fourteen paths from $(0, 0)$ to $(8, 3)$ of Theorem 4.18: the paths A_1, A_3, A_5, A_7 start with two north steps on the lines $x = 0$ and $x = 1$, the paths C_1, C_3, C_5, C_7 end with north steps on the lines $x = 7$ and $x = 8$, and B_4, B_6, D_2, D_4 are their analogues two units further inside; O and O' run along the boundary of the rectangle.

Theorem 4.19. *Let $n \geq 1$. For a noncrossing perfect matching M of $[2n + 2]$ put*

$$R_M = \{x \in L(2n + 1) : \ x1 \in \mathcal{F}_M\}.$$

Then R_M is an odd family of 2^n paths from $(0, 0)$ to $(n + 1, n)$, and the C_{n+1} families R_M are pairwise different. Moreover

$$2^n \leq \mathrm{M}_{\mathrm{od}}(n + 1, n) \leq \sqrt{\frac{(2n + 1)^{2n+1}}{(n + 1)^{n+1}\, n^n}} < 2^{\,n+1/2}.$$

Proof. Let $\{p_0, 2n + 2\}$ be the arc of M at the last point. A word of $\mathcal{F}_M$ ends in 1 exactly when its letter at p_0 is 0, so R_M consists of the words $x \in L(2n + 1)$ with $x_{p_0} = 0$ and $x_p \neq x_q$ for the n arcs $\{p, q\}$ of M inside $[2n + 1]$. It has 2^n elements, all of weight n. For distinct $x, y \in R_M$, Lemma 2.1(b) and Theorem 3.3 give $0 = B_{2n+2}(x1, y1) = B_{2n+1}(x, y) + 1$, so R_M is an odd family.

The matching is determined by R_M: the point p_0 is the only position at which all words of R_M carry the same letter, and two positions of $[2n + 1] \setminus \{p_0\}$ carry different letters in every word of R_M exactly when they form an arc of M. So $M \mapsto R_M$ is injective. The upper bound is (4.1). □

The upper bound exceeds the lower bound by a factor less than $\sqrt{2}$. Figure 9 illustrates the construction for $n = 2$.

4.10. **Small values and conjectures.** Table 1 lists the exact values of $M(a, b)$ and $M^*(a, b)$ for $1 \leq b \leq a \leq 7$ and $a + b \leq 12$, together with the integer part of the box bound. They were obtained by an exhaustive search for a largest set of pairwise compatible paths among the

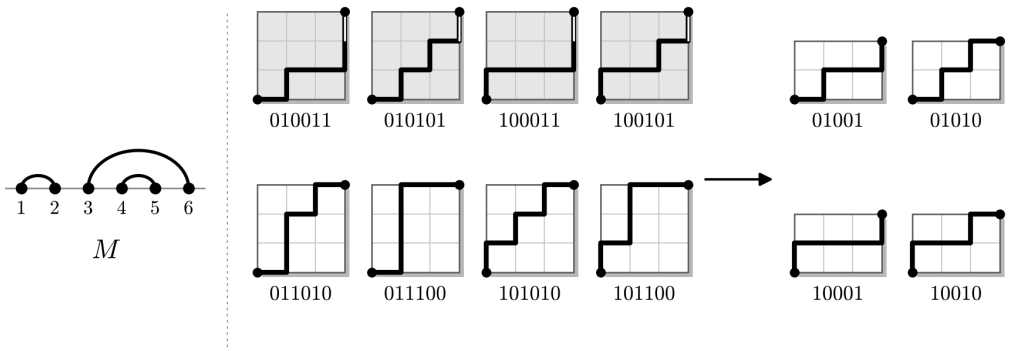


FIGURE 9. The construction of Theorem 4.19 for $n = 2$, with the matching M of $[6]$ whose arcs are $\{1,2\}$, $\{3,6\}$ and $\{4,5\}$. Of the eight paths of $\mathcal{F}_M$ from $(0,0)$ to $(3,3)$, four end with a north step; removing this step gives the odd family R_M of four paths from $(0,0)$ to $(3,2)$, any two of which share one or three edges.

$\binom{a+b}{b}$ paths of each rectangle, recorded in the verification archive [8]; no proof in this section depends on them.

TABLE 1. Exact values of $M(a,b)$ (matching parity) and $M^*(a,b)$ (mismatched parity), and the integer part of the box bound $\beta = \sqrt{m^m/(a^a b^b)}$ for $M(a,b)$. For even $a+b$ the columns M, M^* are $\mathrm{M_{ev}}$, $\mathrm{M_{od}}$; for odd $a+b$ they are $\mathrm{M_{od}}$, $\mathrm{M_{ev}}$.

(a,b)	M	M^*	$\lfloor\beta\rfloor$	(a,b)	M	M^*	$\lfloor\beta\rfloor$	(a,b)	M	M^*	$\lfloor\beta\rfloor$
$(1,1)$	2	1	2	$(4,4)$	16	6	16	$(6,4)$	21	11	28
$(2,1)$	2	2	2	$(5,1)$	2	3	3	$(6,5)$	32	16	44
$(2,2)$	4	2	4	$(5,2)$	5	6	8	$(6,6)$	64	16	64
$(3,1)$	2	2	3	$(5,3)$	10	7	14	$(7,1)$	2	4	4
$(3,2)$	4	3	5	$(5,4)$	16	10	22	$(7,2)$	6	8	10
$(3,3)$	8	3	8	$(5,5)$	32	10	32	$(7,3)$	12	11	21
$(4,1)$	2	3	3	$(6,1)$	2	4	4	$(7,4)$	21	15	36
$(4,2)$	5	4	6	$(6,2)$	6	7	9	$(7,5)$	42	18	58
$(4,3)$	8	6	10	$(6,3)$	10	10	17				

The entries $(n+1,n)$ of the table equal 2^n, the lower bound of Theorem 4.19.

Conjecture 4.20. *For every $n \geq 1$, $\mathrm{M_{od}}(n+1,n) = 2^n$, and the odd families of 2^n paths from $(0,0)$ to $(n+1,n)$ are exactly the C_{n+1} families R_M of Theorem 4.19.*

The exhaustive search confirms the value for $n \leq 5$ and, by listing all odd families of size 2^n, the classification for $n \leq 4$, where their numbers are $2, 5, 14, 42$ [8]. The first part can be restated for admissible systems. Splitting an odd family $\mathcal{F}$ by its last letter as $\mathcal{F} = X0 \cup Y1$, Lemma 2.1(b) shows that X and Y are even families of words of length $2n$ and of weights n and $n-1$, with $B_{2n} = 1$ on $X \times Y$, and conversely every such pair gives an odd family. So $\mathrm{M_{od}}(n+1,n) = 2^n$ means $|X| + |Y| \leq 2^n$ for these two-class systems, whereas Theorem 2.7 gives $|X|^2 + |Y|^2 \leq 4^n$, with strict inequality when both sets are nonempty by Theorem 3.2.[17]

[17]A single quadratic inequality cannot give the linear one: the families R_M split by their last letter as $(|X|,|Y|) = (2^n, 0)$ when the arc at $2n+2$ is $\{2n+1, 2n+2\}$, and as $(2^{n-1}, 2^{n-1})$ otherwise.

The value in Conjecture 4.20 would follow from a statement in which the endpoints are not prescribed.

Conjecture 4.21. *Let m be odd and let $\mathcal{F} \subseteq L(m)$ be a set of words with $B_m(x,y) = 1$ for all distinct $x, y \in \mathcal{F}$; that is, a set of lattice paths of length m from $(0,0)$, with arbitrary endpoints, any two of which share an odd number of edges. Then $|\mathcal{F}| \le 2^{(m-1)/2}$.*

The bound is attained by the families R_M. An exhaustive search confirms Conjecture 4.21 for $m \le 9$ [8]. Applied with $m = 2n+1$ it contains the inequality $\mathrm{M}_{\mathrm{od}}(n+1,n) \le 2^n$; the next proposition shows that the length $2n-1$ already suffices.

Proposition 4.22. *For $k \ge 0$ let $\phi(k)$ be the largest size of a set $\mathcal{F} \subseteq L(2k+1)$ whose words have weights of one parity and satisfy $B_{2k+1}(x,y) = 1$ for $x \ne y$, and let $\eta(k)$ be the largest size of a set $\mathcal{G} \subseteq L(2k)$ with*

$$B_{2k}(u,v) = [\,\mathrm{w}(u) \ne \mathrm{w}(v)\,] \qquad \text{for all distinct } u, v \in \mathcal{G}.$$

Then $\phi(k) = \eta(k)$. If Conjecture 4.21 holds for $m = 2k-1$, then $\eta(k) \le 2^k$. Consequently Conjecture 4.21 for $m = 2n-1$ implies $\mathrm{M}_{\mathrm{od}}(n+1,n) = 2^n$.

Proof. Let $\mathcal{F}$ be as in the definition of $\phi(k)$ and write $\mathcal{F} = X0 \cup Y1$ with $X, Y \subseteq L(2k)$. The weights of the words of X and those of the words of Y have opposite parities, so $X \cap Y = \varnothing$. For distinct $u, v \in X$, Lemma 2.1(b) gives $1 = B_{2k+1}(u0, v0) = B_{2k}(u,v) + [\,\mathrm{w}(u) = \mathrm{w}(v)\,]$, that is $B_{2k}(u,v) = [\,\mathrm{w}(u) \ne \mathrm{w}(v)\,]$; the same holds in Y. For $u \in X$ and $v \in Y$, $B_{2k}(u,v) = B_{2k+1}(u0, v1) = 1$, and $\mathrm{w}(u) \ne \mathrm{w}(v)$. So $\mathcal{G} = X \cup Y$ is as in the definition of $\eta(k)$, and $|\mathcal{G}| = |\mathcal{F}|$. Conversely, given such a set $\mathcal{G}$, append 0 to its words of even weight and 1 to its words of odd weight; the resulting words have even weight, and the same computations read backwards give $B_{2k+1} = 1$ on distinct pairs. Hence $\phi(k) = \eta(k)$.

Let $k \ge 1$, let $\mathcal{G}$ be as in the definition of $\eta(k)$, and put $\mathcal{G}^e = \{v \in L(2k-1) : ve \in \mathcal{G}\}$ for $e \in \{0,1\}$. For distinct $u, v \in \mathcal{G}^e$,

$$B_{2k-1}(u,v) = B_{2k}(ue, ve) + [\,\mathrm{w}(u) = \mathrm{w}(v)\,] = [\,\mathrm{w}(u) \ne \mathrm{w}(v)\,] + [\,\mathrm{w}(u) = \mathrm{w}(v)\,] = 1.$$

So $\mathcal{G}^e$ satisfies the hypothesis of Conjecture 4.21 with $m = 2k-1$, and $|\mathcal{G}| = |\mathcal{G}^0| + |\mathcal{G}^1| \le 2 \cdot 2^{k-1}$. Finally an odd family of paths from $(0,0)$ to $(n+1,n)$ is a set of words of constant weight in $L(2n+1)$, so $\mathrm{M}_{\mathrm{od}}(n+1,n) \le \phi(n) = \eta(n) \le 2^n$, and Theorem 4.19 gives equality. □

Conjecture 4.23. *The lower bounds of Theorem 4.16 are equalities for every $a \ge 2$:*

$$M(a,2) = \Big\lfloor \frac{a}{2} \Big\rfloor + 3, \qquad M^*(a,2) = \Big\lfloor \frac{5a-1}{4} \Big\rfloor \ \ (a \text{ odd}), \qquad M^*(a,2) = \Big\lfloor \frac{5a-2}{4} \Big\rfloor \ \ (a \text{ even}).$$

As stated above, this holds for $2 \le a \le 22$. For the matching part, Lemma 4.15(B) and (C) show that no three paths of a matching family share their first north step, and no three share their second;[18] deleting the first and the last column of the grid then gives $M(a,2) \le M(a-2,2) + 4$, which is weaker than the conjectured increment 1 and weaker than Corollary 4.17. For fixed $b \ge 3$ the order of growth of $M(a,b)$ is not known: Lemma 4.2 and Theorem 4.16 give a linear lower bound, and Theorem 4.5 gives the upper bound $O(a^{b/2})$.

[18] Three paths (i, i'_1), (i, i'_2), (i, i'_3) with $i'_1 < i'_2 < i'_3$ and $i < i'_1$ pairwise fall under (C) with $d \equiv i'_s + i'_t + 1$, so the three sums $i'_s + i'_t$ would all be odd, which is impossible. If $i'_1 = i$, the pairs containing the first path fall under (B) with $d \equiv i + i'_t + 1$, and the remaining pair under (C); the three conditions are again contradictory.

5. Regions, Dyck paths and meanders

The classification of Section 3 describes all even families of the largest size 2^n in the $n \times n$ square: they are the families $\mathcal{F}_M$, where M runs over the noncrossing perfect matchings of $[2n]$ and $\mathcal{F}_M$ consists of the words $w \in \{E,N\}^{2n}$ with $w_p \neq w_q$ for every arc $\{p,q\}$ of M. This section draws consequences. We determine when a region of the square still carries a family of size 2^n, we settle the case of Dyck paths, and we study how the C_n extremal families sit inside the set of all paths: how they intersect, that they are linearly independent, and how many of them contain a given path.

5.1. **Heights.** For a word $w \in \{E,N\}^{2n}$ and $0 \le i \le 2n$ the *height* $h_i(w)$ is the number of letters N minus the number of letters E among $w_1, \dots, w_i$. After i steps the path is at the point (x,y) with $x+y=i$ and $y-x=h_i(w)$. The two *staircases* are $(NE)^n$ and $(EN)^n$; their heights are $i \bmod 2$ and $-(i \bmod 2)$.

For a noncrossing perfect matching M of $[2n]$ let

$$d_M(i) = \#\{\{p,q\} \in M : \ p \le i < q\} \qquad (0 \le i \le 2n)$$

be the number of arcs open after the point i, and let D_M be the word with N at the left end and E at the right end of every arc. The word D_M is a Dyck word, $h_i(D_M) = d_M(i)$, and $M \mapsto D_M$ is the standard bijection between noncrossing perfect matchings of $[2n]$ and Dyck words of length $2n$ [38]. Since $d_M(i) \equiv i \pmod 2$ and $d_M(i) \ge 0$, we have $d_M(i) \ge i \bmod 2$ for all i, and equality holds for all i exactly when M is the matching $M_0 = \{\{1,2\},\{3,4\},\dots,\{2n-1,2n\}\}$. Its family $\mathcal{F}_{M_0}$ consists of the 2^n words in $\{NE,EN\}^n$.

Lemma 5.1. *Let M be a noncrossing perfect matching of $[2n]$, let $0 \le i \le 2n$ and $d = d_M(i)$. For $w \in \mathcal{F}_M$ and an arc $a = \{p < q\}$ of M put $\varepsilon_a(w) = +1$ if $w_p = N$ and $\varepsilon_a(w) = -1$ if $w_p = E$.*

(i) *$h_i(w) = \sum_a \varepsilon_a(w)$, the sum being over the d arcs $a = \{p<q\}$ with $p \le i < q$.*

(ii) *The number of $w \in \mathcal{F}_M$ with $h_i(w) = h$ is $\binom{d}{(d+h)/2} 2^{n-d}$; it is zero unless $|h| \le d$ and $h \equiv d \pmod 2$.*

(iii) *$D_M \in \mathcal{F}_M$ has height $d_M(i)$ at every i, and its complement has height $-d_M(i)$ at every i. Hence $-d_M(i) \le h_i(w) \le d_M(i)$ for all $w \in \mathcal{F}_M$, and each of the two bounds is attained by a single word for all i at once.*

Proof. An arc with both ends at most i contributes one N and one E to the first i letters of w. An arc with both ends larger than i contributes nothing. An arc $a = \{p<q\}$ with $p \le i < q$ contributes the letter w_p, that is, $\varepsilon_a(w)$ to the height. This is (i). A word of $\mathcal{F}_M$ is given by a free choice of the signs ε_a on the n arcs, which gives (ii). In D_M all signs are $+1$ and in its complement all are -1, which gives (iii). □

In coordinates, the number of paths of $\mathcal{F}_M$ through the lattice point (x,y) is $\binom{d}{(d+y-x)/2} 2^{n-d}$ with $d = d_M(x+y)$.

5.2. **Regions.** For a set R of paths from $(0,0)$ to (n,n) let $\mathrm{M}_{\mathrm{ev}}(R)$ be the largest size of an even family contained in R.

Theorem 5.2. (i) *For every set R of paths from $(0,0)$ to (n,n) we have $\mathrm{M}_{\mathrm{ev}}(R) \le 2^n$. The even families of size 2^n in R are exactly the families $\mathcal{F}_M$ with $\mathcal{F}_M \subseteq R$.*

(ii) *Let $L_i \le U_i$ $(0 \le i \le 2n)$ be integers, let*

$$R(L,U) = \{w : \ L_i \le h_i(w) \le U_i \text{ for } 0 \le i \le 2n\},$$

and put $V_i = \min(U_i, -L_i)$. *Then* $\mathcal{F}_M \subseteq R(L,U)$ *if and only if* $d_M(i) \le V_i$ *for all* i. *Consequently the following are equivalent:*
(a) $\mathrm{M}_{\mathrm{ev}}(R(L,U)) = 2^n$;
(b) $V_i \ge i \bmod 2$ *for all* i;
(c) *both staircases lie in* $R(L,U)$.
The number of even families of size 2^n *in* $R(L,U)$ *is the number of Dyck paths* D *of semilength* n *with* $h_i(D) \le V_i$ *for all* i.

Proof. (i) An even family in R is an even family in the square, so the statement is the classification of Section 3.

(ii) By Lemma 5.1(iii), the heights of the words of $\mathcal{F}_M$ at i fill the range from $-d_M(i)$ to $d_M(i)$, with both ends attained. So $\mathcal{F}_M \subseteq R(L,U)$ if and only if $L_i \le -d_M(i)$ and $d_M(i) \le U_i$ for all i, that is, $d_M(i) \le V_i$. Because $d_M(i) \ge i \bmod 2$ with equality for M_0, a matching with $d_M \le V$ exists if and only if (b) holds, and by (i) this is equivalent to (a). The staircases have heights $\pm(i \bmod 2)$, so they lie in $R(L,U)$ exactly when $L_i \le -(i \bmod 2)$ and $i \bmod 2 \le U_i$ for all i, which is (b). The count follows from the bijection $M \mapsto D_M$, under which $d_M(i) = h_i(D_M)$. □

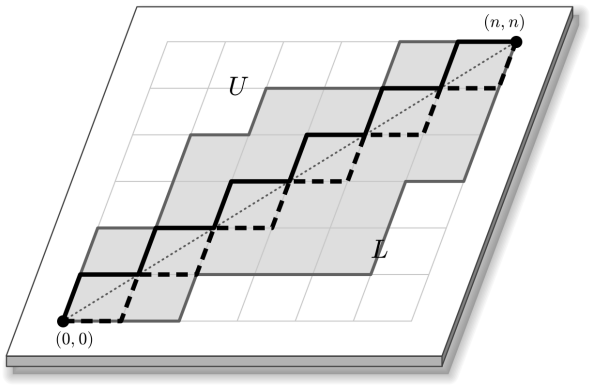


FIGURE 10. A region bounded by two lattice paths. It carries an even family of size 2^n exactly when it contains both staircases, and the extremal families correspond to the Dyck paths that stay below the upper boundary and below the mirror image of the lower boundary.

The theorem says that the two staircases are the only obstruction, and that once they fit the extremal families are counted by a classical lattice path problem.[19] We record three cases.

Corollary 5.3. *Let* $s \ge 1$. *For the strip* $\{w : |h_i(w)| \le s \text{ for all } i\}$ *and for the one-sided region* $\{w : h_i(w) \ge -s \text{ for all } i\}$ *we have* $\mathrm{M}_{\mathrm{ev}} = 2^n$, *and the number of even families of size* 2^n *is the number of Dyck paths of semilength* n *and height at most* s. *This number is* 1 *for* $s = 1$ *and* 2^{n-1} *for* $s = 2$.[20]

Proof. In both cases $V_i = s$ for all i, so Theorem 5.2(ii) applies. The only Dyck path of height at most 1 is $(NE)^n$. □

[19]When V is itself the height sequence of a Dyck path, the Dyck paths below V are counted by a determinant of binomial coefficients; we shall not need it.

[20]A Dyck path of height at most 2 is a sequence of factors $N(EN)^jE$ with $j \ge 0$, so these paths correspond to the compositions of n, of which there are 2^{n-1}. For $s = 3$ one obtains the Fibonacci numbers of odd index $1, 2, 5, 13, 34, \ldots$, and for $s \ge n$ the Catalan number C_n.

Corollary 5.4. *Let U be a path from $(0,0)$ to (n,n) and let R be the set of paths that stay weakly below U, that is, with $h_i(w) \le h_i(U)$ for all i. Then $\mathrm{M}_{\mathrm{ev}}(R) = 2^n$ if and only if $h_i(U) \ge i \bmod 2$ for all i, and in that case the even families of size 2^n in R are counted by the Dyck paths weakly below U.*

Proof. Here $U_i = h_i(U)$ and $L_i = -2n$, so $V_i = h_i(U)$. □

Corollary 5.5. (i) *Let $0 = i_0 < i_1 < \cdots < i_r < i_{r+1} = n$. For the paths through the diagonal points $(i_1,i_1),\dots,(i_r,i_r)$ we have $\mathrm{M}_{\mathrm{ev}} = 2^n$, and there are exactly $\prod_{t=0}^{r} C_{i_{t+1}-i_t}$ even families of size 2^n.*

(ii) *For the paths through a point (x,y) with $x \ne y$ we have $\mathrm{M}_{\mathrm{ev}} < 2^n$.*

(iii) *For the paths avoiding a lattice point (x,y) we have $\mathrm{M}_{\mathrm{ev}} = 2^n$ if and only if $|y-x| \ge 2$, that is, if and only if the point lies on neither staircase.*

Proof. Let $i = x+y$ and $h = y-x$. By Lemma 5.1(ii) all paths of $\mathcal{F}_M$ pass through (x,y) if and only if $d_M(i) = 0 = h$, and no path of $\mathcal{F}_M$ passes through (x,y) if and only if $d_M(i) < |h|$. This gives (ii) at once. For (i), the condition $d_M(2i_t) = 0$ for all t says that no arc of M passes over a point $2i_t$, so M is a union of noncrossing perfect matchings of the $r+1$ intervals between these points. For (iii), since $d_M(i) \ge i \bmod 2$ and $i \equiv h \pmod 2$, a matching with $d_M(i) < |h|$ exists if and only if $|h| \ge 2$, and then M_0 is one. □

5.3. **Dyck paths.** If all paths of a set R begin with N^a and end with E^a, then two of them share the $2a$ edges of the common prefix and suffix. Removing these edges does not change the parity of the number of common edges, and the middle parts are paths in a translate of the square of side $n-a$. So the even and odd families in R correspond to the even and odd families of middle parts. Applied to Dyck paths this gives the following.

Theorem 5.6. *Let $n \ge 1$ and let $\mathcal{D}_n$ be the set of the C_n paths from $(0,0)$ to (n,n) that stay weakly above the diagonal.*

(i) *An even family contained in $\mathcal{D}_n$ has at most 2^{n-1} members.*

(ii) *There is exactly one even family of size 2^{n-1} in $\mathcal{D}_n$: the paths $N\,u_1u_2\cdots u_{n-1}\,E$ with $u_t \in \{NE, EN\}$.*

(iii) *The same two statements hold for the Dyck paths of height at most s, for every $s \ge 2$.*

Proof. A Dyck path begins with N and ends with E. Its middle part $w \in \{E,N\}^{2n-2}$ is a path in the square of side $n-1$, and the original path stays weakly above the diagonal if and only if $h_i(w) \ge -1$ for all i. By the remark before the theorem, the even families in $\mathcal{D}_n$ correspond to the even families in the region $\{w :\ h_i(w) \ge -1\}$ of the square of side $n-1$. By Corollary 5.3 with $s=1$ the largest size is 2^{n-1} and it is attained by one family only, namely $\mathcal{F}_{M_0} = \{NE, EN\}^{n-1}$. This proves (i) and (ii). For Dyck paths of height at most s the region becomes $\{-1 \le h_i \le s-1\}$, for which $V_i = \min(s-1, 1) = 1$ again. □

So the restriction to Dyck paths costs exactly a factor 2, and it removes the Catalan multiplicity: of the C_{n-1} extremal families of paths that begin with N and end with E, only one consists of Dyck paths.[21]

5.4. **Intersections and meanders.** Two noncrossing perfect matchings M, M' of $[2n]$ can be drawn on the same $2n$ collinear points, the arcs of M above the line and those of M' below it. The arcs then form a collection of disjoint closed curves, a *meander system* in the sense of Di Francesco, Golinelli and Guitter [17]; we write $c(M,M')$ for the number of these curves (Figure 11).

[21] For the paths that touch the diagonal only at their two ends ($n \ge 2$) the same argument with $a = 2$ gives the maximum 2^{n-2}, attained only by the paths $NN\,u_1\cdots u_{n-2}\,EE$ with $u_t \in \{NE, EN\}$.

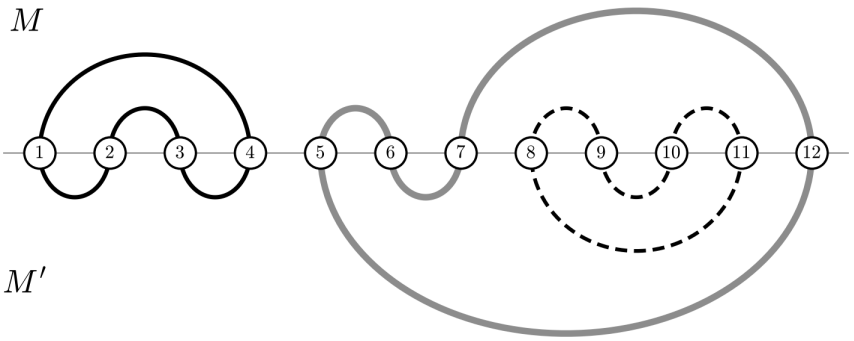


FIGURE 11. Two noncrossing perfect matchings of $[2n]$, one drawn above the line and one below. Their arcs close up into $c(M, M')$ loops, and the families $\mathcal{F}_M$ and $\mathcal{F}_{M'}$ share exactly $2^{c(M,M')}$ paths.

Theorem 5.7. *Let $M_1, \dots, M_k$ be noncrossing perfect matchings of $[2n]$, and let c be the number of connected components of the graph on $[2n]$ with edge set $M_1 \cup \cdots \cup M_k$. Then*

$$|\mathcal{F}_{M_1} \cap \cdots \cap \mathcal{F}_{M_k}| = 2^c.$$

In particular $|\mathcal{F}_M \cap \mathcal{F}_{M'}| = 2^{c(M,M')}$.

Proof. Let G be the graph in question. A word $w \in \{E, N\}^{2n}$ lies in the intersection if and only if $w_p \neq w_q$ for every edge $\{p, q\}$ of G, that is, if and only if w is a proper colouring of G with the two colours E and N. In a noncrossing perfect matching the points strictly between the two ends of an arc are matched among themselves, so their number is even, and every arc joins an odd position to an even one. Hence G is bipartite, and it has exactly 2^c proper 2-colourings. Each of them has n letters N, because it lies in $\mathcal{F}_{M_1}$. For $k = 2$ the components of G are the closed curves of the meander system. □

Corollary 5.8. (i) *Both staircases lie in every family $\mathcal{F}_M$, and for $n \geq 2$ the intersection of all C_n families consists of the two staircases only.*

(ii) *Two different families $\mathcal{F}_M$, $\mathcal{F}_{M'}$ share at most 2^{n-1} paths, and the number of unordered pairs $\{M, M'\}$ with $|\mathcal{F}_M \cap \mathcal{F}_{M'}| = 2^{n-1}$ is $\binom{2n}{n-2}$.*

(iii) *The number of ordered pairs (M, M') with $|\mathcal{F}_M \cap \mathcal{F}_{M'}| = 2^c$ is the number of meander systems of order n with c loops. In particular the ordered pairs of families that share only the two staircases are counted by the meandric numbers $1, 2, 8, 42, 262, 1828, \dots$*

Proof. (i) The staircases are the two colourings by the parity of the position, which are proper for every G as in the proof of Theorem 5.7. For $n \geq 2$ the union of M_0 with the noncrossing matching $\{\{2, 3\}, \{4, 5\}, \dots, \{2n-2, 2n-1\}, \{1, 2n\}\}$ is a Hamiltonian cycle on $[2n]$, so the graph of all matchings is connected and $c = 1$.

(ii) If $M \neq M'$, some component of $M \cup M'$ has more than two points, so $c(M, M') \leq n-1$. Equality means that M and M' have $n-2$ common arcs and that the remaining four points $a < b < c < d$ form one component. On these points one of the matchings is $\{a, b\}, \{c, d\}$ and the other is $\{a, d\}, \{b, c\}$, the third pairing $\{a, c\}, \{b, d\}$ being crossing. Let π be the partition of $[2n]$ into the $n-2$ common arcs and the block $\{a, b, c, d\}$. We claim that π is noncrossing. A common arc $\{e, g\}$ that crossed the block would separate the four points into two nonempty groups that are cyclically consecutive. If one group is a single point, the arc of M at that point crosses $\{e, g\}$. If the groups are $\{a, b\}$ and $\{c, d\}$, the arc $\{a, d\}$ crosses $\{e, g\}$, and if they are $\{b, c\}$ and $\{d, a\}$, the arc $\{a, b\}$ does. Each case contradicts the fact that $\{e, g\}$ is an arc of

both matchings. Conversely, a noncrossing partition of $[2n]$ with one block of size 4 and $n-2$ blocks of size 2 determines such an unordered pair, through the two noncrossing matchings of its block of size 4. By the formula of Kreweras [26, Théorème 4] the number of noncrossing partitions of $[m]$ with b blocks, of which m_j have size j, is $m!/((m-b+1)!\prod_j m_j!)$. For $m=2n$, $b=n-1$, $m_2=n-2$ and $m_4=1$ this is $(2n)!/((n+2)!\,(n-2)!) = \binom{2n}{n-2}$.

(iii) is a restatement of Theorem 5.7; the meandric numbers are by definition the numbers of pairs with one loop [17]. □

Table 2 lists the numbers of ordered pairs by the number of loops for $n \le 6$; its entries were obtained by listing all pairs of matchings [8]. The column $c = n-1$ is $2\binom{2n}{n-2}$.

n	$c=1$	$c=2$	$c=3$	$c=4$	$c=5$	$c=6$
1	1					
2	2	2				
3	8	12	5			
4	42	84	56	14		
5	262	640	580	240	42	
6	1828	5236	5894	3344	990	132

TABLE 2. The number of ordered pairs (M, M') of noncrossing perfect matchings of $[2n]$ whose families share exactly 2^c paths.

Lemma 5.9. *Let M, M' be noncrossing perfect matchings of $[2n]$. If $D_M \in \mathcal{F}_{M'}$, then $d_M(i) \le d_{M'}(i)$ for all i. If moreover $D_{M'} \in \mathcal{F}_M$, then $M = M'$.*

Proof. By Lemma 5.1(iii) for M' we have $d_M(i) = h_i(D_M) \le d_{M'}(i)$. If both memberships hold, then $d_M = d_{M'}$, so $D_M = D_{M'}$ and $M = M'$. □

Theorem 5.10. (i) *The incidence matrix between the C_n families $\mathcal{F}_M$ and the $\binom{2n}{n}$ paths has rank C_n over every field. Thus the indicator vectors of the extremal families are linearly independent; over $\mathbb{F}_2$ this says that for every nonempty set of extremal families some path lies in an odd number of them.*

(ii) *The matrix*

$$G_n = \left(|\mathcal{F}_M \cap \mathcal{F}_{M'}|\right)_{M,M'} = \left(2^{c(M,M')}\right)_{M,M'}$$

is positive definite, and

$$\det G_n = \prod_{i=1}^{n} (i+1)^{a_{n,i}}, \qquad a_{n,i} = \binom{2n}{n-i} - 2\binom{2n}{n-i-1} + \binom{2n}{n-i-2}. \tag{5.1}$$

Proof. (i) Order the matchings by a linear extension of the pointwise order of the functions d_M, and consider the square submatrix whose rows are the families $\mathcal{F}_{M'}$ and whose columns are the paths D_M. Its diagonal entries are 1, and by Lemma 5.9 the entry in row M' and column M vanishes unless $d_M \le d_{M'}$ pointwise. So this submatrix is triangular with ones on the diagonal, and its determinant is 1 in every field.

(ii) The matrix G_n is the Gram matrix of the indicator vectors in $\mathbb{R}^{\binom{2n}{n}}$, which are independent by (i); so it is positive definite. By Theorem 5.7 it is the matrix $\left(q^{c(M,M')}\right)$ at $q=2$. The determinant of this matrix is the meander determinant, the Gram determinant of the standard basis of the Temperley–Lieb algebra with loop value q, and Di Francesco, Golinelli

and Guitter [17] proved that it equals $\prod_{i=1}^{n} U_i(q)^{a_{n,i}}$, where U_i is the Chebyshev polynomial with $U_i(2\cos\theta) = \sin((i+1)\theta)/\sin\theta$. At $q = 2$, that is $\theta \to 0$, we get $U_i(2) = i+1$. □

The Gram determinants of the Temperley–Lieb algebra and of its cell modules were studied earlier by Martin [28] and Westbury [42]; the closed product formula for the full matrix $(q^{c(M,M')})$ is from [17]. Formula (5.1) gives $\det G_n = 2,\ 12,\ 5184,\ 8358844170240$ for $n = 1, 2, 3, 4$.[22] The meander determinant vanishes at $q = 1$ for $n \geq 2$, since $U_2(1) = 0$; at $q = 2$ no factor vanishes.

5.5. **The families containing a given path.** For a word $p \in \{E, N\}^{2n}$ with n letters of each kind let $f(p)$ be the number of noncrossing perfect matchings M of $[2n]$ with $p \in \mathcal{F}_M$, that is, with $p_a \neq p_b$ for every arc $\{a, b\}$ of M. We call such a matching *bichromatic* for p. By the classification of Section 3, $f(p)$ is the number of even families of size 2^n that contain the path p. Counting the pairs (M, p) with $p \in \mathcal{F}_M$ in two ways gives $\sum_p f(p) = 2^n C_n$, so a path lies in $2^n C_n / \binom{2n}{n} = 2^n/(n+1)$ extremal families on average.[23]

Lemma 5.11. (i) *For the empty word $f = 1$, and for $n \geq 1$*

$$f(p) = \sum_b f(p_2 \cdots p_{b-1})\, f(p_{b+1} \cdots p_{2n}),$$

the sum being over the positions b with $p_b \neq p_1$ for which $p_2 \cdots p_{b-1}$ has as many letters N as E.

(ii) *If M is bichromatic for p and $\{a < b\} \in M$, then $h_{a-1}(p) = h_b(p)$. Thus the steps a and b of p join the same two consecutive heights, in opposite directions.*

(iii) *If $p_i \neq p_{i+1}$ and p' is obtained from p by deleting these two letters, then $f(p) \geq f(p')$.*

Proof. (i) Let b be the partner of the point 1 in a bichromatic matching. Then $p_b \neq p_1$. The points $2, \ldots, b-1$ are matched among themselves by arcs that each carry one N and one E, so the word $p_2 \cdots p_{b-1}$ is balanced, and so is $p_{b+1} \cdots p_{2n}$. Conversely, for such a b every pair of bichromatic matchings of the two words gives a bichromatic matching of p. (ii) The word $p_{a+1} \cdots p_{b-1}$ is balanced, so $h_{b-1}(p) = h_a(p)$, and the steps a and b are opposite; hence $h_b(p) = h_{a-1}(p)$. (iii) Adding the arc $\{i, i+1\}$ to a bichromatic matching for p' gives a bichromatic matching for p, and different matchings give different ones. □

A *cyclic run* of p is a maximal block of equal letters when p is written around a circle. The number of cyclic runs is even.

Theorem 5.12. (i) *The function f is invariant under cyclic rotation of p, under reversal, and under the exchange of E and N. Hence $f(p)$ depends only on the cyclic sequence of the run lengths of p, up to reflection.*

(ii) *$1 \leq f(p) \leq C_n$, and $f(p) = C_n$ if and only if p is a staircase.*

(iii) *$f(p) = 1$ if and only if p has at most two cyclic runs, that is, $p = E^a N^n E^{n-a}$ or $p = N^a E^n N^{n-a}$ with $1 \leq a \leq n$. There are $2n$ such paths. Each lies in exactly one extremal family, and the matchings of these families are the rotations of the rainbow matching $\{\{j, 2n+1-j\} :\ 1 \leq j \leq n\}$, of which there are n when $n \geq 2$.*

(iv) *If $a, b, c, d \geq 0$ and $a + c = b + d = n$, then*

$$f(E^a N^b E^c N^d) = 1 + \min(a, b, c, d).$$

[22] For $n = 2$ the two matchings give $G_2 = \left(\begin{smallmatrix} 4 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, and for $n = 3$ the exponents are $a_{3,1} = a_{3,2} = 4$, $a_{3,3} = 1$, so that $\det G_3 = 2^4\, 3^4\, 4$.

[23] The values of f are far from uniform: the $2n$ rotations of $E^n N^n$ lie in one family each, and the two staircases lie in all C_n of them (Theorem 5.12).

Proof. (i) Place the points $1, \dots, 2n$ on a circle. Whether two arcs cross depends only on the cyclic order of their four ends, so rotations and reflections of the circle permute the noncrossing perfect matchings; and the condition $p_a \neq p_b$ does not change when all letters are exchanged.

(ii) A nonempty balanced word has two adjacent unequal letters. So Lemma 5.11(iii) and induction on n give $f(p) \geq 1$. If $f(p) = C_n$, then p lies in all extremal families, and p is a staircase by Corollary 5.8(i) when $n \geq 2$; for $n = 1$ both paths are staircases.

(iii) Let $p = E^n N^n$. The partner of the point 1 is the jth letter N for some j, and the word $E^{n-1}N^{j-1}$ between them is balanced only for $j = n$. The remaining word is $E^{n-1}N^{n-1}$, so by induction $f(p) = 1$ and the only bichromatic matching is the rainbow. By (i) the same holds for all rotations of $E^n N^n$, which are the $2n$ words listed, and the matching of a rotated word is the rotated rainbow. Rotating the rainbow by r positions produces the arc $\{r, r+1\}$; for $1 \leq r \leq n-1$ this arc is not in the rainbow, and the rotation by n maps the rainbow to itself. So there are n distinct rotations when $n \geq 2$.

Conversely, let p have at least four cyclic runs; we show $f(p) \geq 2$ by induction on n. If p has exactly four runs, this follows from (iv), which is proved below without using the present statement. If p has at least six runs, rotate it so that two unequal letters are adjacent, at the junction of two runs X and Y, and delete them. If neither run disappears, the number of cyclic runs is unchanged. If X disappears, the run before X merges with what is left of Y, or, when Y disappears as well, becomes adjacent to the run after Y, which carries the other letter; in either case the number of cyclic runs drops by 2. So the new word p' has at least four cyclic runs, and $f(p) \geq f(p') \geq 2$ by Lemma 5.11(iii).

(iv) If one of the four numbers is 0, the word has at most two cyclic runs and $f = 1$ by the first part of (iii). Let all four be positive. Rotating the word, and exchanging the letters if necessary, we may assume by (i) that a is the smallest of them. We use Lemma 5.11(i). The partner of the first letter is a letter N. If it is the jth letter of the first run of N, the word between is $E^{a-1}N^{j-1}$, which is balanced only for $j = a$; this is possible because $b \geq a$. Then the inner word $E^{a-1}N^{a-1}$ and the outer word $N^{b-a}E^cN^d$ each have at most two cyclic runs, and the contribution is 1. If the partner is the jth letter of the last run, the word between is $E^{a-1}N^bE^cN^{j-1}$, which is balanced only for $j = a + c - b = d$. Then the outer word is empty, and the inner word $E^{a-1}N^bE^cN^{d-1}$ has $f = 1 + \min(a-1, b, c, d-1) = a$ by induction on n. In total $f(E^aN^bE^cN^d) = a + 1$. □

So a path lies in a single extremal family exactly when it is one of the $2n$ rotations of E^nN^n, and the paths $E^aN^bE^cN^d$ with two steps lie in $1 + \min(a, b, c, d)$ families.

For words with more runs the value of f is governed by noncrossing partitions. We work this out for the words that return to height 0 after every pair of runs.

Theorem 5.13. *Let $a_1, \dots, a_k \geq 1$, let $p = N^{a_1}E^{a_1}N^{a_2}E^{a_2}\cdots N^{a_k}E^{a_k}$ and $t = \max_i a_i$. For $0 \leq \ell < t$ let $K_\ell = \{i : \ a_i > \ell\}$. Then $f(p)$ is the number of sequences $(\pi_0, \pi_1, \dots, \pi_{t-1})$ in which π_ℓ is a noncrossing partition of the ordered set K_ℓ and every block of π_ℓ is contained in a block of $\pi_{\ell-1}$, for $1 \leq \ell \leq t-1$.*

Proof. We call the factor $N^{a_i}E^{a_i}$ the ith pyramid. For $i \in K_\ell$ let u_i^ℓ be the position of the $(\ell+1)$st letter N of the ith pyramid and d_i^ℓ the position of its $(a_i - \ell)$th letter E. These are the steps of p between the heights ℓ and $\ell + 1$. Put $X_\ell = \{u_i^\ell, d_i^\ell : \ i \in K_\ell\}$. The positions satisfy

$$u_i^\ell < u_i^{\ell'} < d_i^{\ell'} < d_i^\ell \quad (\ell < \ell',\ i \in K_{\ell'}), \qquad d_i^\ell < u_j^{\ell'} \quad (i < j). \tag{5.2}$$

Step 1. By Lemma 5.11(ii) every arc of a bichromatic matching M joins two points of the same set X_ℓ, so M is a union of noncrossing perfect matchings M_ℓ of the sets X_ℓ. Conversely,

the points of X_ℓ, in their order, carry the letters $N, E, N, E, \dots$, and an arc of a noncrossing perfect matching of X_ℓ joins two points with an even number of points of X_ℓ between them; so every noncrossing perfect matching of X_ℓ joins letters N to letters E. Hence $f(p)$ is the number of tuples $(M_\ell)_{0\le\ell<t}$ of noncrossing perfect matchings of the sets X_ℓ such that no arc of M_ℓ crosses an arc of $M_{\ell'}$ for $\ell < \ell'$.

Step 2. Let K be a finite ordered set and let $u_i < d_i$ $(i \in K)$ be points with $d_i < u_j$ for $i < j$. For a noncrossing partition π of K let $\varphi(\pi)$ be the matching that contains, for each block $\{i_1 < \cdots < i_r\}$ of π, the arcs

$$\{d_{i_1}, u_{i_2}\},\ \{d_{i_2}, u_{i_3}\},\ \dots,\ \{d_{i_{r-1}}, u_{i_r}\} \quad \text{and} \quad \{u_{i_1}, d_{i_r}\}.$$

This is the standard bijection from the noncrossing partitions of K onto the noncrossing perfect matchings of the $2|K|$ points, obtained by drawing the two sides of each block [29, Lecture 9]. We write $M_\ell = \varphi(\pi_\ell)$ with π_ℓ a noncrossing partition of K_ℓ.

Step 3. Let $\ell < \ell'$, $\alpha = \pi_\ell$ and $\beta = \pi_{\ell'}$. We claim that no arc of $\varphi(\alpha)$ crosses an arc of $\varphi(\beta)$ if and only if every block of β is contained in a block of α.

Consider an arc of $\varphi(\beta)$; it joins points of two pyramids $m \le m'$ in one block of β. If $m = m'$ the arc is $\{u^{\ell'}_m, d^{\ell'}_m\}$, by (5.2) no point of X_ℓ lies between its ends, and no arc of $\varphi(\alpha)$ crosses it. If $m < m'$ the arc is $\{d^{\ell'}_m, u^{\ell'}_{m'}\}$ or $\{u^{\ell'}_m, d^{\ell'}_{m'}\}$, and by (5.2) the set of points of X_ℓ strictly between its ends is in both cases

$$I = \{d^\ell_m\} \cup \{u^\ell_q, d^\ell_q :\ m < q < m',\ q \in K_\ell\} \cup \{u^\ell_{m'}\}.$$

An arc of $\varphi(\alpha)$ crosses the given arc if and only if exactly one of its ends lies in I. Note that $d^\ell_i \in I$ if and only if $m \le i < m'$, and $u^\ell_j \in I$ if and only if $m < j \le m'$.

Suppose first that m and m' lie in one block S' of α, and let S be a block of α. If $S = S'$, then $\min S \le m$ and $\max S \ge m'$, so neither end of the arc $\{u^\ell_{\min S}, d^\ell_{\max S}\}$ lies in I. For consecutive elements $i < j$ of S, the conditions $m \le i < m'$ and $m < j \le m'$ are equivalent, because $m, m' \in S$ and no element of S lies strictly between i and j; so the arc $\{d^\ell_i, u^\ell_j\}$ has both ends in I or none. If $S \ne S'$, then, since α is noncrossing and $m, m' \in S'$, either S lies in the open interval between m and m', and all points of the pyramids of S are in I, or S is disjoint from $[m, m']$, and none is. In all cases there is no crossing. This proves that containment of blocks implies the absence of crossings.

Suppose now that there are no crossings, and that an arc of $\varphi(\beta)$ joins pyramids $m < m'$ lying in different blocks $S \ni m$ and $S' \ni m'$ of α. The arc $\{u^\ell_{\min S}, d^\ell_{\max S}\}$ of $\varphi(\alpha)$ has its first end outside I, because $\min S \le m$. So its second end is outside I, which means $\max S \ge m'$, and then $\max S > m'$ because $m' \notin S$. In the same way the arc $\{u^\ell_{\min S'}, d^\ell_{\max S'}\}$ has its second end outside I, because $\max S' \ge m'$; so its first end is outside I, which gives $\min S' < m$. Now $\min S' < m < m' < \max S$ with $\min S', m' \in S'$ and $m, \max S \in S$, so the blocks S and S' cross, a contradiction. Hence any two pyramids joined by an arc of $\varphi(\beta)$ lie in one block of α. Consecutive elements of a block of β are joined by an arc of $\varphi(\beta)$, so every block of β lies in a block of α.

Step 4. The relation "every block of $\pi_{\ell'}$ lies in a block of π_ℓ" is transitive, so it holds for all $\ell < \ell'$ as soon as it holds for consecutive levels. With Steps 1 to 3 this proves the theorem. □

Corollary 5.14. *For $t, k \ge 1$,*

$$f\big((N^tE^t)^k\big) = \frac{1}{tk+1}\binom{(t+1)k}{k}.$$

Thus the path with k steps of height and width t lies in a Fuss–Catalan number of extremal families: C_k for $t = 1$, $\frac{1}{2k+1}\binom{3k}{k}$ for $t = 2$, and $t+1$ for $k = 2$. The same holds for every rotation of the word, for instance for $(E^tN^t)^k$.

Proof. Here $K_\ell = [k]$ for all $\ell < t$, so by Theorem 5.13 f is the number of multichains $\pi_{t-1} \le \cdots \le \pi_0$ of t elements in the lattice of noncrossing partitions of $[k]$ ordered by refinement. By the results of Kreweras [26] and Edelman [18] this number is $\frac{1}{k}\binom{(t+1)k}{k-1}$, which equals $\frac{1}{tk+1}\binom{(t+1)k}{k}$. The last statement is Theorem 5.12(i). □

For $k = 2$ the corollary agrees with Theorem 5.12(iv), since $(N^tE^t)^2$ is a rotation of $E^tN^tE^tN^t$.

5.6. **Paths that determine a family.** By Theorem 5.12(iii) exactly n of the C_n extremal families (for $n \ge 2$) are determined by one of their paths. In general, for a noncrossing perfect matching M of $[2n]$ let $\sigma(M)$ be the least size of a set $P \subseteq \mathcal{F}_M$ that is contained in no other extremal family.

Proposition 5.15. *For every noncrossing perfect matching M of $[2n]$,*

$$\sigma(M) \le 1 + \lceil \log_2 n \rceil,$$

and for $n \ge 2$, $\sigma(M) = 1$ if and only if M is a rotation of the rainbow matching.

Proof. Let $s = 1 + \lceil \log_2 n \rceil$, so that $2^{s-1} \ge n$, and choose distinct vectors $v_a \in \{0\} \times \{0,1\}^{s-1}$ for the n arcs a of M. For $1 \le r \le s$ let $w^{(r)}$ be the word that has the letter $(v_a)_r$ at the left end of every arc a and the other letter at its right end, with $0 = E$ and $1 = N$. These s words lie in $\mathcal{F}_M$. Write them as the rows of an array. The column at the left end of a is v_a, and the column at the right end of a is the complementary vector, whose first coordinate is 1. If the s words lie in $\mathcal{F}_{M'}$, then every arc of M' joins two positions with complementary columns. The only position whose column is complementary to v_a is the right end of a, because the vectors v_a are distinct and have first coordinate 0. So M' contains every arc of M, and $M' = M$. The second statement is Theorem 5.12(iii). □

For $n = 1$ there is only one family, and $\sigma = 0$. For $n = 2, \ldots, 6$ an exhaustive computation gives $\max_M \sigma(M) = 1, 2, 2, 3, 3$ [8].[24]

Question 5.16. *What is $\max_M \sigma(M)$ as a function of n?*

5.7. **Regions without both staircases.** When a region does not contain both staircases, Theorem 5.2 gives only $\mathrm{M}_{\mathrm{ev}} < 2^n$, and the true value is not determined by the families $\mathcal{F}_M$. The simplest case is that of the paths through one point next to the diagonal.

Conjecture 5.17. *Let $0 \le i \le n-1$. An even family of paths from $(0,0)$ to (n,n) that all pass through the point $(i, i+1)$ has at most 2^{n-1} members.*

The value 2^{n-1} is attained: take a matching M with $d_M(2i+1) = 1$, for instance M_0, and the 2^{n-1} words of $\mathcal{F}_M$ in which the arc over the point $2i+1$ has N at its left end. By Lemma 5.1(i) these words have height 1 after $2i+1$ steps, so they pass through $(i, i+1)$. An exhaustive search confirms the conjecture for $n \le 5$ and all i [8]. For $i = 0$ the paths in question begin with N; deleting this letter turns the conjecture into the statement that an odd family of paths from $(0,0)$ to $(n, n-1)$ has at most 2^{n-1} members, which is the first part of Conjecture 4.20 for $n-1$. The argument of Theorem 5.6 does not apply, because only one end of the paths is forced.

[24] For $n = 6$ the distribution is: 6 matchings with $\sigma = 1$, 112 with $\sigma = 2$ and 14 with $\sigma = 3$.

6. Odd families in the square

We now turn to the other parity in the square. A family of paths from $(0,0)$ to (n,n) is *odd* if any two distinct members share an odd number of edges, and $\mathrm{M}_{\mathrm{od}}(n,n)$ is the largest size of an odd family. Shankar [35] posed the problem of determining $\mathrm{M}_{\mathrm{od}}(n,n)$ and proved that it grows quadratically; his construction has about $n^2/4$ paths. In contrast with the even problem, where the answer 2^n is exponential, the odd families are confined to a space of dimension about n^2, and the question is the constant in front of n^2. We prove

$$\tfrac{1}{2}n^2 - 3n \;\le\; \mathrm{M}_{\mathrm{od}}(n,n) \;\le\; n^2 - n + 1,$$

with a refinement of the upper bound that depends on n modulo 8. Throughout, a path is written as a word over $\{0,1\}$ with $0 = E$ and $1 = N$, as in Section 2, and $\mathrm{M}_{\mathrm{ev}}(a,b)$ denotes the largest size of an even family of paths from $(0,0)$ to (a,b), as in Section 4. Reflection in the diagonal gives $\mathrm{M}_{\mathrm{ev}}(a,b) = \mathrm{M}_{\mathrm{ev}}(b,a)$.

6.1. **Windows.** The lower bound rests on a way of placing even families of tall narrow rectangles side by side in the square (Figure 12).

Lemma 6.1 (windows)**.** *Let $n \ge 2$ be even and let $w_1, \dots, w_k$ be odd positive integers with $\sum_{i=1}^{k}(w_i+1) \le n+1$. Then*

$$\mathrm{M}_{\mathrm{od}}(n,n) \;\ge\; \sum_{i=1}^{k} \mathrm{M}_{\mathrm{ev}}(w_i,\, n-2).$$

Proof. Put $j_1 = 0$ and $j_{i+1} = j_i + w_i + 1$. Every j_i is even, because every $w_i + 1$ is even, and $j_k + w_k = \sum_i (w_i+1) - 1 \le n$. Let $\mathcal{G}_i$ be an even family of $\mathrm{M}_{\mathrm{ev}}(w_i, n-2)$ paths from $(0,0)$ to $(w_i, n-2)$, and for $\tau \in \mathcal{G}_i$ let

$$P(i,\tau) = E^{\,j_i}\, N\, \tau\, N\, E^{\,n-w_i-j_i}.$$

This is a path from $(0,0)$ to (n,n). It follows the bottom side of the square to $(j_i, 0)$, makes a north step, follows a translate of τ from $(j_i, 1)$ to $(j_i + w_i, n-1)$, makes a north step to $(j_i + w_i, n)$, and follows the top side. We call the part from $(j_i, 0)$ to $(j_i + w_i, n)$ its *climb*. The climb lies in the strip $j_i \le x \le j_i + w_i$, and since it begins and ends with a north step it contains no edge of the bottom side and no edge of the top side of the square. The word $P(i,\tau)$ determines j_i, as the number of letters E before its first letter N, and then τ; so the paths $P(i,\tau)$ are distinct.

Consider two paths of the same window, $P(i,\tau)$ and $P(i,\tau')$ with $\tau \ne \tau'$ in $\mathcal{G}_i$. They share the j_i edges on the bottom side, the two north steps, the $n - w_i - j_i$ edges on the top side, and $|E(\tau) \cap E(\tau')|$ further edges. The total $n - w_i + 2 + |E(\tau) \cap E(\tau')|$ is odd, because n is even, w_i is odd and $\mathcal{G}_i$ is an even family.

Consider two paths of different windows $i < i'$, and write $j = j_i$, $w = w_i$, $j' = j_{i'}$, $w' = w_{i'}$; then $j' \ge j + w + 1$. The two climbs lie in the disjoint strips $j \le x \le j + w$ and $j' \le x \le j' + w'$, so they share no edge. A climb has no edge on the bottom or the top side, so it shares no edge with the horizontal runs of the other path. The two bottom runs cover $[0, j]$ and $[0, j']$ and share j edges; the two top runs cover $[j + w, n]$ and $[j' + w', n]$ and share $n - j' - w'$ edges; a bottom run and a top run share nothing. The total $j + n - j' - w'$ is odd, because j, j' and n are even and w' is odd. □

The simplest choice is $n/2$ windows of width 1. A path from $(0,0)$ to $(1, n-2)$ is $N^k E N^{n-2-k}$, and for even $k < k'$ two such paths share the $k + (n - 2 - k')$ edges on the left and right sides

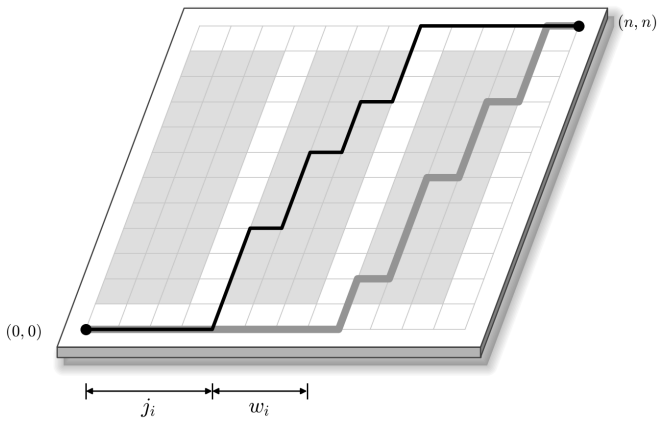


FIGURE 12. The window construction. A path runs along the bottom side to the left edge of its window, climbs inside the window along a path of an even family, and runs along the top side. Consecutive windows are separated by one empty column.

of the window and nothing else, an even number. So

$$\mathrm{M}_{\mathrm{ev}}(1, n-2) \geq \frac{n}{2} \qquad \text{and} \qquad \mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{n^2}{4} \qquad (n \text{ even}), \tag{6.1}$$

which is the order of Shankar's construction. Wider windows do better, because the even families of a rectangle with three columns already have twice as many members per unit of width.

6.2. **The lower bound.**

Theorem 6.2. (i) *If* $n \equiv 0 \pmod 4$ *and* $n \geq 12$, *then* $\mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{n^2}{2} - \frac{3n}{2}$.
(ii) *If* $n \equiv 2 \pmod 4$ *and* $n \geq 10$, *then* $\mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{n^2}{2} - 2n + 3$.
(iii) $\mathrm{M}_{\mathrm{od}}(n,n) \geq \mathrm{M}_{\mathrm{od}}(n-1,n-1)$ *for every* $n \geq 2$.
(iv) $\mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{n^2}{2} - 3n$ *for every* $n \geq 1$. *In particular* $\liminf_{n\to\infty} \mathrm{M}_{\mathrm{od}}(n,n)/n^2 \geq \frac{1}{2}$.

Proof. We use the bound $\mathrm{M}_{\mathrm{ev}}(a,3) \geq 2a-2$ for even $a \geq 8$, which is part of Theorem 4.18; by reflection, $\mathrm{M}_{\mathrm{ev}}(3,a) \geq 2a-2$.

(i) Apply Lemma 6.1 with $n/4$ windows of width 3, for which $\sum(w_i+1) = n$. Here $a = n-2 \geq 10$ is even, so each window contributes at least $2(n-2)-2 = 2n-6$ paths, and $\frac{n}{4}(2n-6) = \frac{n^2}{2} - \frac{3n}{2}$.

(ii) Apply Lemma 6.1 with $(n-2)/4$ windows of width 3 followed by one window of width 1; again $\sum(w_i+1) = n$. Here $a = n-2 \geq 8$ is even, and the window of width 1 contributes at least $n/2$ paths by (6.1). The total is $\frac{n-2}{4}(2n-6) + \frac{n}{2} = \frac{n^2}{2} - 2n + 3$.

(iii) If $\mathcal{F}$ is an odd family in the square of side $n-1$, then $\{1x0 : \ x \in \mathcal{F}\}$ is an odd family in the square of side n. Indeed, the path $1x0$ consists of the edge from $(0,0)$ to $(0,1)$, the path x moved up by one unit, and the edge from $(n-1,n)$ to (n,n); so two such paths share two more edges than the original ones.

(iv) For even $n \geq 10$ the bound is contained in (i) and (ii). For odd $n \geq 11$ we combine (iii) with (i) or (ii) for $n-1$: if $n \equiv 1 \pmod 4$ we get $\mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{(n-1)^2}{2} - \frac{3(n-1)}{2} = \frac{n^2}{2} - \frac{5n}{2} + 2$, and if $n \equiv 3 \pmod 4$ we get $\mathrm{M}_{\mathrm{od}}(n,n) \geq \frac{(n-1)^2}{2} - 2(n-1) + 3 = \frac{n^2}{2} - 3n + \frac{11}{2}$. For $n \leq 6$ the number $\frac{n^2}{2} - 3n$ is not positive. For $n = 8$ it is 8, and $\mathrm{M}_{\mathrm{od}}(8,8) \geq 16$ by (6.1). For $n = 9$ it is 13.5,

and $\mathrm{M}_{\mathrm{od}}(9,9) \geq \mathrm{M}_{\mathrm{od}}(8,8) \geq 16$ by (iii). For $n = 7$ it is 3.5, and $\mathrm{M}_{\mathrm{od}}(7,7) \geq \mathrm{M}_{\mathrm{od}}(6,6) \geq 9$ by (iii) and (6.1). □

For $n = 12$ the family of (i) has 54 paths and for $n = 20$ it has 170, against 36 and 100 from windows of width 1.[25] The paths of the family are easy to describe: each makes its first north step at a point $(j, 0)$ with $j \equiv 0 \pmod 4$, crosses the three columns to its right along a path of the three-column family, and finishes along the top side.

6.3. **The upper bound.** The upper bound comes from a quadratic form on the cycle space of the grid. The *cells* are the n^2 unit squares of the $n \times n$ square; two cells are *adjacent* if they share an edge. Let A be the adjacency matrix of the cells over $\mathbb{F}_2$, a symmetric $n^2 \times n^2$ matrix with zero diagonal. We need its kernel, which is the case $a = b = n$ of Theorem 4.6; we state it in the form used below.

Lemma 6.3. *Regard a vector $x \in \mathbb{F}_2^{n^2}$ as an array of n rows $x^{(1)}, \dots, x^{(n)} \in \mathbb{F}_2^n$. The map $x \mapsto x^{(1)}$ is an isomorphism from $\ker A$ onto $\mathbb{F}_2^n$. In particular $\operatorname{rank} A = n^2 - n$.*

Proof. By the proof of Theorem 4.6, the map $x \mapsto x^{(1)}$ is an isomorphism from $\ker A$ onto $\ker f_{n+1}(P_n)$. By the proof of Lemma 4.8, f_{n+1} is the characteristic polynomial of P_n, so $f_{n+1}(P_n) = 0$ by the Cayley–Hamilton theorem, and the kernel is all of $\mathbb{F}_2^n$. □

Let Z be the cycle space of the grid graph of the square, that is, the space of all edge sets in which every vertex has even degree, with the symmetric difference as addition. The boundaries of the cells form a basis of Z, so each element of Z is ∂D for a unique set D of cells, where ∂D is the sum of the boundaries of the cells of D. For edge sets X, Y write $\langle X, Y\rangle = |X \cap Y| \bmod 2$. On Z define

$$\omega(D, D') = \langle \partial D, \partial D'\rangle, \qquad q(D) = \tfrac{1}{2}|\partial D| \bmod 2.$$

An edge lies in ∂D when exactly one of the cells containing it is in D, so $|\partial D| = 4|D| - 2e(D)$, where $e(D)$ is the number of pairs of adjacent cells in D. Thus $|\partial D|$ is even, q is well defined, and $q(D) \equiv e(D) \pmod 2$. Two different cells share one edge if they are adjacent and none otherwise, and a cell has four edges; so ω is the bilinear form with matrix A in the basis of cells. From $|X + Y| = |X| + |Y| - 2|X \cap Y|$ we get

$$q(D + D') = q(D) + q(D') + \omega(D, D'), \tag{6.2}$$

so q is a quadratic form with polar form ω.

Lemma 6.4. *Let $g = \binom{n}{2}$. The form q vanishes on $\ker A$, so q and ω are defined on $V = Z/\ker A$, a space of dimension $2g$ on which ω is nondegenerate. The number of vectors $v \in V$ with $q(v) = 0$ is $2^{2g-1} + 2^{g-1}$.*

Proof. Colour the cell in row i and column j black if $i + j$ is even and white otherwise, and let Z_b and Z_w be the spans of the black and of the white cells. Two cells of the same colour are never adjacent, so q vanishes on Z_b and on Z_w, and A maps Z_b into Z_w and Z_w into Z_b. Hence if $x = x_b + x_w$ lies in $\ker A$, with $x_b \in Z_b$ and $x_w \in Z_w$, then $Ax_b = Ax_w = 0$, and by (6.2)

$$q(x) = q(x_b) + q(x_w) + \omega(x_b, x_w) = 0.$$

For $x \in \ker A$ and any D, (6.2) now gives $q(D + x) = q(D)$. So q is defined on V, as is ω, whose radical is $\ker A$; and $\dim V = n^2 - n = 2g$ by Lemma 6.3.

Let L be the image of Z_b in V. We compute its dimension. If $x \in \ker A$ has its first row supported on black cells, then $x_b \in \ker A$ has the same first row, so $x = x_b$ by Lemma 6.3.

[25] The families of (i) and (ii) have $33, 54, 73, 104, 129, 170$ members for $n = 10, 12, \dots, 20$. They were also checked pair by pair by machine for all even $n \leq 30$ [8].

Together with the surjectivity in that lemma, this shows that $\ker A \cap Z_b$ is mapped isomorphically onto the vectors of $\mathbb{F}_2^n$ supported on the $\lceil n/2 \rceil$ black cells of the first row. There are $\lceil n^2/2 \rceil$ black cells in all, so

$$\dim L = \lceil n^2/2 \rceil - \lceil n/2 \rceil = g,$$

as one checks separately for even and odd n. The form q vanishes on L, and so does ω, by (6.2). Thus $L \subseteq L^\perp$, and $L = L^\perp$ because $\dim L^\perp = 2g - g$.

We count the zeros of q on the cosets of L. On L there are 2^g. For $v \notin L$ we have $q(v+l) = q(v)+\omega(v,l)$ for $l \in L$, and $l \mapsto \omega(v,l)$ is a nonzero linear form on L, because $v \notin L^\perp$; so the coset $v + L$ contains 2^{g-1} zeros. In total there are $2^g + (2^g - 1)2^{g-1} = 2^{2g-1} + 2^{g-1}$ zeros. □

In the language of quadratic forms over $\mathbb{F}_2$, the lemma says that (V, q) has Arf invariant 0: it is the hyperbolic space of dimension $2g$, and L is a maximal totally singular subspace.

Theorem 6.5. *Let $n \geq 1$. Then*

$$\mathrm{M}_{\mathrm{od}}(n,n) \leq \begin{cases} n(n-1)+1, & n \equiv 0,1 \pmod 8, \\ n(n-1), & n \equiv 2,7 \pmod 8, \\ n(n-1)-1, & n \equiv 3,4,5,6 \pmod 8. \end{cases}$$

Proof. Let $g = \binom{n}{2}$ and let S be the staircase $(NE)^n$, regarded as a set of $2n$ edges.

Paths give singular vectors. Let P be a path from $(0,0)$ to (n,n). The staircase contains the vertical edges from (i,i) to $(i,i+1)$ and the horizontal edges from $(i,i+1)$ to $(i+1,i+1)$. So P contains a vertical edge of S exactly when it makes a north step from a point with $y-x=0$, and a horizontal edge of S exactly when it makes an east step from a point with $y-x=1$. Hence $|P \cap S|$ is the number of steps of P at which $y-x$ passes between the values 0 and 1. Since $y-x$ is 0 at both ends of P, this number is even.

For paths P and P' the sets $P+S$ and $P'+S$ lie in Z, and

$$\langle P+S, P'+S \rangle = \langle P, P' \rangle + \langle P, S \rangle + \langle P', S \rangle + \langle S, S \rangle = \langle P, P' \rangle,$$

because $|S| = 2n$. Moreover $|P+S| = 4n - 2|P \cap S|$, so $q(P+S) \equiv |P \cap S| \equiv 0 \pmod 2$. Let $x_P \in V$ be the class of $P+S$. If $P_1, \ldots, P_m$ is an odd family and $x_i = x_{P_i}$, then

$$q(x_i) = 0 \quad (1 \leq i \leq m), \qquad \omega(x_i, x_j) = 1 \quad (i \neq j). \tag{6.3}$$

Independence. Take an even number $2k$ of the vectors x_i. Their Gram matrix with respect to ω is $J+I$, where J is the all-ones matrix of size $2k$, and $(J+I)^2 = 2kJ + I = I$ over $\mathbb{F}_2$. So the Gram matrix is invertible and the $2k$ vectors are linearly independent. Since $\dim V = 2g$, this gives $m \leq 2g+1$; and if $m \geq 2g$, then $x_1, \ldots, x_{2g}$ is a basis of V.

The case $m \geq 2g$. For $T \subseteq [2g]$, (6.2) and (6.3) give $q(\sum_{i \in T} x_i) = \binom{|T|}{2} \bmod 2$, which is 0 exactly when $|T| \equiv 0, 1 \pmod 4$. So the number of zeros of q on V is

$$\sum_{t \equiv 0,1 \pmod 4} \binom{2g}{t} = 2^{2g-1} + \tfrac{1}{2}\operatorname{Re}\big((1-\mathrm{i})(1+\mathrm{i})^{2g}\big) = 2^{2g-1} + 2^{g-1}\operatorname{Re}\big((1-\mathrm{i})\,\mathrm{i}^{\,g}\big),$$

where we used that $\frac{1}{2} + \frac{1}{2}\operatorname{Re}\big((1-\mathrm{i})\mathrm{i}^t\big)$ is 1 for $t \equiv 0, 1$ and 0 for $t \equiv 2, 3 \pmod 4$, and that $(1+\mathrm{i})^2 = 2\mathrm{i}$. The result is $2^{2g-1} + 2^{g-1}$ for $g \equiv 0, 1 \pmod 4$ and $2^{2g-1} - 2^{g-1}$ for $g \equiv 2, 3 \pmod 4$. By Lemma 6.4 the second value is impossible. So $m \leq 2g-1$ when $g \equiv 2, 3 \pmod 4$.

The case $m = 2g+1$. The vector $s = \sum_{i=1}^m x_i$ satisfies $\omega(s, x_j) = m - 1 = 0$ for all j. Since $x_1, \ldots, x_{2g}$ span V and ω is nondegenerate, $s = 0$. Then $0 = q(s) = \binom{2g+1}{2} = g(2g+1) \equiv g \pmod 2$. So $m = 2g+1$ forces g to be even, and then $g \equiv 0 \pmod 4$ by the previous case.

In summary $m \le 2g+1$ if $g \equiv 0$, $m \le 2g$ if $g \equiv 1$, and $m \le 2g-1$ if $g \equiv 2, 3 \pmod 4$. The residues of $g = \binom{n}{2}$ modulo 4 for $n \equiv 0, 1, \dots, 7 \pmod 8$ are $0, 0, 1, 3, 2, 2, 3, 1$, and $2g = n(n-1)$. □

For $n = 2$ the theorem gives $\mathrm{M}_{\mathrm{od}}(2,2) \le 2$, which is the true value.[26] The argument uses only the configuration (6.3) in a hyperbolic quadratic space, and the leading constant 1 is the dimension count $\dim V = n^2 - n$; an improvement of the constant has to use more of the geometry of paths.

6.4. **Small squares and a conjecture.** Exhaustive search gives $\mathrm{M}_{\mathrm{od}}(n,n) = 1, 2, 3, 6, 10, 16$ for $n = 1, \dots, 6$ [8]. For $7 \le n \le 9$ the graph of all paths is too large for the exhaustive search we ran, and a randomised local search produced the odd families counted in Table 3; each was verified pair by pair [8]. The family for $n = 7$ is listed in Table 4.

n	6	7	8	9
largest odd family found	16	23	31	36
$\binom{n}{2}$	15	21	28	36
upper bound of Theorem 6.5	29	42	57	73

TABLE 3. Odd families in small squares. The value for $n = 6$ is exact; the others are lower bounds: $\mathrm{M}_{\mathrm{od}}(7,7) \ge 23$, $\mathrm{M}_{\mathrm{od}}(8,8) \ge 31$, $\mathrm{M}_{\mathrm{od}}(9,9) \ge 36$.

01001101101010	01111001100010	01111100110000
00000011011111	01010001111010	00001101010111
00000001111111	01111100100100	11111001000001
11011001100001	00000011110111	00000110111011
10100011010101	00011011110100	00000110101111
01001100111010	01111001101000	01111011100000
11110001100001	10100110100101	11101100100001
01101000101011	01100110001011	

TABLE 4. An odd family of 23 paths in the 7×7 square, as words with $0 = E$ and $1 = N$.

So $\mathrm{M}_{\mathrm{od}}(n,n)$ exceeds $\binom{n}{2}$ for $n = 6, 7, 8$, and the lower bound of Theorem 6.2 is not sharp for small n. To see what the window construction can give in the limit, recall from Definition 4.1 the largest size $M^*(a,b)$ of a mismatched family, which is $\mathrm{M}_{\mathrm{ev}}(a,b)$ if $a+b$ is odd and $\mathrm{M}_{\mathrm{od}}(a,b)$ if $a+b$ is even; this is the parity for which the families of the rectangle have polynomial size. The window lemma uses exactly these numbers, since $\mathrm{M}_{\mathrm{ev}}(w, n-2) = M^*(n-2, w)$ for odd w and even n. Exhaustive search gives the following values [8]; the row $b = 1$ is Proposition 4.13, and the row $b = 2$ agrees with Theorem 4.16.

The growth per unit of a is $\frac{1}{2}$ for $b = 1$, $\frac{5}{4}$ for $b = 2$, 2 for $b = 3$, and $\frac{11}{4}$ between $a = 6$ and $a = 10$ for $b = 4$.

Conjecture 6.6. *For every fixed $b \ge 1$,*

$$M^*(a,b) = \frac{3b-1}{4}\, a + O_b(1) \qquad (a \to \infty).$$

[26] The paths $NNEE$ and $NEEN$ share only their first edge.

b	range of a	$M^*(a,b)$
1	$1,\dots,10$	$1,2,2,3,3,4,4,5,5,6$
2	$2,\dots,16$	$2,3,4,6,7,8,9,11,12,13,14,16,17,18,19$
3	$3,\dots,13$	$3,6,7,10,11,14,15,18,18,22,23$
4	$5,\dots,10$	$10,11,15,17,20,22$

TABLE 5. Exact values of $M^*(a,b)$ for $b \le 4$.

The conjecture bears on the square through Lemma 6.1. For an odd width w and even n, the lemma with $\lfloor n/(w+1) \rfloor$ windows of width w gives $\mathrm{M}_{\mathrm{od}}(n,n) \ge \lfloor n/(w+1) \rfloor\, M^*(n-2,w)$. If Conjecture 6.6 holds for $b=w$, this is $\frac{3w-1}{4(w+1)}n^2 + O_w(n)$; the case $w=3$ is Theorem 6.2, the case $w=5$ would give the constant $\frac{7}{12}$, and if the conjecture holds for all odd b then

$$\liminf_{n\to\infty} \frac{\mathrm{M}_{\mathrm{od}}(n,n)}{n^2} \;\ge\; \frac{3}{4}.$$

Question 6.7. *Does* $\mathrm{M}_{\mathrm{od}}(n,n)/n^2$ *converge, and is the limit* $\frac{3}{4}$*?*

7. Lattice paths in higher dimensions

The parity problem makes sense for monotone lattice paths in $\mathbb{Z}^d$. In this section we extend Shankar's inequality to d letters, in a weighted form, and obtain an upper bound for every box. We show by an explicit construction that the bound has the right exponential growth rate, which is $\sqrt{d}$ per step in the cube. We then prove that for $d \ge 3$ the inequality is strict for every system other than the trivial one. This is a sharp difference from the plane, where the extremal systems are the skeleton systems of Section 3 and the bound 2^n is attained.

7.1. **Words, paths and admissible systems.** Let $d \ge 2$, let $[d] = \{0,1,\dots,d-1\}$ and let $L_d(m) = [d]^m$ be the set of words of length m. For a word x let $P(x) \in \mathbb{N}^d$ be the vector whose coordinate c is the number of letters c in x, and let $x_{<i} = x_1 \cdots x_{i-1}$. The path of x starts at the origin of $\mathbb{Z}^d$, and its ith step is the edge from $P(x_{<i})$ to $P(x_{<i}) + e_{x_i}$; it ends at $P(x)$. We write $E(x)$ for its set of edges. The ith edge of every path joins a point with coordinate sum $i-1$ to a point with coordinate sum i, so

$$|E(x) \cap E(y)| = \#\{i :\; x_i = y_i,\; P(x_{<i}) = P(y_{<i})\},$$

and we let $B_m(x,y) \in \mathbb{F}_2$ be the parity of this number, for $x,y \in L_d(m)$. Then $B_m(x,x) = m$, B_m is symmetric, and for $x,y \in L_d(m-1)$ and $a,b \in [d]$

$$B_m(xa,yb) = B_{m-1}(x,y) + [\,a=b\,]\,[\,P(x)=P(y)\,], \tag{7.1}$$

because the last position counts exactly when the last letters agree and the prefixes end at the same point. All equalities between values of B and integers are meant modulo 2.

Definition 7.1. An *admissible system* of length m over d letters is a finite collection $\mathcal{A}$ of pairwise disjoint nonempty subsets of $L_d(m)$, its *classes*, such that

(A1) P is constant on every class A; we write $P(A)$ for its value;
(A2) $B_m(x,y) = m$ for all x,y in the same class;
(A3) B_m is constant on $A \times A'$ for any two distinct classes A, A'.

For $\omega = (\omega_0,\dots,\omega_{d-1})$ with all $\omega_c > 0$ and $p \in \mathbb{N}^d$ put $\omega^p = \prod_c \omega_c^{p_c}$ and $|\omega| = \sum_c \omega_c$, and let

$$\|\mathcal{A}\|_\omega = \sum_{A\in\mathcal{A}} |A|^2\, \omega^{P(A)}.$$

For $d = 2$ this extends the notion of Section 2, and for $\omega = (1, 1)$ the quantity $\|\mathcal{A}\|_\omega$ is the sum $\sum |A|^2$ of Shankar's inequality. For a *box* $p \in \mathbb{N}^d$ with $m = p_0 + \cdots + p_{d-1}$ we consider families of words x with $P(x) = p$, that is, of paths from the origin to p. Let $\mathrm{M_{ev}}(p)$ and $\mathrm{M_{od}}(p)$ be the largest sizes of families in which any two distinct members share an even, respectively odd, number of edges. Let $M_{\mathrm{adm}}(p)$ be the one of the two for which the parity is that of m, and $M_{\mathrm{oth}}(p)$ the other one. A family counted by $M_{\mathrm{adm}}(p)$ is an admissible system with one class.

7.2. **The weighted inequality.** Let $\mathcal{A} = \{A_1, \dots, A_r\}$ be an admissible system of length $m \geq 1$. For $a \in [d]$ put

$$B_i^a = \{u \in L_d(m-1) :\ ua \in A_i\}, \qquad S_i = \{a \in [d] :\ B_i^a \neq \varnothing\}.$$

We call S_i the *support* of A_i; thus $|A_i| = \sum_{a \in S_i} |B_i^a|$. For $e \in [d]$ let $\mathcal{G}_e$ be the collection of the sets B_i^a over all pairs (i, a) with $e \in S_i$ and $a \in S_i$.

Lemma 7.2. *If $B_i^a \cap B_j^b \neq \varnothing$ and $a \neq b$, then $i \neq j$, $S_i = \{a\}$ and $S_j = \{b\}$.*

Proof. Let $u \in B_i^a \cap B_j^b$. The words $ua \in A_i$ and $ub \in A_j$ end at different points, so $i \neq j$ by (A1). By (7.1), $B_m(ua, ub) = B_{m-1}(u, u) = m - 1$, so $B_m = m - 1$ on $A_i \times A_j$ by (A3). Suppose that $g \in S_i$ and $g \neq a$, and take $vg \in A_i$. Since $P(vg) = P(ua)$, we have $P(v) = P(u) + e_a - e_g \neq P(u)$. Then (A2) and (7.1) give $m = B_m(ua, vg) = B_{m-1}(u, v)$, and (7.1) again gives $B_m(ub, vg) = B_{m-1}(u, v) = m$. This contradicts $B_m = m - 1$ on $A_j \times A_i$. So $S_i = \{a\}$, and $S_j = \{b\}$ by symmetry. □

Theorem 7.3. *Let $\mathcal{A}$ be an admissible system of length m over d letters and let $\omega \in \mathbb{R}^d_{>0}$.*

(i) *If $m \geq 1$, then for every $e \in [d]$ the sets listed in $\mathcal{G}_e$ are nonempty and pairwise disjoint, and $\mathcal{G}_e$ is an admissible system of length $m - 1$.*

(ii) *If $m \geq 1$, then*

$$\|\mathcal{A}\|_\omega \leq \sum_{e \in [d]} \omega_e \, \|\mathcal{G}_e\|_\omega,$$

with equality if and only if, for every i, the numbers $|B_i^a|/\omega_a$ $(a \in S_i)$ are equal.

(iii) $\|\mathcal{A}\|_\omega \leq |\omega|^m$. *In particular* $\sum_{A \in \mathcal{A}} |A|^2 \leq d^m$.

Proof. (i) Let $(i, a) \neq (j, b)$ be two pairs with $e \in S_i \cap S_j$, $a \in S_i$, $b \in S_j$. If $a = b$, then $i \neq j$ and $B_i^a \cap B_j^a = \varnothing$ because $A_i \cap A_j = \varnothing$. If $a \neq b$ and the two sets met, Lemma 7.2 would give $S_i = \{a\}$ and $S_j = \{b\}$, which is impossible since e lies in both. So the sets are pairwise disjoint. For (A1), $P(u) = P(A_i) - e_a$ for $u \in B_i^a$. For (A2), if $u, v \in B_i^a$ then by (7.1) $m = B_m(ua, va) = B_{m-1}(u, v) + 1$. For (A3), let $u \in B_i^a$ and $v \in B_j^b$ with $(i, a) \neq (j, b)$. By (7.1),

$$B_{m-1}(u, v) = B_m(ua, vb) + [\, a = b \,]\, [\, P(A_i) - e_a = P(A_j) - e_b \,].$$

The last term depends only on the two pairs. The first term is constant by (A3) for $\mathcal{A}$ if $i \neq j$, and it equals m by (A2) if $i = j$.

(ii) By the Cauchy–Schwarz inequality,

$$|A_i|^2 = \Big(\sum_{a \in S_i} \sqrt{\omega_a} \cdot \frac{|B_i^a|}{\sqrt{\omega_a}} \Big)^2 \leq \Big(\sum_{e \in S_i} \omega_e \Big) \Big(\sum_{a \in S_i} \frac{|B_i^a|^2}{\omega_a} \Big),$$

with equality if and only if $|B_i^a|/\omega_a$ is constant on S_i. Multiply by $\omega^{P(A_i)}$ and use $\omega^{P(A_i)}/\omega_a = \omega^{P(B_i^a)}$:

$$|A_i|^2 \, \omega^{P(A_i)} \leq \sum_{e \in S_i} \omega_e \sum_{a \in S_i} |B_i^a|^2 \, \omega^{P(B_i^a)}.$$

Summing over i and collecting the terms with a given e, the right-hand side becomes

$$\sum_{e\in[d]} \omega_e \sum_{(i,a)} |B_i^a|^2\, \omega^{P(B_i^a)},$$

the inner sum being over the pairs (i,a) that define $\mathcal{G}_e$. By (i) the inner sum is $\|\mathcal{G}_e\|_\omega$.

(iii) We use induction on m. For $m = 0$ there is at most one class, consisting of the empty word, and $\|\mathcal{A}\|_\omega \le 1$. For $m \ge 1$, (i), (ii) and the induction hypothesis give $\|\mathcal{A}\|_\omega \le \sum_e \omega_e\, |\omega|^{m-1} = |\omega|^m$. □

For $d = 2$ and $\omega = (1,1)$ this is Shankar's inequality [35], and the collections $\mathcal{G}_0$, $\mathcal{G}_1$ are the two systems of his induction. The bound is attained for every ω by the system of all singletons, since $\sum_{x\in L_d(m)} \omega^{P(x)} = |\omega|^m$. The weights make the inequality useful for boxes that are not cubes.

Corollary 7.4. *Let $p \in \mathbb{N}^d$ and $m = p_0 + \cdots + p_{d-1}$. Then*

$$M_{\mathrm{adm}}(p) \le \sqrt{\frac{m^m}{\prod_c p_c^{p_c}}} \qquad (0^0 = 1).$$

In particular $M_{\mathrm{adm}}(n,\dots,n) \le d^{\,dn/2}$. More generally, if any two distinct words of a set $\mathcal{F} \subseteq L_d(m)$ share a number of edges congruent to m modulo 2, and N_p is the number of words of $\mathcal{F}$ ending at p, then $\sum_p N_p^2\, \omega^p \le |\omega|^m$ for all $\omega \in \mathbb{R}^d_{>0}$.

Proof. The nonempty sets $\{x \in \mathcal{F} :\ P(x) = p\}$ form an admissible system, in which B_m is constant equal to m; Theorem 7.3(iii) gives the last statement. For a single box, suppose first that all p_c are positive, and take $\omega_c = p_c$. Then $|\omega| = m$ and $N_p^2 \prod_c p_c^{p_c} \le m^m$. If some p_c vanish, the words ending at p use only the letters with $p_c > 0$, and the same argument applies over this smaller alphabet. □

The choice $\omega = p$ minimises $|\omega|^m/\omega^p$. For $d = 2$ the corollary bounds the families of the rectangle with sides a and b by $\sqrt{(a+b)^{a+b}/(a^a b^b)}$, which equals $2^{(a+b)/2}$ only for $a = b$ and is smaller otherwise.[27] Rectangles are treated in Section 4.

7.3. **Block families.** We now construct large families. Fix a partition of the alphabet $[d]$ into k two-element sets $\{a_j, b_j\}$, the *pairs*, and s one-element sets $\{c_i\}$, the *singles*; thus $2k + s = d$. The *blocks* are the d words of length 2

$$a_j b_j, \quad b_j a_j \quad (1 \le j \le k), \qquad c_i c_i \quad (1 \le i \le s),$$

and the *type* of a block is the pair or single from which it is formed. Let $\mathcal{B}_r \subseteq L_d(2r)$ be the set of all concatenations of r blocks, so $|\mathcal{B}_r| = d^r$ (Figure 13).

Theorem 7.5. (i) *Any two words of $\mathcal{B}_r$ share an even number of edges.*

(ii) *Let $p \in \mathbb{N}^d$ satisfy $p_{a_j} = p_{b_j}$ for all pairs and $p_{c_i} \in 2\mathbb{N}$ for all singles, and let $r = \frac{1}{2}\sum_c p_c$. Then*

$$\mathrm{M}_{\mathrm{ev}}(p) \ \ge\ \frac{r!}{\prod_j p_{a_j}!\, \prod_i (p_{c_i}/2)!}\, 2^{\sum_j p_{a_j}}.$$

[27] Since $(a+b)^{a+b}/(a^a b^b) = ((a+b)/b)^b (1+b/a)^a < (e(a+b)/b)^b$, the bound is polynomial in a for fixed b: at most $(e(a+b)/b)^{b/2}$.

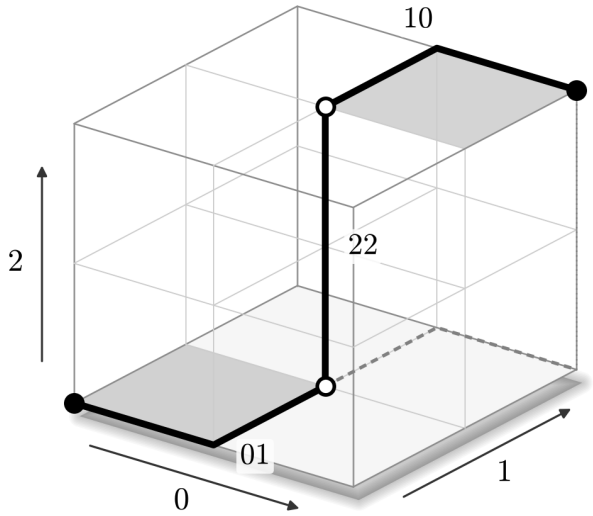


FIGURE 13. A monotone path in $\mathbb{Z}^3$ built from blocks: with the pair $\{0,1\}$ and the single $\{2\}$, each block is a corner 01 or 10 in a horizontal plane or a double step 22.

(iii) *Let $k = \lfloor d/2 \rfloor$, $s = d - 2k \in \{0,1\}$ and $t = k + s$, and let $n \geq 1$, with n even if d is odd. With $r = dn/2$,*

$$\frac{d^{\,dn/2}}{\binom{r+t-1}{t-1}} \;\leq\; \frac{r!\,2^{nk}}{(n!)^k\,((n/2)!)^s} \;\leq\; \mathrm{M}_{\mathrm{ev}}(n,\dots,n) \;\leq\; d^{\,dn/2}.$$

(iv) *For all $d \geq 2$, $\lim_{n\to\infty} M_{\mathrm{adm}}(n,\dots,n)^{1/(dn)} = \sqrt{d}$.*

Proof. (i) A block is determined by its first letter, and also by its second letter. Let $x, y \in \mathcal{B}_r$, consider their ith blocks, which occupy the positions $2i-1$ and $2i$, and let $\Delta = P(x_{<2i-1}) - P(y_{<2i-1})$. We show that the two positions contribute 0 or 2 common edges. Suppose $\Delta = 0$. If the two blocks are equal, both positions count. If they are different, their first letters differ, so position $2i-1$ does not count, and their second letters differ, so position $2i$ does not count. Suppose $\Delta \neq 0$. Then position $2i-1$ does not count. If position $2i$ counted, the second letters would agree, the two blocks would be equal, and the two prefixes of length $2i-1$ would differ by $\Delta \neq 0$, so the position would not count after all.

(ii) A word of $\mathcal{B}_r$ ends at p if and only if it has p_{a_j} blocks of the type $\{a_j, b_j\}$ and $p_{c_i}/2$ blocks of the type $\{c_i\}$. Such a word is given by the sequence of the types and by the choice of one of the two blocks for each pair type. By (i) these words form an even family.

(iii) The upper bound is Corollary 7.4, since $m = dn$ is even and $M_{\mathrm{adm}} = \mathrm{M}_{\mathrm{ev}}$. The middle inequality is (ii) for the partition into k pairs and s singles, with n blocks of each pair type and $n/2$ blocks of the single type. For the left inequality give each type τ the weight $w_\tau = 2$ if it is a pair and $w_\tau = 1$ if it is a single. Sorting the words of $\mathcal{B}_r$ by the numbers N_τ of blocks of each type gives

$$d^r = \sum_N T(N), \qquad T(N) = \frac{r!}{\prod_\tau N_\tau!} \prod_\tau w_\tau^{N_\tau},$$

where N runs over the $\binom{r+t-1}{t-1}$ vectors of nonnegative integers indexed by the t types with sum r. Let N maximise T. For types $\tau \neq \sigma$ with $N_\tau \geq 1$, the inequality $T(N - e_\tau + e_\sigma) \leq T(N)$

reads $N_\tau w_\sigma \le (N_\sigma + 1)w_\tau$, that is,

$$\frac{N_\tau}{w_\tau} \le \frac{N_\sigma + 1}{w_\sigma},$$

and this holds trivially when $N_\tau = 0$. Let $N^*_\tau = nw_\tau/2$; these are integers with sum $nd/2 = r$. If $N \ne N^*$, there are types with $N_\sigma < N^*_\sigma$ and $N_\tau > N^*_\tau$. Then $N_\tau/w_\tau > n/2$, while $(N_\sigma + 1)/w_\sigma \le N^*_\sigma/w_\sigma = n/2$, a contradiction. So the largest term is $T(N^*)$, which is the middle quantity, and the largest term is at least the average $d^r/\binom{r+t-1}{t-1}$.

(iv) The upper bound $M_{\mathrm{adm}}(n, \dots, n) \le d^{dn/2}$ is Corollary 7.4. If d is even, or d is odd and n is even, (iii) gives $M_{\mathrm{adm}}(n, \dots, n) \ge d^{dn/2}/\binom{r+t-1}{t-1}$, and the binomial coefficient is a polynomial in n of degree $t-1$. Let d and n be odd, so that $m = dn$ is odd and $M_{\mathrm{adm}} = \mathrm{M}_{\mathrm{od}}$. Take an even family in the cube of side $n-1$ and append the word $01\cdots(d-1)$ to each of its members. All members end at the same point, so every pair gains exactly d common edges, and the new family is an odd family in the cube of side n. Hence $\mathrm{M}_{\mathrm{od}}(n, \dots, n) \ge \mathrm{M}_{\mathrm{ev}}(n-1, \dots, n-1)$, and the limit follows in all cases. □

For $d = 2$ the three quantities in (iii) are all equal to 2^n, and $\mathcal{B}_n$ restricted to the square is the family $\mathcal{F}_{M_0} = \{NE, EN\}^n$ of Section 5. For $d \ge 3$ the lower and upper bounds in (iii) differ by a polynomial factor, and the next subsection shows that the upper bound is never attained.

7.4. **Rigidity in dimension at least three.** In the plane the systems with $\sum |A|^2 = 2^m$ are the skeleton systems of Section 3, of which there are many. Their existence rests on pairs of words that no third word of a different endpoint can tell apart. With three letters there are no such pairs beyond the obvious ones.

For a lattice point $v \in \mathbb{N}^d$ and letters $a \ne b$ let $Q(v; a, b)$ be the unit square with corners v, $v + e_a$, $v + e_b$, $v + e_a + e_b$, and let ∂Q be the set of its four edges. The corner $v + e_a + e_b$ is its *top*. The *level* of a lattice point is the sum of its coordinates.

Lemma 7.6. *Let $d \ge 3$ and $k \ge 1$, and let $x \ne y$ in $L_d(k)$ satisfy $B_k(x, y) = k + 1$ and*

$$B_k(x, w) = B_k(y, w) \quad \text{for every } w \in L_d(k) \text{ with } P(w) \notin \{P(x), P(y)\}. \tag{7.2}$$

Then x and y differ only in the last letter.

Proof. If $w = zabz'$ and $w' = zbaz'$ with $P(z) = v$, the paths of w and w' differ only around the square $Q = Q(v; a, b)$, so $E(w) + E(w') = \partial Q$ as sets modulo 2, and for every word u of the same length

$$B(u, w) + B(u, w') = |E(u) \cap \partial Q| \bmod 2. \tag{7.3}$$

Let D be the symmetric difference of $E(x)$ and $E(y)$. Call a square $Q(v; a, b)$ whose top has level at most k *good* if there is a word $w = zabz' \in L_d(k)$ with $P(z) = v$ and $P(w) \notin \{P(x), P(y)\}$. Then $P(w') = P(w)$ as well, and (7.3) for $u = x$ and for $u = y$ together with (7.2) gives $|E(x) \cap \partial Q| \equiv |E(y) \cap \partial Q|$. Hence

$$|D \cap \partial Q| \text{ is even for every good square } Q. \tag{7.4}$$

Which squares are good? Let T be the top of Q. If the level of T is at most $k-1$, the $d \ge 3$ words $z' = g^j$, with $g \in [d]$ and j equal to k minus the level of T, lead to d distinct endpoints, and one of them is neither $P(x)$ nor $P(y)$; so Q is good. If the level of T is k, then $w = zab$ ends at T, and Q is good exactly when $T \notin \{P(x), P(y)\}$.

Let t be the first position with $x_t \ne y_t$, and write $r = x_{<t} = y_{<t}$, $v_0 = P(r)$, $\alpha = x_t$ and $\beta = y_t$. We assume $t \le k-1$ and derive a contradiction. The two paths coincide up to the point v_0 of level $t-1$. Between the levels $t-1$ and t the set D consists of the two edges from

v_0 to $v_0 + e_\alpha$ and to $v_0 + e_\beta$. Between the levels t and $t+1$ it consists of the edge of x that leaves $v_0 + e_\alpha$ in the direction x_{t+1} and the edge of y that leaves $v_0 + e_\beta$ in the direction y_{t+1}.

Step A. Let $\gamma \notin \{\alpha, \beta\}$ and $Q = Q(v_0; \alpha, \gamma)$, with top $T_\gamma = v_0 + e_\alpha + e_\gamma$ of level $t+1 \le k$. Of its four edges, the edge from v_0 to $v_0 + e_\alpha$ lies in D; the edge from v_0 to $v_0 + e_\gamma$ does not; the edge from $v_0 + e_\gamma$ to T_γ does not, since neither path visits $v_0 + e_\gamma$; and the edge from $v_0 + e_\alpha$ to T_γ lies in D exactly when $x_{t+1} = \gamma$. So $|D \cap \partial Q| = 1 + [\, x_{t+1} = \gamma \,]$, and by (7.4)

$$Q(v_0; \alpha, \gamma) \text{ good } \implies x_{t+1} = \gamma. \tag{7.5}$$

Suppose $t + 1 \le k - 1$. Then all these squares are good, so $x_{t+1} = \gamma$ for every $\gamma \notin \{\alpha, \beta\}$. Since there is at least one such γ and x_{t+1} is a single letter, $d = 3$ and x_{t+1} is the third letter γ_0. Exchanging the roles of x and y gives $y_{t+1} = \gamma_0$.

Suppose $t + 1 = k$. If $Q(v_0; \alpha, \gamma)$ were good, (7.5) would give $x_k = \gamma$, so $T_\gamma = P(x)$ and the square would not be good. Hence $T_\gamma \in \{P(x), P(y)\}$ for every $\gamma \notin \{\alpha, \beta\}$. Now $P(y) = v_0 + e_\beta + e_{y_k}$ is different from $v_0 + e_\alpha + e_\gamma$, because $\beta \notin \{\alpha, \gamma\}$. So $T_\gamma = P(x)$, that is, $x_k = \gamma$, for every $\gamma \notin \{\alpha, \beta\}$. Again $d = 3$ and $x_k = \gamma_0$, and by symmetry $y_k = \gamma_0$. But then $x = r\alpha\gamma_0$ and $y = r\beta\gamma_0$ share exactly their first $k - 2$ edges, since at the last position the prefixes end at different points. So $B_k(x, y) = k$, which contradicts $B_k(x, y) = k + 1$.

Therefore $t \le k - 2$, $d = 3$, and $x = r\alpha\gamma\cdots$, $y = r\beta\gamma\cdots$, where γ is the third letter.

Step B. Let $v_1 = v_0 + e_\alpha + e_\gamma$ and $v_2 = v_0 + e_\beta + e_\gamma$ be the points of x and of y at level $t + 1$. For $b \in \{\alpha, \beta\}$ consider the square $Q_b = Q(v_0 + e_\alpha; \gamma, b)$, with top $T_b = v_1 + e_b$ of level $t + 2 \le k$. The edge from $v_0 + e_\alpha$ to v_1 is an edge of x, and y does not visit $v_0 + e_\alpha$; so it lies in D. The edge from $v_0 + e_\alpha$ to $v_0 + e_\alpha + e_b$ does not lie in D, because x leaves $v_0 + e_\alpha$ in the direction $\gamma \ne b$. The edge from $v_0 + e_\alpha + e_b$ to T_b does not lie in D: at level $t + 1$ the path x is at $v_1 \ne v_0 + e_\alpha + e_b$, because $b \ne \gamma$, and y is at $v_2 \ne v_0 + e_\alpha + e_b$, because $\gamma \notin \{\alpha, b\}$. The edge from v_1 to T_b lies in D exactly when $x_{t+2} = b$, because y is at $v_2 \ne v_1$. So $|D \cap \partial Q_b| = 1 + [\, x_{t+2} = b \,]$, and

$$Q_b \text{ good } \implies x_{t+2} = b.$$

If $t+2 \le k-1$, both Q_α and Q_β are good, and $x_{t+2} = \alpha = \beta$, which is false. If $t+2 = k$, a good Q_b would give $x_k = b$ and $T_b = P(x)$, so it would not be good. Hence $T_\alpha, T_\beta \in \{P(x), P(y)\}$, and since $T_\alpha \ne T_\beta$,

$$\{P(x), P(y)\} = \{v_1 + e_\alpha,\ v_1 + e_\beta\}.$$

The same argument with the roles of x and y exchanged gives $\{P(x), P(y)\} = \{v_2 + e_\alpha,\ v_2 + e_\beta\}$. Subtracting $v_0 + e_\gamma$ from both descriptions gives $\{2e_\alpha,\ e_\alpha + e_\beta\} = \{e_\alpha + e_\beta,\ 2e_\beta\}$, which is false.

So $t \le k - 1$ is impossible, and $t = k$. □

For $d = 2$ the conclusion fails: the words $x = 110$ and $y = 001$ share no edge, each shares an even number of edges with 000 and with 111, and so they satisfy the hypotheses with $k = 3$. In the proof the third letter enters through the squares $Q(v_0; \alpha, \gamma)$, which do not exist over two letters.

Theorem 7.7. *Let $d \ge 3$, let $\omega \in \mathbb{R}^d_{>0}$, and let $\mathcal{A}$ be an admissible system of length m over d letters with $\|\mathcal{A}\|_\omega = |\omega|^m$. Then $\mathcal{A}$ is the system of all singletons $\{x\}$, $x \in L_d(m)$.*

Proof. We use induction on m; the case $m = 0$ is clear. Let $m \ge 1$. By Theorem 7.3,

$$|\omega|^m = \|\mathcal{A}\|_\omega \le \sum_e \omega_e \|\mathcal{G}_e\|_\omega \le \sum_e \omega_e |\omega|^{m-1},$$

so $\|\mathcal{G}_e\|_\omega = |\omega|^{m-1}$ for every e, and by the induction hypothesis every $\mathcal{G}_e$ is the system of all singletons of $L_d(m-1)$. This has two consequences. First, every nonempty B_i^a is a singleton, because it is listed in $\mathcal{G}_a$. Second,

(7.6) for every $w \in L_d(m-1)$ and $e \in [d]$ there are i and a with $wa \in A_i$ and $e \in S_i$,

because $\{w\}$ is a class of $\mathcal{G}_e$.

Suppose that some class, say A_1, has $|S_1| \geq 2$, and let $ua, vb \in A_1$ with $a \neq b$. By (A1), $P(v) \neq P(u)$, and by (A2) and (7.1), $m = B_m(ua, vb) = B_{m-1}(u, v)$.

We claim that no word ug with $g \neq a$ lies in a class of $\mathcal{A}$. It does not lie in A_1, by (A1). If $ug \in A_j$ with $j \neq 1$, then (A3) gives $B_m(ua, ug) = B_m(vb, ug)$. By (7.1) the left-hand side is $B_{m-1}(u, u) = m-1$, and the right-hand side is $B_{m-1}(v, u) = m$, since $P(v) \neq P(u)$. This is a contradiction.

By (7.6) with $w = u$, for each $e \in [d]$ there are i and a' with $ua' \in A_i$ and $e \in S_i$. By the claim $a' = a$, hence $i = 1$. So $S_1 = [d]$, and since the sets B_1^c are singletons,

$$A_1 = \{u_c\, c : \ c \in [d]\}, \qquad P(u_c) = P(A_1) - e_c,$$

and $B_{m-1}(u_c, u_{c'}) = B_m(u_c c, u_{c'} c') = m$ for $c \neq c'$, by (7.1) and (A2).

Fix $c \neq c'$ and put $k = m-1$, $x = u_c$, $y = u_{c'}$. Then $x \neq y$, so $k \geq 1$, and $B_k(x, y) = k+1$. We verify (7.2). Let $w \in L_d(k)$ with $P(w) \notin \{P(x), P(y)\}$. By (7.6) some word wa lies in a class A_j. If $j = 1$, then $w = u_a$ with $a \notin \{c, c'\}$, and $B_k(x, w) = m = B_k(y, w)$. If $j \neq 1$, (A3) gives $B_m(u_c c, wa) = B_m(u_{c'} c', wa)$, that is,

$$B_k(x, w) + [\, c = a\,][\, P(x) = P(w)\,] = B_k(y, w) + [\, c' = a\,][\, P(y) = P(w)\,],$$

and both products vanish. So Lemma 7.6 applies: $u_c = r\alpha$ and $u_{c'} = r\beta$ for a word r and letters $\alpha \neq \beta$. Comparing endpoints, $e_\alpha - e_\beta = P(u_c) - P(u_{c'}) = e_{c'} - e_c$, so $\alpha = c'$. Thus u_c ends with the letter c', for every $c' \neq c$. Since $d \geq 3$ there are two different letters $c' \neq c$, a contradiction.

Hence $|S_i| = 1$ for all i, and every class is a singleton. Then $\|\mathcal{A}\|_\omega = \sum_{\{x\} \in \mathcal{A}} \omega^{P(x)}$, and this equals $|\omega|^m = \sum_{x \in L_d(m)} \omega^{P(x)}$ only if every word occurs. □

Corollary 7.8. *Let $p \in \mathbb{N}^d$ have at least three positive coordinates and let $m = \sum_c p_c$. Then*

$$M_{\mathrm{adm}}(p) < \sqrt{\frac{m^m}{\prod_c p_c^{p_c}}}.$$

In particular $M_{\mathrm{adm}}(n, \ldots, n) < d^{dn/2}$ for all $d \geq 3$ and $n \geq 1$.

Proof. Discarding the letters with $p_c = 0$, we may assume that all p_c are positive and $d \geq 3$. A family of N paths counted by $M_{\mathrm{adm}}(p)$ is an admissible system with one class, and for $\omega = p$ its norm is $N^2 \prod_c p_c^{p_c}$. If this were equal to $|\omega|^m = m^m$, the system would consist of all singletons by Theorem 7.7, whereas it has a single class and $L_d(m)$ has more than one word. □

So the plane is exceptional: for $d = 2$ the cube bound 2^n is attained by the C_n families of Section 3, and for $d \geq 3$ it is never attained. The exact value of $\mathrm{M}_{\mathrm{ev}}(n, \ldots, n)$ for $d \geq 3$ lies between the block family and the bound, and Table 6 shows that for $d = 3$ it is strictly between them already for $n = 2$.

7.5. **The other parity and free endpoints.**

Proposition 7.9. *Let $p \in \mathbb{N}^d$, $m = \sum_c p_c$, and let G_p be the grid graph of the box $\prod_c \{0, \dots, p_c\}$, with $|V| = \prod_c (p_c + 1)$ vertices and $|E| = \sum_c p_c \prod_{c' \neq c} (p_{c'} + 1)$ edges. Let $R(p)$ be the rank over $\mathbb{F}_2$ of the matrix $\big(B_m(x, y)\big)$ indexed by all paths from the origin to p. Then*

$$M_{\mathrm{oth}}(p) \le R(p) + [\, m \text{ even}\,] \le |E| - |V| + 2 + [\, m \text{ even}\,].$$

Proof. Let $\mathcal{F}$ be a family of N paths with $B_m(x, y) = m + 1$ for $x \neq y$. The corresponding principal submatrix is the identity if m is odd, and $J + I$ if m is even, where J is the all-ones matrix; $J + I$ has rank at least $N - 1$ over $\mathbb{F}_2$, since $(J + I)^2 = I$ when N is even and a principal submatrix of size $N - 1$ is of this kind when N is odd. So $N \le R(p) + [\, m \text{ even}\,]$. The full matrix is the Gram matrix of the indicator vectors of the sets $E(x)$ in $\mathbb{F}_2^E$, so $R(p)$ is at most the dimension of their span. For two paths x, y with the same ends, every vertex has even degree in $E(x) + E(y)$, so the sum of the two indicator vectors lies in the cycle space of the connected graph G_p, which has dimension $|E| - |V| + 1$. Hence the span has dimension at most $|E| - |V| + 2$. □

For the cube the bound is $(d - 1)n^d + O(n^{d-1})$, so as in the plane one parity is exponential and the other polynomial.[28]

Proposition 7.10. *Let $\mathcal{F} \subseteq L_d(m)$ be a set of words in which any two distinct members share a number of edges congruent to m modulo 2. Then*

$$|\mathcal{F}| \le \sqrt{\tbinom{m+d-1}{d-1}}\, d^{\,m/2},$$

and there is such a set with $d^{\lfloor m/2 \rfloor}$ elements.

Proof. By Corollary 7.4 with $\omega = (1, \dots, 1)$ we have $\sum_p N_p^2 \le d^m$, where p runs over the $\binom{m+d-1}{d-1}$ possible endpoints, and the Cauchy–Schwarz inequality gives $|\mathcal{F}|^2 = (\sum_p N_p)^2 \le \binom{m+d-1}{d-1} d^m$. For $m = 2r$ the set $\mathcal{B}_r$ of Theorem 7.5 has d^r elements. For $m = 2r + 1$ take $\{0z : \ z \in \mathcal{B}_r\}$: a common first letter adds one common edge to every pair. □

7.6. **Computed values and conjectures.** Table 6 lists exact values for 26 boxes, found by exhaustive search for a maximum clique in the graph of all paths of the box [8]. The column "block" is the largest lower bound given by Theorem 7.5(ii) over all partitions of the alphabet; it is defined only for even m, and a dash means that m is odd or that no partition satisfies the hypothesis. The column "bound" is the integer part of the bound of Corollary 7.4, and $R(p)$ and $|E| - |V| + 2$ are as in Proposition 7.9.

The cube of side 2 in $\mathbb{Z}^3$ has $\mathrm{M}_{\mathrm{ev}}(2, 2, 2) = 15$, strictly between the block value 12 and the bound 27; an extremal family is[29]

$$001221,\ 002121,\ 012201,\ 020211,\ 022011,\ 100122,\ 101022,\ 110202,$$
$$112002,\ 120012,\ 201120,\ 212010,\ 212100,\ 221010,\ 221100.$$

One more value follows from the table.

Proposition 7.11. $\mathrm{M}_{\mathrm{ev}}(1, 1, 1, 1, 1, 1) = 48$.

[28] For $d = 2$ and $p = (a, b)$ the bound is $ab + 1 + [\, a + b \text{ even}\,]$; the exact rank is computed in Section 4.

[29] The extremal families are not all of block type. For the box $(1, 1, 1, 1)$ the search returned the family 0132, 0312, 1023, 1203, 2301, 2310, 3201, 3210 of size 8, which is not a set of concatenations of blocks for any partition of the alphabet.

box p	m	paths	M_{adm}	block	bound	M_{oth}	$R(p)$	$\lvert E\rvert - \lvert V\rvert + 2$
$(1,1)$	2	2	2	2	2	1	0	2
$(2,2)$	4	6	4	4	4	2	2	5
$(3,3)$	6	20	8	8	8	3	6	10
$(4,2)$	6	15	5	3	6	4	8	9
$(5,1)$	6	6	2	–	3	3	4	6
$(6,2)$	8	28	6	4	9	7	12	13
$(1,1,1)$	3	6	2	–	5	3	4	6
$(2,1,1)$	4	12	5	4	8	3	8	10
$(2,2,1)$	5	30	5	–	13	7	14	17
$(3,1,1)$	5	20	5	–	10	6	12	14
$(2,2,2)$	6	90	15	12	27	7	22	29
$(3,2,1)$	6	60	12	–	20	7	24	24
$(4,1,1)$	6	30	7	6	13	6	16	18
$(3,3,1)$	7	140	12	–	33	13	28	34
$(3,2,2)$	7	210	15	–	43	16	38	41
$(4,2,1)$	7	105	12	–	28	11	31	31
$(5,1,1)$	7	42	7	–	16	9	20	22
$(4,2,2)$	8	420	30	24	64	16	50	53
$(3,3,2)$	8	560	37	32	75	16	54	58
$(4,3,1)$	8	280	26	–	49	13	44	44
$(1,1,1,1)$	4	24	8	8	16	3	12	18
$(2,1,1,1)$	5	60	8	–	27	10	28	30
$(2,2,1,1)$	6	180	24	24	54	10	42	50
$(3,1,1,1)$	6	120	14	–	41	10	36	42
$(1,1,1,1,1)$	5	120	8	–	55	15	36	50
$(2,1,1,1,1)$	6	360	31	24	108	15	76	82

TABLE 6. Exact values of $M_{\mathrm{adm}}(p)$ and $M_{\mathrm{oth}}(p)$ for small boxes in dimensions 2 to 5.

Proof. For every box p we have $M_{\mathrm{adm}}(p) \le \sum_c M_{\mathrm{adm}}(p - e_c)$, the sum being over the letters with $p_c > 0$. Indeed, sort a family counted by $M_{\mathrm{adm}}(p)$ by the last letter. If xc and yc are two of its members, then $P(x) = P(y)$, and (7.1) gives $B_{m-1}(x, y) = B_m(xc, yc) + 1 = m - 1$; so the words x with xc in the family form a family counted by $M_{\mathrm{adm}}(p - e_c)$. For $p = (1, 1, 1, 1, 1, 1)$ each of the six boxes $p - e_c$ is a copy of $(1, 1, 1, 1, 1)$, for which $M_{\mathrm{adm}} = \mathrm{M}_{\mathrm{od}} = 8$ by Table 6. So $\mathrm{M}_{\mathrm{ev}}(p) \le 48$. The block family with three pairs has $3! \cdot 2^3 = 48$ members. □

For words with free endpoints, exhaustive search gives the largest size of a set as in Proposition 7.10 for (d, m) with $d = 2$, $m \le 8$; $d = 3$, $m \le 5$; $d = 4$, $m \le 4$; and $d = 5$, $m \le 3$. In all these cases it is exactly $d^{\lfloor m/2 \rfloor}$ [8].

Conjecture 7.12. *Let $d = 2k$ be even. Then for all $n \ge 1$*

$$\mathrm{M}_{\mathrm{ev}}(n, \ldots, n) = \frac{(nk)!}{(n!)^k}\, 2^{nk},$$

the size of the block family with k pairs.

For $d = 2$ this is the theorem of Shankar [35]. For $d = 4$, $n = 1$ and $d = 6$, $n = 1$ it holds by Table 6 and Proposition 7.11. The first open case is the box $(2, 2, 2, 2)$, with 2520 paths and conjectured value 96.

Conjecture 7.13. *For all $d \geq 2$, $M_{\mathrm{adm}}(1, \ldots, 1) = 2^{\lfloor d/2 \rfloor} \lfloor d/2 \rfloor !$.*

The words of the box $(1, \ldots, 1)$ are the permutations of $[d]$. The conjecture holds for $d \leq 6$, and for even d the right-hand side is the block family. By the argument of Proposition 7.11, it would follow for all d from the inequality $\mathrm{M}_{\mathrm{od}}(1^{2k+1}) \leq \mathrm{M}_{\mathrm{ev}}(1^{2k})$, where 1^j is the box with j sides equal to 1; the reverse inequality holds by appending the missing letter.

Conjecture 7.14. *A set of words of length m over d letters in which any two distinct members share a number of edges congruent to m modulo 2 has at most $d^{\lfloor m/2 \rfloor}$ elements.*

Proposition 7.10 proves this up to the factor $\sqrt{\binom{m+d-1}{d-1}}\, d^{\,m/2-\lfloor m/2 \rfloor}$, which is polynomial in m for fixed d. For odd d the exact value of $\mathrm{M}_{\mathrm{ev}}(n, \ldots, n)$ is open even conjecturally: the value 15 for the cube $(2, 2, 2)$ matches no construction known to us.

8. Signed Eulerian polynomials are real-rooted

The second part of the paper treats conjectures and questions on permutations stated by Shankar and by his coauthors and colleagues. This section settles two of them, both about the zeros of Eulerian polynomials restricted to the alternating group.

8.1. **The polynomials and the two conjectures.** For a permutation $\pi = \pi_1 \cdots \pi_n$ of $[n]$ let $\mathrm{des}(\pi)$ be the number of $i \in [n-1]$ with $\pi_i > \pi_{i+1}$, and let $\mathrm{exc}(\pi)$ be the number of $i \in [n]$ with $\pi_i > i$. The Eulerian polynomial is

$$A_n(t) = \sum_{\pi \in S_n} t^{\mathrm{des}(\pi)} = \sum_{\pi \in S_n} t^{\mathrm{exc}(\pi)},$$

so that $A_1 = 1$, $A_2 = 1 + t$, $A_3 = 1 + 4t + t^2$, the degree of A_n is $n - 1$, and A_n is monic with $A_n(0) = 1$. The equality of the two sums is MacMahon's equidistribution of descents and excedances [30]. We write

$$A_n^{\pm}(t) = \sum_{\substack{\pi \in S_n \\ \operatorname{sgn} \pi = \pm 1}} t^{\mathrm{des}(\pi)}, \qquad E_n^{\pm}(t) = \sum_{\substack{\pi \in S_n \\ \operatorname{sgn} \pi = \pm 1}} t^{\mathrm{exc}(\pi)}.$$

Over the alternating group the two statistics are no longer equidistributed, and the four polynomials are different.[30]

Conjecture 8.1 (Fulman, Kim, Lee and Petersen [23, Conjecture 1.3])**.** *"The $\pm$-Eulerian polynomials have only real roots: $A_n^+(t)$ has real roots for $n \geq 1$ and $A_n^-(t)$ has real roots for $n \geq 2$."*

Conjecture 8.2 (Shankar [32, Open problems])**.** *"The polynomials $\sum_k P_{n,k} t^k$ and $\sum_k Q_{n,k} t^k$ are all real-rooted for $n \geq 5$." Here $P_{n,k}$ and $Q_{n,k}$ are the numbers of even and of odd permutations in S_n with k excedances, so the two polynomials are E_n^+ and E_n^-.*

[30] Fulman, Kim, Lee and Petersen [23] use the generating functions of des $+1$, which are $tA_n^{\pm}(t)$ in the present notation; the extra factor t does not affect the reality of the zeros. In [32] the coefficients of $A_n^+, A_n^-, E_n^+, E_n^-$ are denoted $B_{n,k}, C_{n,k}, P_{n,k}, Q_{n,k}$.

The authors of [23] verified Conjecture 8.1 for $n \le 100$ and reduced it to even n. For $n \ge 4$ with $n \equiv 0, 1 \pmod 4$ the same statement had been conjectured earlier by Dey and Sivasubramanian [16], in their study of gamma-positivity of these polynomials. Shankar [32] proved that for $n \ge 5$ the four coefficient sequences are ultra-log-concave, which confirms and strengthens two conjectures of Dey [12, Conjectures 32 and 33] and is the inequality that Newton's theorem would deduce from real zeros. As far as the author could determine, both conjectures were open; Alexandersson [1] has recently proved real-rootedness for several other families of Eulerian type, by methods that do not cover the alternating group. The results of this section are as follows.

Theorem 8.3. *Let $n \ge 50$, let $1 \le m \le \lceil n/2 \rceil$ and let c be a real number with $|c| \le 1$. Then the polynomial*

$$F(t) = A_n(t) + c\,(1-t)^{n-m} A_m(t)$$

is not identically zero, and all its zeros are real, simple and nonpositive.

Theorem 8.4 (The conjecture of Fulman, Kim, Lee and Petersen)**.** *For every $n \ge 2$ the polynomials $A_n^+(t)$ and $A_n^-(t)$ have only real zeros.*

Theorem 8.5 (Shankar's excedance conjecture)**.** *The polynomial $E_5^+(t) = 1 + 11t + 36t^2 + 11t^3 + t^4$ has no real zero, so Conjecture 8.2 fails at $n = 5$. For every $n \ge 6$ the polynomials $E_n^+(t)$ and $E_n^-(t)$ have only real zeros.*

Theorems 8.4 and 8.5 follow from Theorem 8.3 for $n \ge 50$ and from an exact finite computation for $n \le 49$; see Subsection 8.6. The case $c = 0$ of Theorem 8.3 is the classical theorem of Frobenius that A_n has real simple zeros [30], here for $n \ge 50$.

8.2. **Two signed identities.**

Lemma 8.6 (Mantaci [27])**.** *For every $n \ge 1$, $\displaystyle\sum_{\pi \in S_n} \operatorname{sgn}(\pi)\, t^{\operatorname{exc}(\pi)} = (1-t)^{n-1}$. Consequently*

$$2E_n^{\pm}(t) = A_n(t) \pm (1-t)^{n-1}. \tag{8.1}$$

Proof. The signed sum is the determinant of the $n \times n$ matrix M with entries $M_{ij} = t$ for $j > i$ and $M_{ij} = 1$ for $j \le i$, because the product $\prod_i M_{i\pi(i)}$ equals $t^{\operatorname{exc}(\pi)}$. For $i = 1, \dots, n-1$ subtract row $i+1$ from row i. The new row i has the single nonzero entry $t-1$, in column $i+1$, and the last row consists of ones. The only permutation contributing to the determinant is therefore the cycle $1 \mapsto 2 \mapsto \cdots \mapsto n \mapsto 1$, of sign $(-1)^{n-1}$, and $\det M = (-1)^{n-1}(t-1)^{n-1} = (1-t)^{n-1}$. Adding and subtracting A_n gives (8.1). □

Lemma 8.7 (Désarménien and Foata [11])**.** *For every $n \ge 1$, $\displaystyle\sum_{\pi \in S_n} \operatorname{sgn}(\pi)\, t^{\operatorname{des}(\pi)} = (1-t)^{\lfloor n/2 \rfloor} A_{\lceil n/2 \rceil}(t)$. Consequently*

$$2A_n^{\pm}(t) = A_n(t) \pm (1-t)^{\lfloor n/2 \rfloor} A_{\lceil n/2 \rceil}(t). \tag{8.2}$$

This identity was conjectured by Loday and proved in [11]; a proof by a sign-reversing involution is due to Wachs [41], and [23, Theorem 1.1] is an equivalent statement.[31] By (8.1) and (8.2), each of $2E_n^{\pm}$ and $2A_n^{\pm}$ is a polynomial F of the form considered in Theorem 8.3, with $c = \pm 1$ and with $m = 1$, respectively $m = \lceil n/2 \rceil$.

[31] For $n = 3$ the even permutations $123, 231, 312$ give $A_3^+ = 1 + 2t$ and the odd ones $132, 213, 321$ give $A_3^- = 2t + t^2$; the difference is $1 - t^2 = (1-t)A_2(t)$.

8.3. **Derivatives of the logistic function.** Let $L(s) = 1/(1+e^{-s})$ for real s. The proof of Theorem 8.3 rests on three descriptions of the derivatives $L^{(n)}$, $n \geq 1$.

Series. For $s > 0$ we have $L(s) = \sum_{k\geq 0}(-1)^k e^{-ks}$, and termwise differentiation gives

$$L^{(n)}(s) = (-1)^n \sum_{k\geq 1}(-1)^k k^n e^{-ks} \qquad (s>0). \tag{8.3}$$

Eulerian polynomials. By the classical identity $\sum_{k\geq 1} k^n x^k = xA_n(x)/(1-x)^{n+1}$ for $|x|<1$ [30], applied with $x = -e^{-s}$, the right side of (8.3) equals $(-1)^n tA_n(t)/(1-t)^{n+1}$ with $t = -e^{-s}$. Both sides are real-analytic functions of $s \in \mathbb{R}$, hence

$$L^{(n)}(s) = (-1)^n \frac{t\,A_n(t)}{(1-t)^{n+1}}, \qquad t = -e^{-s}, \quad s \in \mathbb{R}. \tag{8.4}$$

Partial fractions. The function L is meromorphic with simple poles of residue 1 at the points $\pm(2j+1)\pi i$, $j \geq 0$, and $L(s) = \frac{1}{2} + \sum_{j\geq 0}\big[(s-(2j+1)\pi i)^{-1} + (s+(2j+1)\pi i)^{-1}\big]$.[32] Differentiating $n \geq 1$ times and writing $s + (2j+1)\pi i = \rho_j e^{i\varphi_j}$ with $\rho_j > 0$ gives, for real s,

$$L^{(n)}(s) = 2(-1)^n\, n! \sum_{j\geq 0} \rho_j^{-n-1} \cos\big((n+1)\varphi_j\big). \tag{8.5}$$

Two consequences will be used. Since $L(s) + L(-s) = 1$,

$$L^{(k)}(-s) = (-1)^{k+1} L^{(k)}(s) \qquad (k \geq 1). \tag{8.6}$$

Secondly, let F be as in Theorem 8.3 and put

$$\gamma = (-1)^{n-m} c, \qquad g(s) = L^{(n)}(s) + \gamma\, L^{(m)}(s). \tag{8.7}$$

Then (8.4), applied to n and to m, yields

$$\frac{t\,F(t)}{(1-t)^{n+1}} = (-1)^n\, g(s), \qquad t = -e^{-s}, \tag{8.8}$$

so the negative zeros of F correspond bijectively to the real zeros of g under $t = -e^{-s}$.

8.4. **Test points and the dominance lemma.** Fix $n \geq 50$ and put $K = \lfloor \sqrt{n}/2 \rfloor$, so that $K \geq 3$ and $4K^2 \leq n$. Define points $s_2, \ldots, s_{n-1}$ by

$$s_i = \frac{n}{i} \quad (2 \leq i \leq K), \qquad s_i = \pi \cot \frac{i\pi}{n+1} \quad \Big(K < i \leq \frac{n+1}{2}\Big), \qquad s_{n+1-i} = -s_i.$$

Since $\cot x < 1/x$ on $(0, \pi/2]$, we have $s_{K+1} < \frac{n+1}{K+1} < \frac{n}{K} = s_K$, and therefore

$$s_2 > s_3 > \cdots > s_{n-1};$$

for odd n the middle point is $s_{(n+1)/2} = 0$. The first family of points sits at the maxima of the summands of (8.3), the second where the leading term of (8.5) is extremal.

Lemma 8.8 (Dominance)**.** *Let $n \geq 50$ and $1 \leq m \leq \lceil n/2 \rceil$. For every i with $2 \leq i \leq n-1$,*

$$(-1)^{n+i}\, L^{(n)}(s_i) > \big|L^{(m)}(s_i)\big|.$$

Proof. By (8.6) the left side at $s_{n+1-i} = -s_i$ equals

$$(-1)^{n+(n+1-i)}(-1)^{n+1} L^{(n)}(s_i) = (-1)^{n+i} L^{(n)}(s_i),$$

and the right side is unchanged. It therefore suffices to treat $i \leq (n+1)/2$. There are two regimes.

[32] This is the expansion $\tanh(s/2) = \sum_{j\geq 0} 4s/(s^2 + (2j+1)^2\pi^2)$ together with $L(s) = \frac{1}{2} + \frac{1}{2}\tanh(s/2)$.

Regime A: $s = n/k$ with $2 \le k \le K$. Here $n \ge 4k^2$. Put $a_j = j^n e^{-jn/k}$ for $j \ge 1$. By (8.3),

$$(-1)^{n+k} L^{(n)}(s) = a_k + \sum_{j \ne k} (-1)^{j+k} a_j \;\ge\; a_k\,(1 - S), \qquad S = \sum_{j \ne k} \frac{a_j}{a_k},$$

and $a_j/a_k = \exp(n\psi(j/k))$ with $\psi(x) = \log x - x + 1$. We use the inequalities $\log(1+x) \le x - \frac{x^2}{2(1+x)}$ for $x \ge 0$ and $\log(1-x) \le -x - \frac{x^2}{2}$ for $0 \le x < 1$.[33] Three ranges of j occur.

(i) $j = k + d$ with $1 \le d \le k$. Then $x = d/k \le 1$ and $n\psi \le -\frac{nd^2}{2k^2(1+d/k)} \le -\frac{nd^2}{4k^2} \le -d^2$.

(ii) $j = k+d$ with $d > k$. For $x = d/k \ge 1$ we have $\frac{x^2}{2(1+x)} \ge \frac{x}{4}$, so $n\psi \le -\frac{nd}{4k} \le -kd \le -2d$; moreover $d \ge k + 1 \ge 3$.

(iii) $j = k - d$ with $1 \le d \le k - 1$. Then $n\psi \le -\frac{nd^2}{2k^2} \le -2d^2$.

Therefore

$$S \le \sum_{d \ge 1} e^{-d^2} + \frac{e^{-6}}{1 - e^{-2}} + \sum_{d \ge 1} e^{-2d^2} < 0.38632 + 0.00287 + 0.13568 < 0.5249,$$

and by (iii) with $d = k - 1$, $a_1/a_k \le e^{-2} < 0.1354$. On the other hand, (8.3) for m gives

$$\begin{aligned} \bigl|L^{(m)}(s)\bigr| &\le \sum_{j \ge 1} j^m e^{-jn/k} = \sum_{j \ge 1} j^{m-n} a_j \le a_1 + 2^{m-n} \sum_{j \ge 2} a_j \\ &\le a_k\bigl(0.1354 + 2^{-(n-m)} \cdot 1.5249\bigr) \le 0.3261\, a_k, \end{aligned}$$

because $n - m \ge \lfloor n/2 \rfloor \ge 3$. Since $0.3261 < 0.4751 < 1 - S$, the lemma holds at $s = s_k$, with the sign $(-1)^{n+k}$.

Regime B: $s = \pi \cot\theta$ with $\theta = i\pi/(n+1)$ and $K < i \le (n+1)/2$. Then $s + \pi i = \frac{\pi}{\sin\theta} e^{i\theta}$, so in (8.5) we have $\varphi_0 = \theta$ and $\cos((n+1)\varphi_0) = \cos(i\pi) = (-1)^i$. Put $X = \pi^2 + s^2 = \rho_0^2$; then $\rho_j^2 = X + 4j(j+1)\pi^2$. Bernoulli's inequality gives $(1+y)^{-(n+1)/2} \le \frac{2}{(n+1)y}$ for $y > 0$, and $(1+y)^{-(m+1)/2} \le \frac{1}{y}$. With $y = 4j(j+1)\pi^2/X$ and $\sum_{j \ge 1} \frac{1}{j(j+1)} = 1$ we obtain from (8.5)

$$(-1)^{n+i} L^{(n)}(s) \ge 2\, n!\, X^{-\frac{n+1}{2}} \Bigl(1 - \frac{X}{2\pi^2(n+1)}\Bigr), \qquad \bigl|L^{(m)}(s)\bigr| \le 2\, m!\, X^{-\frac{m+1}{2}} \Bigl(1 + \frac{X}{4\pi^2}\Bigr). \tag{8.9}$$

Hence it suffices to prove

$$\frac{n!}{m!}\Bigl(1 - \frac{X}{2\pi^2(n+1)}\Bigr) > X^{\frac{n-m}{2}} \Bigl(1 + \frac{X}{4\pi^2}\Bigr). \tag{8.10}$$

The left side decreases and the right side increases with X, so X may be replaced by an upper bound. From $\cot\theta < 1/\theta$ and $i > \sqrt{n}/2$ we get $s < \frac{n+1}{i} < \frac{2(n+1)}{\sqrt{n}}$, hence

$$X < \pi^2 + 4n + 8 + \frac{4}{n} < Y := 4n + 18 \qquad (n \ge 31).$$

For $n \ge 50$ we have $\frac{Y}{2\pi^2(n+1)} \le \frac{218}{2\pi^2 \cdot 51} < 0.22$ and $1 + \frac{Y}{4\pi^2} < 1.457 + 0.1014\, n \le 0.131\, n$.[34] Since $0.131/0.78 < 1/5$, inequality (8.10) follows from

$$\frac{n!}{m!} \;\ge\; \frac{n}{5}\, Y^{\frac{n-m}{2}}. \tag{8.11}$$

[33] The first holds at $x = 0$, and the derivative of its right side, $\frac{x^2+2x+2}{2(1+x)^2}$, is at least $\frac{1}{1+x}$. The second is the series of $\log(1-x)$.

[34] The quotient $Y/(n+1) = 4 + 14/(n+1)$ decreases with n; $4/(4\pi^2) < 0.1014$, $1 + 18/(4\pi^2) < 1.457$, and $1.457 \le 0.0296\, n$ for $n \ge 50$.

The function $m \mapsto \log(n!/m!) - \frac{n-m}{2}\log Y$ is concave on the integers, so (8.11) needs to be checked only at $m = 1$ and at $m = M := \lceil n/2 \rceil$.

For $m = 1$: Stirling's lower bound gives $n! \geq \sqrt{2\pi n}\,(n/e)^n = \sqrt{2\pi n}\,\frac{n}{e}\big(\frac{n^2}{e^2}\big)^{\frac{n-1}{2}}$, and $n^2 \geq e^2 Y$ for $n \geq 34$. Hence $n!\,Y^{-(n-1)/2} \geq \sqrt{2\pi n}\,n/e > n/5$.

For $m = M$: let $h = n - M = \lfloor n/2 \rfloor$. Each of the h factors of $n!/M!$ exceeds $n/2$, so $n!/M! > (n/2)^h$. Further $\frac{n^2}{4Y} \geq \frac{2500}{872} > 2.866$ for $n \geq 50$, and $h \geq (n-1)/2$, so

$$\frac{n!}{M!}\,Y^{-h/2} > 2.866^{\,h/2} \geq e^{0.263\,(n-1)} > \frac{n}{5},$$

the last step because $e^x > x^2/2$ and $(0.263 \cdot 49)^2/2 > 83$.[35] This proves (8.11), hence the lemma. □

8.5. **Proof of Theorem 8.3.** Let g and γ be as in (8.7). Since $|\gamma| \leq 1$, Lemma 8.8 gives

$$(-1)^{n+i} g(s_i) \;\geq\; (-1)^{n+i} L^{(n)}(s_i) - \big|L^{(m)}(s_i)\big| \;> 0 \qquad (2 \leq i \leq n-1).$$

In particular g is not identically zero, so $F \neq 0$ by (8.8), and g changes sign on each of the $n-3$ intervals (s_{i+1}, s_i), $2 \leq i \leq n-2$. This gives $n-3$ distinct negative zeros of F. Two more zeros come from the ends of the real line.

The end $s \to +\infty$. By (8.3), $L^{(k)}(s) \sim (-1)^{k+1} e^{-s}$, so $g(s) \sim (-1)^{n+1}(1+c)\,e^{-s}$. The sign of $g(s_2)$ is $(-1)^n$. If $c > -1$, then g has a further zero in (s_2, ∞). If $c = -1$, then $F(0) = A_n(0) + cA_m(0) = 0$, so $t = 0$ is a zero of F.

The end $s \to -\infty$. Here $L(s) = e^s/(1+e^s)$ gives $L^{(k)}(s) \sim e^s$, so $g(s) \sim (1+\gamma)\,e^s$. The sign of $g(s_{n-1})$ is $(-1)^{2n-1} = -1$. If $\gamma > -1$, then g has a further zero in $(-\infty, s_{n-1})$. If $\gamma = -1$, then the coefficient of t^{n-1} in F, which is $1 + (-1)^{n-m}c = 1 + \gamma$, vanishes, and $\deg F \leq n-2$.

Thus F has at least $n-1$ distinct zeros in $(-\infty, 0]$ when $\gamma > -1$, and at least $n-2$ when $\gamma = -1$. In both cases this number is an upper bound for the degree of F. Hence F has exactly this degree, and all its zeros are real, simple and nonpositive. □

8.6. **The two conjectures.**

Proof of Theorems 8.4 and 8.5. For $n \geq 50$, formulas (8.2) and (8.1) present $2A_n^{\pm}$ and $2E_n^{\pm}$ as the polynomial F of Theorem 8.3 with $(m, c) = (\lceil n/2 \rceil, \pm 1)$ and $(m, c) = (1, \pm 1)$, so these four polynomials have only real zeros.

For $n = 5$ we have $A_5 = 1 + 26t + 66t^2 + 26t^3 + t^4$ and $(1-t)^4 = 1 - 4t + 6t^2 - 4t^3 + t^4$, hence $E_5^+ = 1 + 11t + 36t^2 + 11t^3 + t^4$ by (8.1). With $u = t + 1/t$,

$$t^{-2} E_5^+(t) = u^2 + 11u + 34,$$

a quadratic of discriminant $121 - 136 < 0$. A real zero $t \neq 0$ of E_5^+ would give a real zero u of this quadratic, so E_5^+ has no real zero.

The remaining cases are finite. For $2 \leq n \leq 60$ the number of real zeros of each of $A_n^+, A_n^-, E_n^+, E_n^-$, counted with multiplicity, was determined by Sturm sequences in exact integer arithmetic. The count equals the degree for $A_n^{\pm}$ for all $2 \leq n \leq 60$ and for $E_n^{\pm}$ for all $6 \leq n \leq 60$. The computation is in the verification archive [8]. Only the range $n \leq 49$ is needed here. □

Remark 8.9. The same computation shows that among the polynomials $E_n^{\pm}$ with $n \geq 2$ exactly four have a nonreal zero: $E_3^+ = 1 + t + t^2$, $E_4^+ = 1 + 4t + 7t^2$, $E_4^- = t(7 + 4t + t^2)$ and E_5^+. The polynomial $E_5^- = 15t(1+t)^2$ has a double zero, so the simplicity of the zeros in Theorem 8.3 does not extend to small n. The threshold $n \geq 6$ in Theorem 8.5 is therefore the correct

[35] $\log 2.866 > 1.052$, and $\frac{1}{2}(0.263(n-1))^2 \geq n/5$ for all $n \geq 50$ since the left side grows quadratically.

one, and Shankar's theorem that the coefficients of E_5^+ are ultra-log-concave [32] is not a consequence of real zeros at $n = 5$.

Remark 8.10. Theorem 8.3 allows any $m \le \lceil n/2 \rceil$ and any real $c \in [-1, 1]$, which is more than the two conjectures need. For instance every polynomial $A_n^+ + \lambda A_n^-$ with $\lambda \ge 0$, and every $E_n^+ + \lambda E_n^-$ with $\lambda \ge 0$, has only real zeros when $n \ge 50$: it is a positive multiple of $A_n + c\,(1-t)^{n-m} A_m$ with $c = (1-\lambda)/(1+\lambda) \in (-1, 1]$. The analogous conjecture of [23] for the hyperoctahedral group is not treated here. Figure 14 shows the zeros for small n in the coordinate $s = -\log(-t)$ of the proof.

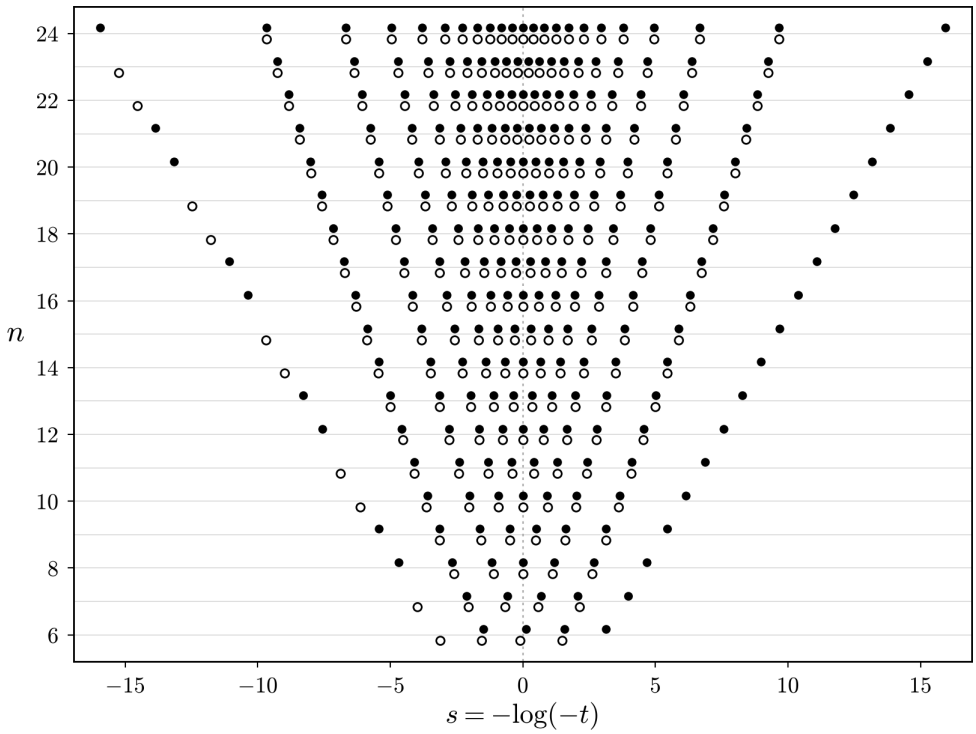


FIGURE 14. The negative zeros t of A_n^+ (filled) and A_n^- (open) for $6 \le n \le 24$, drawn in the coordinate $s = -\log(-t)$. A zero at $t = 0$ is not shown.

9. Canon permutations

A *canon permutation* is an arrangement of the multiset $\{1^k, \ldots, n^k\}$ in which, for every $j \le k$, the jth copies of the letters appear in one and the same relative order. Shankar [33] classified the canon permutations with $n = 3$ that avoid a set of patterns from S_3, and closed his paper with four conjectures and two questions. This section proves two of the conjectures, proves one half of a third and disproves its other half, and gives partial answers to the two questions. One conjecture remains open.

9.1. **Lattice words and the statements.** A word on $[n]$ is a *k-regular lattice word* if each letter occurs k times and every prefix contains at least as many letters i as letters $i+1$, for each $i < n$. The set of these words is $\mathcal{L}_n^k$. A word *contains* a pattern $\tau \in S_r$ if it has a subsequence order-isomorphic to τ, and $\mathcal{L}_n^k(\Omega)$ is the set of words of $\mathcal{L}_n^k$ that contain no pattern of Ω. The number of descents of a word $w = w_1 \cdots w_N$ is $\mathrm{des}(w) = \#\{i : w_i > w_{i+1}\}$.

By [33, Theorem 2.4], the canon permutations with underlying permutation $\sigma \in S_3$ that avoid $\Lambda \subseteq S_3$ are in bijection with $\mathcal{L}_3^k(\sigma^{-1}\Lambda)$, so the enumeration reduces to lattice words; and by [33, Theorem 3.4] only six counting sequences occur, namely 0, 1, the Catalan numbers C_k, the numbers $|\mathcal{L}_3^k|$ of standard Young tableaux of shape (k,k,k), and the two sequences

$$B_k = |\mathcal{L}_3^k(231)|, \qquad Q_k = |\mathcal{L}_3^k(321)|, \qquad B_0 = Q_0 = 1.$$

By [33, Theorem 3.2], a word of $\mathcal{L}_3^k$ avoids 231 if and only if its last 1 precedes its first 3. The descent refinements and the generating function of (Q_k) are

$$\mathsf{B}_k(x) = \sum_{w \in \mathcal{L}_3^k(231)} x^{\mathrm{des}(w)}, \qquad \mathsf{Q}_k(x) = \sum_{w \in \mathcal{L}_3^k(321)} x^{\mathrm{des}(w)}, \qquad Q(z) = \sum_{k \geq 0} Q_k z^k.$$

For instance $\mathsf{B}_3 = 1 + 6x + 10x^2 + 4x^3$ and $\mathsf{Q}_3 = 1 + 10x + 16x^2 + 6x^3$. The six items of [33, Section 9] read as follows.

Conjecture 9.1 ([33, Conjecture 9.1])**.** *"For every $k \geq 1$, the polynomial $\mathsf{B}_k(x)$ is real-rooted."*

Conjecture 9.2 ([33, Conjecture 9.2])**.** *"The D-finite series $Q(z)$ is transcendental over $\mathbb{Q}(z)$."*

Conjecture 9.3 ([33, Conjecture 9.3])**.** *"For every $k \geq 1$, the coefficients of $\mathsf{Q}_k(x)$ are log-concave and therefore unimodal."*

Conjecture 9.4 ([33, Conjecture 9.5])**.** *"Both $(B_k)_{k\geq 0}$ and $(Q_k)_{k\geq 0}$ are Stieltjes moment sequences; equivalently, their infinite Hankel matrices are totally nonnegative."*

Question 9.5 ([33, Question 9.4])**.** *"Is there a finite-class generating tree, or a model by a fixed number of classical path or tree objects with local compatibility conditions, whose natural parameters recover (b, c, r) in Theorem 7.1?"*

Question 9.6 ([33, Question 9.6])**.** *"Which parts of the relabeling and class-reduction mechanism survive for fixed alphabet size $n = 4$? A first target might be to classify those forbidden sets whose lattice-word components reduce to forced blocks and binary or ternary Catalan objects, before attempting the complete classification of the 2^{24} forbidden subsets of $\mathfrak{S}_4$."*

The results: Conjecture 9.1 holds (Theorem 9.10); Conjecture 9.2 holds (Theorem 9.14); Conjecture 9.4 holds for (B_k) and fails for (Q_k) (Theorems 9.15 and 9.16); Conjecture 9.3 is open and is confirmed for $k \leq 110$ (Subsection 9.5). Partial answers to the two questions are in Subsections 9.6 and 9.7. Throughout, $\binom{n}{j} = 0$ unless $0 \leq j \leq n$.

9.2. **Real zeros of B_k.** A *ballot word* is a word in the letters U, D in which every prefix has at least as many U as D. For $u \geq d \geq 0$ with $u \geq 1$ let $\mathcal{U}_{u,d}(x)$, respectively $\mathcal{D}_{u,d}(x)$, be the generating polynomial, by the number of factors DU, of the ballot words with u letters U and d letters D whose last letter is U, respectively D.

Lemma 9.7. *For $u \geq d \geq 0$, $u \geq 1$ and all $j \geq 0$,*

$$[x^j]\,\mathcal{U}_{u,d} = \binom{u-1}{j}\binom{d}{j} - \binom{u}{j}\binom{d-1}{j}, \qquad [x^j]\,\mathcal{D}_{u,d} = \binom{u-1}{j}\binom{d-1}{j} - \binom{u-1}{j-1}\binom{d-1}{j+1}.$$

Proof. Write $U_j(u,d)$ and $D_j(u,d)$ for the two right sides. Deleting the last letter of a ballot word gives

$$\mathcal{D}_{u,d} = \mathcal{U}_{u,d-1} + \mathcal{D}_{u,d-1} \quad (u \geq d \geq 1), \qquad \mathcal{U}_{u,d} = \mathcal{U}_{u-1,d} + x\,\mathcal{D}_{u-1,d} \quad (u-1 \geq d \geq 1),$$

together with $\mathcal{U}_{u,0} = 1$, $\mathcal{D}_{u,0} = 0$ and $\mathcal{U}_{u,u} = 0$. These relations determine all the polynomials by induction on $u + d$, so it suffices to check them for U_j, D_j. The boundary values are

immediate: $U_j(u,0) = \binom{u-1}{j}\binom{0}{j}$ is 1 for $j=0$ and 0 otherwise, $D_j(u,0)=0$, and $U_j(u,u) = \binom{u-1}{j}\binom{u}{j} - \binom{u}{j}\binom{u-1}{j} = 0$. The first relation with $d=1$ reads $D_j(u,1) = U_j(u,0)$, which holds. For $d \geq 2$, Pascal's rule applied to $\binom{d-1}{j}$ and $\binom{d-1}{j+1}$ gives

$$D_j(u,d) - D_j(u,d-1) = \tbinom{u-1}{j}\tbinom{d-2}{j-1} - \tbinom{u-1}{j-1}\tbinom{d-2}{j},$$

and the same expression results from $U_j(u,d-1)$ after writing $\binom{d-1}{j} = \binom{d-2}{j} + \binom{d-2}{j-1}$ and $\binom{u}{j} = \binom{u-1}{j} + \binom{u-1}{j-1}$. For the second relation, Pascal's rule in the upper index u gives

$$U_j(u,d) - U_j(u-1,d) = \tbinom{u-2}{j-1}\tbinom{d}{j} - \tbinom{u-1}{j-1}\tbinom{d-1}{j},$$

while $D_{j-1}(u-1,d) = \binom{u-2}{j-1}\binom{d-1}{j-1} - \binom{u-2}{j-2}\binom{d-1}{j}$. The difference of the two is $\binom{u-2}{j-1}\big[\binom{d}{j} - \binom{d-1}{j-1}\big] - \binom{d-1}{j}\big[\binom{u-1}{j-1} - \binom{u-2}{j-2}\big] = \binom{u-2}{j-1}\binom{d-1}{j} - \binom{d-1}{j}\binom{u-2}{j-1} = 0$. □

Proposition 9.8. *For $k \geq 2$ and $0 \leq m \leq k$,*

$$[x^m]\,\mathsf{B}_k(x) = \binom{2k}{m}\binom{k}{m}\frac{q_k(m)}{2k(2k-1)(m+1)(m+2)}, \tag{9.1}$$

where $q_k(m) = (k^2-k+4)\,m^2 + (3k^2-13k+2)\,m + 8k^2 - 4k$.

Proof. Let $w \in \mathcal{L}_3^k(231)$ and cut $w = w'w''$ after its last 1. The word w' uses the letters $1,2$, contains k ones and some number $b \leq k-1$ of twos, ends in 1, and is a ballot word for $U=1$, $D=2$. The word w'' uses the letters $2,3$, with $k-b$ twos and k threes, and the lattice condition says that b plus the number of twos is at least the number of threes in each of its prefixes. Read backwards with $U=3$ and $D=2$, the word w'' is a ballot word with k letters U and $k-b$ letters D: a suffix of w'' with p_3 threes and p_2 twos missing satisfies $k-p_3 \geq k-b-p_2$. Every such pair (w',w'') arises. The junction of w' and w'' is an ascent, the descents of w' are its factors $21 = DU$, and the descents of w'' are its factors 32, which become factors DU of the reversed word. Hence

$$\mathsf{B}_k(x) = \sum_{b=0}^{k} \mathcal{U}_{k,b}(x)\,\big(\mathcal{U}_{k,k-b}(x) + \mathcal{D}_{k,k-b}(x)\big), \tag{9.2}$$

where the term $b=k$ vanishes because $\mathcal{U}_{k,k}=0$. Insert Lemma 9.7. The coefficient of $x^i x^j$ in the summand is a sum of eight products, and in each of them the dependence on b has the form $\binom{b+\alpha}{i}\binom{k-b+\beta}{j'}$ with $\alpha,\beta \in \{0,-1\}$. By the Chu–Vandermonde identity $\sum_b \binom{b+\alpha}{i}\binom{k-b+\beta}{j'} = \binom{k+\alpha+\beta+1}{i+j'+1}$, and the remaining sum over $i+j=m$ is an ordinary Vandermonde convolution. The result is

$$\begin{aligned}[x^m]\mathsf{B}_k = {} & \tbinom{2k-2}{m}\tbinom{k+1}{m+1} - 2\tbinom{2k-1}{m}\tbinom{k}{m+1} + \tbinom{2k-2}{m}\tbinom{k}{m+1} - \tbinom{2k-2}{m-1}\tbinom{k}{m+2} \\ & + \tbinom{2k}{m}\tbinom{k-1}{m+1} - \tbinom{2k-1}{m}\tbinom{k-1}{m+1} + \tbinom{2k-1}{m-1}\tbinom{k-1}{m+2}.\end{aligned}$$

Divide by $\binom{2k}{m}\binom{k}{m}$. With $a = 2k-m$ and $e = k-m$ the seven quotients are

$$\tfrac{a(a-1)(k+1)}{2k(2k-1)(m+1)},\ \tfrac{a\,e}{k(m+1)},\ \tfrac{a(a-1)e}{2k(2k-1)(m+1)},\ \tfrac{m\,a\,e(e-1)}{2k(2k-1)(m+1)(m+2)},\ \tfrac{e(e-1)}{k(m+1)},\ \tfrac{a\,e(e-1)}{2k^2(m+1)},\ \tfrac{m\,e(e-1)(e-2)}{2k^2(m+1)(m+2)},$$

taken with the signs $+,-,+,-,+,-,+$. Their signed sum, brought to the denominator $2k^2(2k-1)(m+1)(m+2)$, has numerator $k\,q_k(m)$; this is an identity between polynomials of degree four in m with coefficients in $\mathbb{Z}[k]$.[36] This is (9.1). □

[36] At $m=0$ the seven quotients are $k+1$, $2k$, k, 0, $k-1$, $k-1$, 0, with signed sum $1 = q_k(0)/(4k(2k-1))$, in agreement with $\mathsf{B}_k(0)=1$.

For $0 \le n \le k$ put

$$p_n(x) = \sum_m \binom{n+k}{m}\binom{n+2}{n-m} x^m = (x-1)^n\, P_n^{(k,2)}\Big(\frac{x+1}{x-1}\Big), \tag{9.3}$$

where $P_n^{(\alpha,\beta)}(y) = \sum_s \binom{n+\alpha}{n-s}\binom{n+\beta}{s}\big(\frac{y-1}{2}\big)^s\big(\frac{y+1}{2}\big)^{n-s}$ is the Jacobi polynomial [40, (4.3.2)]; the second equality in (9.3) is the substitution $\frac{y-1}{2} = \frac{1}{x-1}$, $\frac{y+1}{2} = \frac{x}{x-1}$.

Proposition 9.9. *For every $k \ge 2$,*

$$\mathsf{B}_k(x) = c_0\, p_k(x) + c_1\,(x-1)\, p_{k-1}(x) + c_2\,(x-1)^2\, p_{k-2}(x), \tag{9.4}$$

with

$$c_0 = \frac{k^2+k+2}{(3k+1)(3k+2)}, \qquad c_1 = \frac{2(k-2)(k+2)}{3k(3k+2)}, \qquad c_2 = \frac{k^2-k+4}{3k(3k+1)}.$$

Proof. The coefficients of x^m in p_k, $(x-1)p_{k-1}$ and $(x-1)^2 p_{k-2}$ are

$$\tbinom{2k}{m}\tbinom{k+2}{m+2}, \qquad \tbinom{2k-1}{m-1}\tbinom{k+1}{m+1} - \tbinom{2k-1}{m}\tbinom{k+1}{m+2},$$
$$\tbinom{2k-2}{m-2}\tbinom{k}{m} - 2\tbinom{2k-2}{m-1}\tbinom{k}{m+1} + \tbinom{2k-2}{m}\tbinom{k}{m+2}.$$

Multiply them by $2k(2k-1)(m+1)(m+2)/\big[\binom{2k}{m}\binom{k}{m}\big]$. The three results are the polynomials

$$2k(2k-1)(k+1)(k+2), \qquad (2k-1)(k+1)\big[(3k+2)m - 2k^2\big],$$
$$3k(3k+1)m^2 - (3k+1)(4k^2-3k+2)m + 2k^2(k-1)(2k-1)$$

in m.[37] By (9.1), identity (9.4) is therefore equivalent to

$$q_k(m) = 2k(2k-1)(k+1)(k+2)\,c_0 + (2k-1)(k+1)\big[(3k+2)m - 2k^2\big]c_1 + \big[3k(3k+1)m^2 - (3k+1)(4k^2-3k+2)m + 2k^2(k-1)(2k-1)\big]c_2.$$

Comparing the coefficients of m^2, m and 1 gives a triangular linear system: the first equation is $3k(3k+1)c_2 = k^2-k+4$, the second then gives c_1, and the third c_0. The solution is the stated triple. □

Theorem 9.10. *For every $k \ge 1$ all zeros of $\mathsf{B}_k(x)$ are real and negative, so Conjecture 9.1 holds. For $k \ge 3$ the zeros are simple, and they strictly interlace the $k-1$ zeros of $p_{k-1}(x) = \sum_m \binom{2k-1}{m}\binom{k+1}{m+2}x^m$. In particular the coefficients of B_k form a log-concave sequence.*

Proof. We have $\mathsf{B}_1 = 1$ and $\mathsf{B}_2 = (1+x)^2$. Let $k \ge 3$, write $P_n = P_n^{(k,2)}$, and put $y = (x+1)/(x-1)$. By (9.3) and (9.4),

$$(x-1)^{-k}\,\mathsf{B}_k(x) = g_k(y) := c_0 P_k(y) + c_1 P_{k-1}(y) + c_2 P_{k-2}(y).$$

The polynomials P_n are orthogonal on $(-1,1)$ for the weight $(1-y)^k(1+y)^2$. Let $1 > \xi_1 > \cdots > \xi_{k-1} > -1$ be the zeros of P_{k-1}. The three-term recurrence [40, (4.5.1)] with $n = \alpha = k$ and $\beta = 2$, evaluated at a zero ξ_i of P_{k-1}, reads $2k(2k+2)\,3k\,P_k(\xi_i) = -2(2k-1)(k+1)(3k+2)\,P_{k-2}(\xi_i)$, that is

$$P_k(\xi_i) = -\kappa_k\, P_{k-2}(\xi_i), \qquad \kappa_k = \frac{(2k-1)(3k+2)}{6k^2} > 0.$$

Therefore

$$g_k(\xi_i) = \Big(c_0 - \frac{c_2}{\kappa_k}\Big)P_k(\xi_i), \qquad c_0\kappa_k - c_2 = \frac{(k^2+k+2)(2k-1) - 2k(k^2-k+4)}{6k^2(3k+1)} = \frac{k-2}{6k^2} > 0.$$

[37] For the third one the quotient is $m(m-1)(m+1)(m+2) - 2m(2k-m)(k-m)(m+2) + (2k-m)(2k-m-1)(k-m)(k-m-1)$, in which the terms of degree four and three in m cancel.

The zeros of P_k interlace those of P_{k-1}, and $P_k(1)=\binom{2k}{k}>0$; hence the sign of $g_k(\xi_i)$ is that of $P_k(\xi_i)$, which is $(-1)^i$. Next, $g_k(1)=c_0\binom{2k}{k}+c_1\binom{2k-1}{k-1}+c_2\binom{2k-2}{k-2}>0$, since $c_0,c_2>0$ and $c_1\ge 0$. Finally $y\to -1$ corresponds to $x\to 0$, so $g_k(-1)=(-1)^k\mathsf{B}_k(0)=(-1)^k$. Thus g_k changes sign on each of the k intervals $(\xi_1,1),(\xi_2,\xi_1),\dots,(\xi_{k-1},\xi_{k-2}),(-1,\xi_{k-1})$. Its degree is at most k, so it has exactly k zeros, all simple and all in $(-1,1)$. The map $y\mapsto x=(y+1)/(y-1)$ sends $(-1,1)$ onto $(-\infty,0)$, the zeros of g_k to zeros of B_k, and the ξ_i to the zeros of p_{k-1}. This gives $k=\deg\mathsf{B}_k$ distinct negative zeros with the stated interlacing. Log-concavity follows from Newton's inequalities. □

Remark 9.11. By (9.1), the polynomial B_k is obtained, up to a constant, from the real-rooted polynomial $\sum_m\binom{2k}{m}\binom{k+2}{m+2}x^m$ by the diagonal operator that multiplies x^m by $q_k(m)$. For $k\ge 2$ the quadratic q_k has negative discriminant, so it is not a product of two factors $m+\theta$ with $\theta>0$ real, and the classical theorem that such factors preserve real zeros does not apply.[38] The expansion (9.4) takes its place: B_k is, in the variable y, a combination of three consecutive orthogonal polynomials whose outer coefficients satisfy $c_0\kappa_k>c_2$.

9.3. **Asymptotics of Q_k and the transcendence of $Q(z)$.** The starting point is Shankar's multisum [33, Theorem 7.1]: for $k\ge 1$,

$$Q_k=C_k+\sum_{(b,c,r)\in I_k}T(b,c,r),\qquad I_k=\{(b,c,r):1\le b\le k-1,\ 0\le c\le b\le r\le k-1\},\tag{9.5}$$

where C_k is the Catalan number and

$$T(b,c,r)=\frac{b-c+1}{k-c+1}\binom{n_1}{k-b}\cdot\frac{r-b+2}{r+1}\binom{n_2}{b-1}\cdot\binom{n_3}{c},$$
$$n_1=2k-b-c,\qquad n_2=r+b-1,\qquad n_3=k-1-r+c.$$

Here $n_1+n_2+n_3=3k-2$.[39] We use the coordinates

$$x=r-b\ge 0,\qquad y=b-c\ge 0,\qquad u=c+r+1-k,$$

for which the three binomial coefficients $\binom{n_i}{m_i}$ have $2m_1-n_1=-y$, $2m_2-n_2=-(x+1)$ and $2m_3-n_3=u$. The map $(b,c,r)\mapsto(x,y,u)$ is injective, with $2b=k-1+u-x+y$.

Lemma 9.12. *For all integers $0\le m\le n$,* $\displaystyle\binom{n}{m}\le\frac{2^{n+1}}{\sqrt{n+1}}\exp\Bigl(-\frac{(2m-n)^2}{4(n+1)}\Bigr)$.

Proof. The case $n=0$ is trivial. Let $n=2M\ge 2$ and $m=M+t$ with $0\le t\le M$. Then $\binom{2M}{M+t}/\binom{2M}{M}=\prod_{i=1}^{t}(1-\frac{2i-1}{M+i})\le\exp(-\sum_{i\le t}\frac{2i-1}{2M})=e^{-t^2/n}$. Let $n=2M+1$ and $m=M+1+t$ with $0\le t\le M$. Then $\binom{n}{m}/\binom{n}{M+1}=\prod_{i=1}^{t}(1-\frac{2i}{M+1+i})\le e^{-t(t+1)/n}=e^{1/(4n)}e^{-(2m-n)^2/(4n)}$. The cases $m<n/2$ follow by symmetry. The central coefficient is at most $2^n/\sqrt{n}$.[40] Hence $\binom{n}{m}\le e^{1/4}2^n n^{-1/2}e^{-(2m-n)^2/(4n)}$ for $n\ge 1$, and $e^{1/4}n^{-1/2}\le 2(n+1)^{-1/2}$. □

Theorem 9.13. *As $k\to\infty$,*

$$Q_k\sim\frac{2}{\pi}\cdot\frac{8^k}{k}.$$

[38] The discriminant of q_k as a polynomial in m is $-23k^4-30k^3+37k^2+12k+4$. It vanishes at $k=1$ and is negative from $k=2$ on; for $k=12$ it equals -523292.

[39] In [33] the middle factor is written $\binom{r+b-1}{b-1}-\binom{r+b-1}{b-2}$; the two forms agree because $\binom{N}{j-1}=\frac{j}{N-j+1}\binom{N}{j}$.

[40] By induction $\binom{2M}{M}\le 4^M/\sqrt{3M+1}$, the induction step being $(2M+1)^2(3M+4)\le 4(M+1)^2(3M+1)$; and $\binom{2M+1}{M}=\frac{1}{2}\binom{2M+2}{M+1}$.

Proof. Step 1: a Gaussian bound. For $(b,c,r) \in I_k$ we have $n_i + 1 \le 2k$ for $i = 1,2,3$, so Lemma 9.12 gives

$$T(b,c,r) \le 2^{3k+1}\,\frac{(y+1)(x+2)}{(k-c+1)(r+1)}\,\prod_{i=1}^{3}(n_i+1)^{-1/2}\,\exp\Big(-\frac{x^2+y^2+u^2}{8k}\Big). \tag{9.6}$$

Call a triple *good* if $r \ge k/4$, $c \le 3k/4$ and $x+y \le k/4$, and *bad* otherwise. For a good triple each of $k-c+1$, $r+1$, $n_1+1 \ge k-c$ and $n_2+1 = r+b$ is at least $k/4$, and $n_3+1 = k-(x+y) \ge 3k/4$. Hence

$$T(b,c,r) \le 150\cdot 8^k\,k^{-7/2}\,(x+2)(y+1)\,\exp\Big(-\frac{x^2+y^2+u^2}{8k}\Big) \qquad \text{(good triples)}. \tag{9.7}$$

For a bad triple with $k \ge 4$: if $r < k/4$, then $c \le b \le r$ gives $u < 1 - k/2$; if $c > 3k/4$, then $r \ge c$ gives $u > k/2$; if $x+y > k/4$, then $x^2+y^2 > k^2/32$. In each case $x^2+y^2+u^2 \ge k^2/32$, and (9.6) with $(y+1)(x+2) \le k^2$ gives $T \le 2^{3k+1}k^2e^{-k/256}$. There are at most k^3 triples, so

$$C_k + \sum_{\text{bad}} T \le 4^k + 2\cdot 8^k\,k^5e^{-k/256} = o(8^k/k). \tag{9.8}$$

For $L \ge 1$ let W_L be the set of triples with $x \le L\sqrt{k}$, $y \le L\sqrt{k}$ and $|u| \le L\sqrt{k}$. Summing (9.7) over all integers $x,y \ge 0$ and u for which one of these three inequalities fails, and comparing the sums with Gaussian integrals, gives

$$\sum_{\text{good},\ \notin W_L} T \le \varepsilon(L)\,\frac{8^k}{k}, \qquad \varepsilon(L) = 10^7\,e^{-L^2/32}, \tag{9.9}$$

for all $k \ge 1$.[41]

Step 2: the central window. Fix L. If k is large, every integer point (x,y,u) with $0 \le x,y \le L\sqrt{k}$, $|u| \le L\sqrt{k}$ and $u \equiv k-1+x-y \pmod 2$ comes from a unique triple of I_k, this triple is good, and $b,c,r = \frac{k}{2} + O(L\sqrt{k})$, $n_i = k + O(L\sqrt{k})$. The local limit theorem of de Moivre and Laplace [19, Section VII.2] states that $\binom{n}{m} = (1+o(1))\,2^n\sqrt{2/(\pi n)}\,e^{-(2m-n)^2/(2n)}$ uniformly for $|2m-n| = O(\sqrt{n})$. Applied to the three binomial coefficients it gives, uniformly on W_L,

$$T(b,c,r) = (1+o(1))\,\frac{(y+1)(x+2)}{(k/2)^2}\,2^{3k-2}\Big(\frac{2}{\pi k}\Big)^{3/2}\exp\Big(-\frac{x^2+y^2+u^2}{2k}\Big).$$

The admissible points form one coset of a sublattice of index 2 in $\mathbb{Z}^3$. Writing $x = X\sqrt{k}$, $y = Y\sqrt{k}$, $u = U\sqrt{k}$, the sum over W_L is a Riemann sum with cells of volume $2k^{-3/2}$, and

$$\sum_{W_L} T = (1+o(1))\,\frac{8^k}{k}\cdot\frac{1}{2}\Big(\frac{2}{\pi}\Big)^{3/2}J_L, \qquad J_L = \int_0^L\!\int_0^L\!\int_{-L}^{L} XY\,e^{-(X^2+Y^2+U^2)/2}\,dU\,dY\,dX.$$

Step 3. By (9.5), (9.8), (9.9) and Step 2,

$$\tfrac{1}{2}(2/\pi)^{3/2}J_L \le \liminf_{k\to\infty}\frac{k\,Q_k}{8^k} \le \limsup_{k\to\infty}\frac{k\,Q_k}{8^k} \le \tfrac{1}{2}(2/\pi)^{3/2}J_L + \varepsilon(L)$$

for every $L \ge 1$. As $L \to \infty$, $J_L \to 1\cdot 1\cdot\sqrt{2\pi}$ and $\varepsilon(L) \to 0$, and $\frac{1}{2}(2/\pi)^{3/2}\sqrt{2\pi} = 2/\pi$. □

[41] Put $G_\lambda = \sum_{u\in\mathbb{Z}} e^{-u^2/\lambda} \le 1 + \sqrt{\pi\lambda}$ and $H_\lambda = \sum_{x\ge 0} e^{-x^2/\lambda} \le 1 + \frac{1}{2}\sqrt{\pi\lambda}$. Since $te^{-t^2/(16k)} \le \sqrt{8k/e}$, we have $(x+2)e^{-x^2/(8k)} \le 4\sqrt{k}\,e^{-x^2/(16k)}$, hence $\sum_{x\ge 0}(x+2)e^{-x^2/(8k)} \le 4\sqrt{k}\,H_{16k} \le 20k$ and $\sum_{x>L\sqrt{k}}(x+2)e^{-x^2/(8k)} \le 4\sqrt{k}\,e^{-L^2/32}H_{32k} \le 28k\,e^{-L^2/32}$; also $G_{8k} \le 7\sqrt{k}$ and $\sum_{|u|>L\sqrt{k}} e^{-u^2/(8k)} \le e^{-L^2/16}G_{16k} \le 9\sqrt{k}\,e^{-L^2/16}$. The sum in (9.9) is at most $150\cdot 8^k k^{-7/2}\big(2\cdot 28\cdot 20\cdot 7 + 400\cdot 9\big)k^{5/2}e^{-L^2/32}$.

Theorem 9.14. *The series $Q(z) = \sum_{k\ge 0} Q_k z^k$ is transcendental over $\mathbb{Q}(z)$. Hence Conjecture 9.2 holds.*

Proof. Suppose that $Q(z)$ is algebraic over $\mathbb{Q}(z)$. By the theorem on the coefficients of algebraic functions [20, Theorem D], [21, Theorem VII.8], there are a real $\beta > 0$, a rational number $s \notin \{-1, -2, -3, \dots\}$, finitely many distinct complex numbers $\omega_1, \dots, \omega_\ell$ of modulus 1 and nonzero constants $d_1, \dots, d_\ell$ such that

$$Q_k = \beta^k\, k^s\, \bigl(h(k) + o(1)\bigr), \qquad h(k) = \sum_{j=1}^{\ell} d_j\, \omega_j^k.$$

The exponent -1 is excluded because a term $(1 - z/\zeta)^\theta$ of a Puiseux expansion at a singularity ζ contributes to the coefficients only if θ is not a nonnegative integer, and then with order $k^{-\theta-1}$. The function h is bounded, and it does not tend to zero, because $\frac{1}{N}\sum_{k<N} |h(k)|^2 \to \sum_j |d_j|^2 > 0$. Choose $\delta > 0$ and an infinite set S of integers with $\delta \le |h(k)|$ for $k \in S$. By Theorem 9.13,

$$\Bigl(\frac{8}{\beta}\Bigr)^k k^{-1-s} = \frac{\pi}{2}\, |h(k)|\, (1 + o(1)) \qquad (k \in S,\ k \to \infty),$$

and the right side stays between two positive constants. This forces $\beta = 8$ and $s = -1$, which is excluded. □

Only the two-sided bound $Q_k = \Theta(8^k/k)$ enters the proof. The asymptotic of Theorem 9.13 is that of the coefficients of $\frac{2}{\pi} \log \frac{1}{1-8z}$.

9.4. **Moment sequences.** A sequence $(a_k)_{k\ge 0}$ is a *Stieltjes moment sequence* if there is a positive measure μ on $[0, \infty)$ with $a_k = \int_0^\infty x^k\, d\mu(x)$ for all k, and a *Hamburger moment sequence* if this holds with a positive measure on $\mathbb{R}$.

Theorem 9.15. *The sequence $(B_k)_{k\ge 0}$ is a Stieltjes moment sequence. Explicitly, for $0 < \varphi < \pi/3$ put $c = \cos\varphi$ and*

$$x(\varphi) = \frac{\sin^3 3\varphi}{\sin\varphi\, \sin^2 2\varphi}, \qquad w(x(\varphi)) = \frac{1}{\pi\, x(\varphi)} \cdot \frac{(4c^2 - 1)\,(64c^6 - 112c^4 + 44c^2 + 7)}{8\, c\, \sin\varphi\, (8c^2 + 1)}.$$

Then $\varphi \mapsto x(\varphi)$ is a bijection from $(0, \pi/3)$ onto $(0, 27/4)$, the function w is positive and continuous on $(0, 27/4)$, and

$$B_k = \int_0^{27/4} x^k\, w(x)\, dx \qquad (k \ge 0).$$

Proof. By [33, Theorems 6.2 and 6.3], $B(z) = \sum_k B_k z^k$ satisfies

$$B = \frac{T(T-2)^2}{3 - 2T}, \qquad T = 1 + zT^3, \quad T(0) = 1, \tag{9.10}$$

where $T(z) = \sum_k \frac{1}{2k+1}\binom{3k}{k} z^k$ has radius of convergence $4/27$. Let $\Omega = \mathbb{C} \setminus [4/27, \infty)$.

Continuation. The discriminant of $zT^3 - T + 1$ with respect to T is $z(4 - 27z)$. The three roots are therefore distinct for $z \notin \{0, 4/27\}$ and continue analytically along every path avoiding these two points. Near $z = 0$ one root is the analytic series $T(z)$, so a small loop around 0 returns it to itself. Since Ω is simply connected, T extends to an analytic function on Ω, and it extends analytically across each point of $(4/27, \infty)$ from above and from below. On Ω we have $T \ne 3/2$, because $T = 3/2$ forces $z = (T-1)/T^3 = 4/27$; so B is analytic on Ω.

Growth. If $|z| > 1$ and $|T| \ge 2$, then $|T|^3 = |T - 1|/|z| < \frac{3}{2}|T|$, which is absurd; so $|T| < 2$ and $|T|^3 \le 3/|z|$ for $|z| > 1$. Hence $B(z) = O(|z|^{-1/3})$ as $z \to \infty$ in Ω. At $z = 4/27$ the cubic

has the double root $3/2$ and the simple root -3. The series T has positive coefficients and increases on $(0,4/27)$ from 1 to $3/2$, so on Ω near $4/27$ the function T is one of the two roots close to $3/2$, and $T-\frac{3}{2}=O(|z-\frac{4}{27}|^{1/2})$ by the Puiseux expansion. Since $3-2T$ vanishes there to order exactly $\frac{1}{2}$, $B(z)=O(|z-\frac{4}{27}|^{-1/2})$.

Boundary values. If $T(z)$ is real, then $z=(T-1)/T^3$ is real. Hence $\operatorname{Im} T$ has no zero in the upper half-plane, and it is positive there because $T(z)=1+z+O(z^2)$. For real $\zeta>4/27$ let $\tau(\zeta)=T(\zeta+i0)$. This is a continuous function with $\operatorname{Im}\tau\geq 0$ whose value is a root of ζT^3-T+1. That cubic has one real root, which is negative,[42] and a pair of conjugate roots. The three roots stay distinct, so τ is either the real root for all ζ or the root of positive imaginary part for all ζ; since $\tau(\zeta)\to 3/2$ as $\zeta\to 4/27$, it is the latter. Write $\tau=\rho e^{i\varphi}$ with $0<\varphi<\pi$. The imaginary and real parts of $\zeta\tau^3-\tau+1=0$ give $\zeta\rho^2\sin 3\varphi=\sin\varphi$ and then $\rho\sin 2\varphi=\sin 3\varphi$, that is

$$\rho=\frac{\sin 3\varphi}{\sin 2\varphi}=\frac{4c^2-1}{2c},\qquad \zeta=\frac{\sin\varphi\,\sin^2 2\varphi}{\sin^3 3\varphi}=\frac{1}{x(\varphi)}.$$

Positivity of ρ and ζ forces $0<\varphi<\pi/3$, and conversely each such φ gives a root with positive imaginary part for $\zeta=1/x(\varphi)>4/27$. As each ζ has exactly one such root, $\varphi\mapsto x(\varphi)$ is a bijection of $(0,\pi/3)$ onto $(0,27/4)$.

The Stieltjes formula. The coefficients of B are real, so $B(\zeta-i0)=\overline{B(\zeta+i0)}$. Cauchy's formula on a keyhole contour around the cut, with the growth bounds above, gives for $z\in\Omega$

$$B(z)=\frac{1}{\pi}\int_{4/27}^{\infty}\frac{\operatorname{Im} B(\zeta+i0)}{\zeta-z}\,d\zeta,$$

and hence

$$B_k=\frac{1}{\pi}\int_{4/27}^{\infty}\operatorname{Im} B(\zeta+i0)\,\zeta^{-k-1}\,d\zeta=\int_0^{27/4}x^k\,\frac{\operatorname{Im} B(1/x+i0)}{\pi x}\,dx.$$

The density. By (9.10), $\operatorname{Im} B=\operatorname{Im}\big[(T^3-4T^2+4T)(3-2\overline{T})\big]/|3-2T|^2$. For $T=\rho e^{i\varphi}$ the numerator is $3\rho^3\sin 3\varphi-12\rho^2\sin 2\varphi+12\rho\sin\varphi-2\rho^4\sin 2\varphi+8\rho^3\sin\varphi$, and with $\rho\sin 2\varphi=\sin 3\varphi=(4c^2-1)\sin\varphi$ and $\rho=(4c^2-1)/(2c)$ this becomes

$$\rho\big[\rho^2(\sin 3\varphi+8\sin\varphi)-12(\sin 3\varphi-\sin\varphi)\big]=\frac{(4c^2-1)\sin\varphi}{8c^3}\,(64c^6-112c^4+44c^2+7),$$

while $|3-2T|^2=9-12\rho c+4\rho^2=(1-c^2)(8c^2+1)/c^2$. The quotient is the stated formula for $\pi x\,w(x)$. For positivity put $v=c^2\in(1/4,1)$ and $f(v)=64v^3-112v^2+44v+7$. Then $f'(v)=4(4v-1)(12v-11)$, so the minimum of f on $[1/4,1]$ is $f(11/12)=68/27>0$. □

Theorem 9.16. *The sequence $(Q_k)_{k\geq 0}$ is not a Stieltjes moment sequence, and not a Hamburger moment sequence. More precisely,*

$$\det\big(Q_{i+j+1}\big)_{0\leq i,j\leq 11}=-16871599970737005465806167716982557747256,$$
$$\det\big(Q_{i+j}\big)_{0\leq i,j\leq 12}=-16923297773141382020735162813385871280086948 2.$$

Hence Conjecture 9.4 fails for (Q_k), and the infinite Hankel matrix of (Q_k) has a negative minor.

Proof. If $Q_k=\int x^k\,d\mu$ for a positive measure μ on $\mathbb{R}$, then $(Q_{i+j})_{0\leq i,j\leq 12}$ is the Gram matrix of $1,x,\dots,x^{12}$ in $L^2(\mu)$ and has a nonnegative determinant. If μ is carried by $[0,\infty)$, then

[42] For $T\geq 0$ the minimum of ζT^3-T+1 is $1-\frac{2}{3\sqrt{3\zeta}}$, positive for $\zeta>4/27$. At $\zeta=4/27$ the real root is -3.

$(Q_{i+j+1})_{0\le i,j\le 11}$ is the Gram matrix of $1, x, \dots, x^{11}$ in $L^2(x\,d\mu)$. The two determinants involve only $Q_0, \dots, Q_{24}$, which are

1, 1, 5, 33, 234, 1706, 12618, 94157, 706878, 5330403, 40332881, 306023196, 2327290506, 17734226120, 135376087792, 1035065812309, 7925664456804, 60771825678389, 466587436417569, 3586731522598141, 27604168028694252, 212685629040409758, 1640473066548280308, 12666272996480868428, 97894846717057492194.

These values were computed from the definition, by a transfer over the prefixes of lattice words, and from (9.5); the two determinants were evaluated in exact integer arithmetic by two independent methods, rational Gaussian elimination and fraction-free elimination [8]. The first determinant is the minor of the infinite Hankel matrix with rows $0, \dots, 11$ and columns $1, \dots, 12$. □

Remark 9.17. The leading Hankel determinants $\det(Q_{i+j})_{0\le i,j<N}$ are positive for $N \le 12$, and $\det(Q_{i+j+1})_{0\le i,j<N}$ is positive for $N \le 11$ [8]. The failure is therefore invisible in the 7×7 matrix examined in [33, Section 9]; it first shows at the moment Q_{23}.

9.5. **Log-concavity of Q_k.** Conjecture 9.3 remains open. The polynomials Q_k were computed exactly for $k \le 110$, and their coefficient sequences are log-concave for all these k [8]; the range tested in [33] is $k \le 25$. The method of Theorem 9.10 is not available: Q_k has 0 nonreal zeros for $k \le 10$, two for $11 \le k \le 19$, four for $20 \le k \le 29$, six for $30 \le k \le 39$ and eight for $40 \le k \le 45$ [8], so log-concavity is not a consequence of Newton's inequalities.

9.6. **A structural model for Q_k.**

Proposition 9.18. *A word $w \in \mathcal{L}_3^k$ avoids* 321 *if and only if it factors as $w = w_{\mathrm{I}}\, w_{\mathrm{II}}\, w_{\mathrm{III}}$ with*

$$w_{\mathrm{I}} \in \{1,2\}^*, \qquad w_{\mathrm{II}} \in 3\{1,3\}^*, \qquad w_{\mathrm{III}} \in \{\varepsilon\} \cup 2\{2,3\}^*,$$

where ε is the empty word; the factorisation is then unique. Consequently Q_k is the number of walks of length $3k$ from $(0,0)$ to $(0,0)$ in the quarter plane $\mathbb{Z}^2_{\ge 0}$ that use the steps $(1,0),(-1,1)$ in a first phase, the steps $(1,0),(0,-1)$ in a second phase that begins with $(0,-1)$, and the steps $(-1,1),(0,-1)$ in a third phase, which is either empty or begins with $(-1,1)$.

Proof. Suppose w has such a factorisation and contains letters $3,2,1$ in this order. The 3 lies in $w_{\mathrm{II}}w_{\mathrm{III}}$, so the later 2 lies in w_{III}, which contains no 1; a contradiction. Conversely let w avoid 321. It contains a 3. Let w_{I} be the prefix before the first 3 and let w_{II} extend from the first 3 up to, and excluding, the next 2. The remainder is empty or starts with a 2 that is preceded by a 3, and then it contains no 1. Uniqueness holds because w_{II} must start at the first 3 and stop before the first later 2. The walk records $(\#1-\#2,\ \#2-\#3)$ along the prefixes, so the letters $1,2,3$ are the steps $(1,0),(-1,1),(0,-1)$ and the lattice condition is the quarter plane. □

The parameters of (9.5) are read off from the three phases. In [33, Theorem 7.1], b and c are the numbers of letters 2 and 3 before the last 1, and r is the number of letters 1 before the last 2 in that prefix. If w_{II} contains a 1, then b is the number of twos of w_{I}, r is the number of ones of w_{I} before its last 2, and c is the number of threes of w_{II} before its last 1; if w_{II} contains no 1, then $c = 0$ and b, r are read in w_{I} alone. So the second alternative of Question 9.5 has a positive answer with three classical two-step paths glued along the two heights. The first alternative has a negative answer.

Corollary 9.19. *There is no generating tree with finitely many labels, and no unambiguous context-free language, in which Q_k is the number of nodes at level $\alpha k+\beta$, respectively the number of words of length $\alpha k+\beta$, for fixed integers $\alpha \ge 1$ and $\beta \ge 0$ and all k.*

Proof. The level sizes of a generating tree with finitely many labels have a rational generating function, and the generating function of an unambiguous context-free language is algebraic [20]. The series $\sum_k a_{\alpha k+\beta} z^k$ extracted from a rational or algebraic series $\sum_N a_N z^N$ is again rational, respectively algebraic.[43] Either model would make $Q(z)$ algebraic, against Theorem 9.14. □

9.7. **Alphabets of size four.** Question 9.6 has three parts: the reduction, the classes that reduce to Catalan objects, and the full classification. We answer the first, give a list of classes for the second, and report a computation on the third.

Proposition 9.20. *Let $n \geq 1$, $\sigma \in S_n$ and $\Lambda \subseteq S_n$. For $w \in \mathcal{L}_n^k$ let $\Phi_\sigma(w)$ be the canon permutation obtained by replacing each letter i by σ_i. Then $\Phi_\sigma(w)$ avoids Λ if and only if w avoids $\sigma^{-1}\Lambda = \{\sigma^{-1} \circ \tau : \tau \in \Lambda\}$. Hence the number of canon permutations of $\{1^k, \dots, n^k\}$ avoiding Λ is $\sum_{\sigma \in S_n} |\mathcal{L}_n^k(\sigma^{-1}\Lambda)|$, and it depends only on the orbit of Λ under left multiplication by S_n.*

Proof. An occurrence of a pattern of S_n in a word on $[n]$ is a subsequence in which every letter appears once, so it is itself a permutation $\rho \in S_n$ in one-line notation. Its image under Φ_σ is $\sigma \circ \rho$, which lies in Λ exactly when $\rho \in \sigma^{-1}\Lambda$. The count follows since Φ_σ is a bijection from $\mathcal{L}_n^k$ onto the canon permutations with underlying permutation σ [33, Section 2], and the orbit statement by reindexing the sum. □

This is the proof of [33, Theorem 2.4], which does not use $n = 3$. For the second part, write $T_k = |\mathcal{L}_3^k|$.

Proposition 9.21. *Let $k \geq 1$ and $w \in \mathcal{L}_4^k$.*

(i) *w contains 1234.*
(ii) *w avoids 1243 if and only if $w = v\,4^k$ with $v \in \mathcal{L}_3^k$; and w avoids 2134 if and only if $w = 1^k v$ with v a lattice word on $\{2,3,4\}$. Each class has T_k elements.*
(iii) *w avoids 1324 if and only if $w = vv'$ with v a lattice word on $\{1,2\}$ and v' a lattice word on $\{3,4\}$. This class has C_k^2 elements.*
(iv) *The classes $\mathcal{L}_4^k(1243, 2134)$, $\mathcal{L}_4^k(1324, 2134)$ and $\mathcal{L}_4^k(1243, 1324)$ have C_k elements each; moreover $|\mathcal{L}_4^k(1243, 1324, 2134)| = 1$ and $|\mathcal{L}_4^k(1324, 2143)| = 2C_k - 1$.*
(v) *$|\mathcal{L}_4^k(\Omega)| = B_k$ when Ω is one of $\{2134, 1342\}$, $\{2134, 1423\}$, $\{1243, 2314\}$, $\{1243, 3124\}$, and $|\mathcal{L}_4^k(\Omega)| = Q_k$ when Ω is $\{2134, 1432\}$ or $\{1243, 3214\}$.*

Proof. A lattice word ends with the letter 4, and the first occurrences of $1, 2, 3, 4$ appear in this order; this gives (i).

(ii) If some 3 follows some 4, then the first 1, the first 2, that 4 and that 3 form 1243. If no 3 follows a 4, then at the first 4 all threes have occurred, hence all twos and all ones, and $w = v4^k$; such a word avoids 1243. Similarly, if a 2 precedes a 1, then fewer than k threes precede that 1, so a 3 follows it, and the final 4 completes a pattern 2134; otherwise $w = 1^k v$.

(iii) If a 3 precedes a 2, the first 1, that 3, that 2 and the final 4 form 1324. If every 2 precedes every 3, then no 1324 occurs, all ones precede the first 3 and all fours follow it, so $w = vv'$.

(iv) The first four classes are the intersections of the classes in (ii) and (iii): the words $1^k v 4^k$, $1^k 2^k v'$, $v 3^k 4^k$ and $1^k 2^k 3^k 4^k$. A word vv' of (iii) contains 2143 if and only if $v \neq 1^k 2^k$ and $v' \neq 3^k 4^k$, which leaves $2C_k - 1$ words.

(v) A word $1^k v$ contains 1342, 1423 or 1432 if and only if v contains 342, 423 or 432, which are the patterns 231, 312, 321 on the alphabet $\{2,3,4\}$; a word $v4^k$ contains 2314, 3124 or

[43] It is a linear combination of the series $f(\omega z^{1/\alpha})$ over the αth roots of unity ω, divided by $z^{\beta/\alpha}$.

3214 if and only if v contains 231, 312 or 321. The counts follow from $\mathcal{L}_3^k(231) = \mathcal{L}_3^k(312)$ [33, Theorem 3.2]. □

For the full classification, the words of $\mathcal{L}_4^k$ were enumerated for $k \le 6$, together with the set $M(w) \subseteq S_4$ of patterns contained in each word [8]. The number of distinct sets $M(w)$ is $1, 14, 54, 92, 92, 92$ for $k = 1, \dots, 6$, and the family of these sets is the same for $k = 4, 5, 6$. Two forbidden sets define the same class of lattice words when they meet the same sets $M(w)$; for each $k \in \{4, 5, 6\}$ this partitions the 2^{24} forbidden sets into 146 classes, and these carry 95 distinct sequences $\left(|\mathcal{L}_4^k(\Omega)|\right)_{1 \le k \le 6}$. For comparison, the same procedure on three letters returns the six sequences of [33, Theorem 3.4]. Among the 146 classes, those of Proposition 9.21 account for the sequences 0, 1, C_k, $2C_k - 1$, C_k^2, T_k, B_k, Q_k and $|\mathcal{L}_4^k|$; the remaining 86 sequences are not identified. The counts for single patterns are in Table 7. Whether the 92 sets $M(w)$ are the same for every $k \ge 4$ is not proved, and a classification for all k remains open. By Theorem 9.14, the two classes of Proposition 9.21(v) counted by Q_k have transcendental generating functions, so a classification in terms of Catalan objects alone does not exist for $n = 4$.

pattern τ	$\lvert\mathcal{L}_4^k(\tau)\rvert$ for $k = 1, \dots, 6$	value
1234	$0, 0, 0, 0, 0, 0$	0
1324	$1, 4, 25, 196, 1764, 17424$	C_k^2
1243, 2134	$1, 5, 42, 462, 6006, 87516$	T_k
2143	$1, 9, 102, 1308, 18463, 281411$	
1342, 1423, 2314, 3124	$1, 10, 168, 3528, 84018, 2170740$	
3142	$1, 13, 319, 9411, 293707, 9361968$	
2341, 2413, 3412, 4123	$1, 13, 341, 12012, 492252, 22064130$	
1432, 3214	$1, 14, 321, 8938, 278891, 9407065$	
3241, 4132	$1, 14, 419, 16950, 785866, 39117335$	
2431, 3421, 4213, 4312	$1, 14, 426, 17782, 862398, 45465438$	
4231	$1, 14, 446, 20507, 1123561, 67346038$	
4321	$1, 14, 462, 22007, 1231705, 74690663$	

TABLE 7. Lattice words on four letters avoiding one pattern of S_4, for $k \le 6$ [8]. The first three rows are Proposition 9.21.

10. Caged permutations and deformed Dumont sets

This section treats two further problems on permutations. The first comes from the paper of Balachandran, Ramdas and Shankar on caged subsequences [4]; we determine all extremal permutations of length $5k + 4$ and bound the extremal value on the class of n-cycles within one unit. The second is the closing question of the paper of Poddar, Sawant and Shankar on deformed Dumont permutations [31], which we answer by an explicit formula.

10.1. **Caged subsequences.** Let $\pi \in S_n$. For positions $1 \le i < j \le n$ let $c_\pi[i, j]$ be the number of k with $i \le k \le j$ such that $\pi(k)$ lies between $\pi(i)$ and $\pi(j)$, the two ends included, and put $c(\pi) = \max_{i<j} c_\pi[i, j]$. The entries $\pi(k)$ counted by $c_\pi[i, j]$ form a subsequence all of whose terms lie between its first and its last term; such a subsequence is called *caged*, and $c(\pi)$ is the length of a longest one.[44] We call a pair (i, j) with $c_\pi[i, j] = c(\pi)$ a *maximal pair*.

[44] A longest caged subsequence that starts at position i and ends at position j must contain every admissible entry in between, so longest caged subsequences correspond bijectively to the pairs (i, j) with $c_\pi[i, j] = c(\pi)$.

It is convenient to identify π with its diagram, the set of points $(i, \pi(i))$ in the square $[1, n]^2$. Then $c_\pi[i, j]$ is the number of points of the diagram in the closed rectangle with sides parallel to the axes and opposite corners $(i, \pi(i))$, $(j, \pi(j))$. Consequently c is invariant under the eight symmetries of the square. We shall use the complement $\overline{\pi} = n + 1 - \pi$ and the rotation

$$\rho\colon (x, y) \longmapsto (y,\ n + 1 - x), \tag{10.1}$$

which sends the diagram of π to the diagram of a permutation that we denote $\rho(\pi)$.

The main theorem of [4] is that

$$\min_{\pi \in S_n} c(\pi) = \left\lfloor \frac{n}{5} \right\rfloor + 2 \qquad (n \geq 5), \tag{10.2}$$

see [4, Theorem 1.1]. For $n = 5k + 4$ the minimum is attained by the permutation π_{ext} that consists of the five monotone runs $3k + 3, \ldots, 4k + 3$; $k + 1, \ldots, 1$; $2k + 3, \ldots, 3k + 2$; $5k + 4, \ldots, 4k + 4$; $k + 2, \ldots, 2k + 2$. We call $\pi \in S_n$ *extremal* if $c(\pi) = \lfloor n/5 \rfloor + 2$. The concluding section of [4] raises the following problems, which we quote.

Question 10.1 (Balachandran, Ramdas and Shankar [4, Concluding remarks])**.** *"The set $\mathcal{E}_{ext}$ of all the extremal permutations that minimize $c(\pi)$ is not yet completely known ... whether there are permutations distinctly different from π_{ext} is not clear." The authors give examples, for other residues of n modulo 5, in which the number of longest caged subsequences is $O(k)$, and continue: "However, whether such examples exist for $n = 5k + 4$ is not yet clear."*

Question 10.2 (Balachandran, Ramdas and Shankar [4, Concluding remarks])**.** *"The problem of determining $c(\pi)$ when we restrict ourselves to the subfamily of n-cycles is still open; it is not clear whether this algebraic imposition changes the answer to something significantly larger than $n/5$."*

Theorem 10.4 answers Question 10.1 for $n = 5k + 4$: the extremal permutations are π_{ext}, its complement, and the permutations obtained from these two by rearranging the middle run, and each of them has $2(k + 1)(k + 3)$ longest caged subsequences. Theorem 10.5 answers Question 10.2 up to an additive constant: the minimum of c over n-cycles exceeds (10.2) by at most one, and by exactly one when $n = 5k + 4$.

10.2. **A five-term inequality.** For the next lemma we extend the notation by $c_\pi[i, i] = 1$.

Lemma 10.3. *Let $\pi \in S_n$, $n \geq 2$, and suppose that $i_0 = \pi^{-1}(1)$ and $j_0 = \pi^{-1}(n)$ satisfy $i_0 < j_0$. Then*

$$c_\pi[i_0, j_0] + c_\pi[1, j_0] + c_\pi[1, i_0] + c_\pi[i_0, n] + c_\pi[j_0, n] \ \geq\ n + 6. \tag{10.3}$$

Equality holds if and only if $\pi(n) < \pi(k) < \pi(1)$ for every k with $i_0 < k < j_0$.

Proof. For each position k we count the terms of (10.3) to which k contributes. Since $\pi(j_0) = n$, a position $k \leq j_0$ contributes to $c_\pi[1, j_0]$ exactly when $\pi(k) \geq \pi(1)$; since $\pi(i_0) = 1$, a position $k \leq i_0$ contributes to $c_\pi[1, i_0]$ exactly when $\pi(k) \leq \pi(1)$. The analogous statements hold for $c_\pi[i_0, n]$ and $c_\pi[j_0, n]$ with $\pi(n)$ in place of $\pi(1)$.

Let $k < i_0$. Then k contributes to exactly one of $c_\pi[1, j_0]$, $c_\pi[1, i_0]$ if $k \neq 1$, to both if $k = 1$, and to no other term. The position i_0 contributes to $c_\pi[i_0, j_0]$, $c_\pi[1, i_0]$ and $c_\pi[i_0, n]$, and if $i_0 = 1$ also to $c_\pi[1, j_0]$. In both cases the positions $k \leq i_0$ contribute $i_0{+}3$ in total. By symmetry the positions $k \geq j_0$ contribute $(n - j_0 + 1) + 3$. A position k with $i_0 < k < j_0$ contributes to $c_\pi[i_0, j_0]$, to $c_\pi[1, j_0]$ exactly when $\pi(k) \geq \pi(1)$, to $c_\pi[i_0, n]$ exactly when $\pi(k) \leq \pi(n)$, and to no other term. Hence the left side of (10.3) is at least $(i_0{+}3){+}(j_0{-}i_0{-}1){+}(n{-}j_0{+}4) = n{+}6$, and it equals $n + 6$ exactly when no position strictly between i_0 and j_0 has $\pi(k) \geq \pi(1)$ or $\pi(k) \leq \pi(n)$. □

If $\pi^{-1}(1) > \pi^{-1}(n)$, the lemma applies to $\overline{\pi}$. Since one of the five terms is at least $(n+6)/5$, and $\lceil (n+6)/5 \rceil = \lfloor n/5 \rfloor + 2$, the lemma gives the lower bound in (10.2) for every $n \geq 5$.[45]

10.3. **The extremal permutations of length** $5k+4$**.** Let $n = 5k+4$ with $k \geq 1$. For an arrangement $\mu = (\mu_1, \ldots, \mu_k)$ of the values $2k+3, \ldots, 3k+2$ let $\beta(\mu) \in S_n$ be the permutation whose one-line notation is the concatenation of the five blocks

$$\begin{aligned} &B_1 = (3k+3,\, 3k+4, \ldots, 4k+3), \quad B_2 = (k+1,\, k, \ldots, 1), \qquad\qquad B_3 = \mu, \\ &B_4 = (5k+4,\, 5k+3, \ldots, 4k+4), \quad B_5 = (k+2,\, k+3, \ldots, 2k+2). \end{aligned} \tag{10.4}$$

The blocks B_1, B_2, B_4, B_5 have $k+1$ entries and B_3 has k. The permutation π_{ext} is $\beta(\mu)$ for the increasing arrangement μ.

Theorem 10.4. *Let $n = 5k+4$ with $k \geq 1$, and let $\pi \in S_n$.*

(i) $c(\pi) = k+2$ if and only if $\pi = \beta(\mu)$ or $\pi = \overline{\beta(\mu)}$ for some arrangement μ of $2k+3, \ldots, 3k+2$. These $2 \cdot k!$ permutations are distinct.

(ii) Every extremal permutation has exactly $2(k+1)(k+3)$ maximal pairs.

(iii) The permutation $\beta(\mu)$ contains the 4-cycle $(1,\, 3k+3,\, n,\, 2k+2)$, and $\overline{\beta(\mu)}$ contains the 4-cycle $(1,\, 2k+2,\, n,\, 3k+3)$. In particular no extremal permutation of length $5k+4$ is an n-cycle.

Proof. Write $B_a(p)$ for the pth entry of the block B_a and also for the corresponding point of the diagram. In order of position the blocks are $B_1, \ldots, B_5$; in order of value they are $B_2 < B_5 < B_3 < B_1 < B_4$.

The values of c on $\beta(\mu)$. Let $\pi = \beta(\mu)$. We determine c_π for every pair of points by listing which points between them, in position, have a value between theirs.

For $B_1(p)$ and $B_2(q)$: the later entries of B_1 are larger than $B_1(p)$, and the $q-1$ earlier entries of B_2 lie between the two values; so $c_\pi = q+1$. For $B_1(p)$ and $B_4(t)$: the $k+1-p$ later entries of B_1 lie between, the blocks B_2, B_3 are below $B_1(p)$ and the earlier entries of B_4 are above $B_4(t)$; so $c_\pi = k+3-p$. For $B_1(p)$ and $B_5(u)$: the later entries of B_1 and all of B_4 are too large, all of B_2 and the earlier entries of B_5 are too small, and all k entries of B_3 lie between; so $c_\pi = k+2$. For $B_2(q)$ and $B_4(t)$: the later entries of B_2 are too small, the earlier entries of B_4 too large, and B_3 lies between; so $c_\pi = k+2$. For $B_2(q)$ and $B_5(u)$: the later entries of B_2 are too small, B_3 and B_4 are too large, and the $u-1$ earlier entries of B_5 lie between; so $c_\pi = u+1$. For $B_4(t)$ and $B_5(u)$: the $k+1-t$ later entries of B_4 lie between and the earlier entries of B_5 are too small; so $c_\pi = k+3-t$. If one of the two points lies in B_3 and the other in a different block, then no point of a block other than B_3 lies between them in position and in value, by the same inspection, so $c_\pi \leq (k-1)+2$. Two points of the same block give $c_\pi \leq k+1$, the size of the block.

Hence $c(\beta(\mu)) = k+2$, and the maximal pairs are the $(k+1)^2$ pairs of type B_1B_5, the $(k+1)^2$ pairs of type B_2B_4, and $k+1$ pairs of each of the types B_1B_2 (with $q = k+1$), B_1B_4 (with $p = 1$), B_2B_5 (with $u = k+1$) and B_4B_5 (with $t = 1$). Their number is $2(k+1)^2 + 4(k+1) = 2(k+1)(k+3)$. Complementation preserves c and the maximal pairs. This proves (ii) and one direction of (i), once the other direction is known. In $\beta(\mu)$ the value 1 stands to the left of the value n, and in $\overline{\beta(\mu)}$ to the right; so the two families are disjoint and (i) lists $2 \cdot k!$ permutations.

Necessity: the four corner points. Let $c(\pi) = k+2$. Replacing π by $\overline{\pi}$ if necessary, we assume $i_0 = \pi^{-1}(1) < j_0 = \pi^{-1}(n)$ and show that $\pi = \beta(\mu)$. Since $n+6 = 5(k+2)$ and no

[45] The proof of the lower bound in [4] distinguishes cases according to the relative position of the entries $\pi(1)$ and $\pi(n)$. The count above treats all of them at once, and it records the equality case, which is what the classification needs.

term of (10.3) exceeds $k+2$, Lemma 10.3 shows that each of the five terms equals $k+2$ and that equality holds in the lemma. In particular $c_\pi[1,i_0] > 1$ and $c_\pi[j_0,n] > 1$, so $i_0 > 1$ and $j_0 < n$. Every position between i_0 and j_0 is counted by $c_\pi[i_0,j_0]$, so $j_0 - i_0 = k+1$, and the k positions strictly between them carry values in $(\pi(n),\pi(1))$; as $k \geq 1$, this gives $\pi(n) < \pi(1)$. By the equality case, $c_\pi[1,j_0]$ counts j_0 and the positions $k < i_0$ with $\pi(k) \geq \pi(1)$, and $c_\pi[1,i_0]$ counts i_0 and the positions $k < i_0$ with $\pi(k) \leq \pi(1)$. Both numbers are $k+2$ and position 1 is counted twice, so $i_0 - 1 = 2k+1$. Thus $i_0 = 2k+2$ and $j_0 = 3k+3$, and among the positions $2,\dots,2k+1$ exactly k carry a value below $\pi(1)$ and exactly k a value above $\pi(1)$.

The diagram of $\rho(\pi)$ consists of the points $(\pi(i), n+1-i)$. Its value 1 stands at position $\pi(n)$ and its value n at position $\pi(1)$, and $\pi(n) < \pi(1)$; moreover $c(\rho(\pi)) = k+2$. So everything proved so far applies to $\rho(\pi)$, and by iteration to $\rho^2(\pi)$ and $\rho^3(\pi)$.[46] Applied to $\rho(\pi)$ it gives $\pi(n) = 2k+2$ and $\pi(1) = 3k+3$, and it shows that the points of π with values in $[2k+3, 3k+2]$ are those with positions in $[2k+3, 3k+2]$. We name the four corner points

$$A = (1, 3k+3), \qquad B = (2k+2, 1), \qquad T = (3k+3, n), \qquad R = (n, 2k+2),$$

and call the k points between B and T the *middle points*.

Necessity: the left part. Among the positions $2,\dots,2k+1$ there are k whose value is below $3k+3$; such a value is not a middle value, not 1 and not $2k+2$, so it lies in $[2, 2k+1]$, and we call the point a *low*. The other k positions carry values in $[3k+4, 5k+3]$, and we call these points *highs*.

Let X be a low, at position p. The rectangle spanned by X and T contains X, T and the k middle points, which makes $k+2$ points. Therefore it contains no other point: every position q with $p < q \leq 2k+2$ carries a value smaller than that of X. Hence no high stands to the right of a low, and the lows are decreasing from left to right. Let X be a high. The rectangle spanned by X and B contains X, B and the k lows, which all stand to the right of X. Therefore it contains no other point, and every high to the right of X has a larger value. Together with A and B this shows that π is increasing on the positions $1,\dots,k+1$, with values $3k+3$ and then k highs, and decreasing on the positions $k+2,\dots,2k+2$, with k lows and then the value 1.

Necessity: the remaining parts. The same statement for $\rho^2(\pi)$, whose diagram is the image of that of π under $(x,y) \mapsto (n+1-x, n+1-y)$, reads: π is decreasing on the positions $3k+3,\dots,4k+3$, with the value n followed by k values in $[3k+4, 5k+3]$, and increasing on the positions $4k+4,\dots,n$, with k values in $[2, 2k+1]$ followed by $2k+2$. The statement for $\rho(\pi)$ says that the points of $\rho(\pi)$ at positions $2,\dots,k+1$ have values in $[3k+4, 5k+3]$, and that the points of $\rho(\pi)$ at positions $3k+4,\dots,4k+3$ have values in $[3k+4, 5k+3]$. A point of $\rho(\pi)$ at position v with value w is the point of π with value v and position $n+1-w$. So the values $2,\dots,k+1$ and the values $3k+4,\dots,4k+3$ of π stand at positions in $[2, 2k+1]$. These are $2k$ values on $2k$ positions; hence the lows on the left are $2,\dots,k+1$, the highs on the left are $3k+4,\dots,4k+3$, and the values on the positions to the right of T are the remaining ones. With the monotonicity already proved this is the block structure (10.4), where μ is the arrangement of the middle values. So $\pi = \beta(\mu)$, and (i) is proved.

The 4-cycle. In $\beta(\mu)$ we have $1 \mapsto 3k+3 \mapsto n \mapsto 2k+2 \mapsto 1$, by the positions of A, T, R, B. In the complement, $1 \mapsto n+1-(3k+3) = 2k+2 \mapsto n \mapsto 3k+3 \mapsto 1$. Since $n > 4$, a permutation that contains a 4-cycle is not an n-cycle. □

For $n = 5k+4$ the theorem settles Question 10.1. Up to the symmetries of the square the extremal permutations are exactly the permutations $\beta(\mu)$; they differ from π_{ext} only inside the middle block, whose arrangement is arbitrary. All of them have $2(k+1)(k+3)$ longest caged

[46] The hypothesis $i_0 < j_0$ is used once, to choose between π and $\overline{\pi}$; after that the four rotations of π all satisfy it, and this is what makes the argument short.

subsequences, a number of order k^2; extremal permutations of length $5k+4$ with $O(k)$ longest caged subsequences do not exist.

For the other residues of n modulo 5 the extremal set is much larger. Table 8 records the number of extremal permutations for $5 \le n \le 14$, the number of their orbits under the eight symmetries, and the number of extremal n-cycles. These three rows are the result of an exhaustive enumeration, which is part of the verification archive [8]; the entries for $n = 9$ and $n = 14$ agree with Theorem 10.4.[47] No classification is claimed for $n \not\equiv 4 \pmod 5$.

n	5	6	7	8	9	10	11	12	13	14
extremal permutations	60	140	230	196	2	42604	64890	50568	788	4
orbits	11	21	33	28	1	5433	8152	6454	100	1
extremal n-cycles	16	30	38	18	0	5178	6706	3856	44	0

TABLE 8. Permutations $\pi \in S_n$ with $c(\pi) = \lfloor n/5 \rfloor + 2$.

10.4. **Cycles of full length.** Let $\mathrm{cyc}(n)$ be the minimum of $c(\pi)$ over all n-cycles $\pi \in S_n$.

Theorem 10.5. *For every $n \ge 5$,*

$$\left\lfloor \frac{n}{5} \right\rfloor + 2 \;\le\; \mathrm{cyc}(n) \;\le\; \left\lfloor \frac{n}{5} \right\rfloor + 3.$$

If $n = 5k+4$ with $k \ge 1$, then $\mathrm{cyc}(n) = k+3$. If $n \in \{5,6,7,8,10,11,12,13,15,16\}$, then $\mathrm{cyc}(n) = \lfloor n/5 \rfloor + 2$.

Proof. The lower bound is (10.2). For $n = 5k+4$ it improves to $k+3$ by Theorem 10.4(iii). The following ten permutations, in one-line notation, are n-cycles with $c = \lfloor n/5 \rfloor + 2$, as one checks directly from the definition:[48]

n	cycle	n	cycle
5	2 3 5 1 4	11	3 7 2 10 11 1 8 9 4 5 6
6	2 3 6 5 1 4	12	4 3 8 2 11 12 1 9 10 5 6 7
7	2 5 7 1 6 3 4	13	5 4 9 3 12 13 6 11 1 2 10 7 8
8	3 6 2 8 1 7 4 5	15	4 3 9 2 10 14 15 1 11 13 12 5 6 7 8
10	2 6 9 10 1 7 8 3 4 5	16	5 4 10 3 2 11 15 16 1 12 14 13 6 7 8 9

This proves the last assertion and the upper bound for these ten values of n. For the remaining n we construct an n-cycle with $c \le \lfloor n/5 \rfloor + 3$. Put

$$K = \left\lfloor \frac{n}{5} \right\rfloor + 1, \qquad m = n - 4K.$$

Writing $n = 5a + b$ with $0 \le b \le 4$ we have $m = a + b - 4 \le K - 1$, and $m \ge 0$ holds for all $n \ge 8$ other than 10, 11, 15; moreover $K \ge 2$. So we assume $K \ge 2$ and $0 \le m \le K - 1$.

[47] For $n = 9$ the two extremal permutations are 6 7 2 1 5 9 8 3 4 and its complement. For $n = 14$ the four permutations $\beta(\mu)$, $\overline{\beta(\mu)}$ form a single orbit: the reversal of $\beta(7,8)$ is its complement, and the inverse of $\beta(7,8)$ is the complement of $\beta(8,7)$.

[48] They were found by a backtracking search. The search also shows that no such cycle exists for $n = 9$ and $n = 14$, in agreement with the theorem. For $n = 17$ and $n = 18$ the search did not terminate, and these two cases are undecided.

The base permutation. Let $\beta \in S_n$ consist of five blocks, given by position and value as follows, where p, q, t, u run over $1, \dots, K$:

$$B_1(p)\colon\ p \mapsto 2K+m+p, \qquad B_2(q)\colon\ K+q \mapsto K+1-q,$$
$$B_4(t)\colon\ 2K+m+t \mapsto n+1-t, \quad B_5(u)\colon\ 3K+m+u \mapsto K+u,$$

and B_3 occupies the positions $2K+1, \dots, 2K+m$ with $2K+i \mapsto 2K+i+1$ for $i < m$ and $2K+m \mapsto 2K+1$. This is the shape (10.4) with outer blocks of size K and a middle block of size m. The middle block is one cycle Z of β of length m, absent if $m = 0$. The other cycles of β are the 4-cycles

$$C_p = (p,\ 2K+m+p,\ n+1-p,\ 2K+1-p), \qquad 1 \le p \le K,$$

because B_1 sends p to $2K+m+p$, B_4 sends this to $n+1-p = 3K+m+(K+1-p)$, B_5 sends this to $2K+1-p = K+(K+1-p)$, and B_2 sends this to p.

The inspection in the proof of Theorem 10.4 applies word for word, with K in place of $k+1$ and m in place of k. It gives $c_\beta = q+1$ for the pair $B_1(p), B_2(q)$; $c_\beta = K+2-p$ for $B_1(p), B_4(t)$; $c_\beta = u+1$ for $B_2(q), B_5(u)$; $c_\beta = K+2-t$ for $B_4(t), B_5(u)$; $c_\beta = m+2$ for the types B_1B_5 and B_2B_4; $c_\beta \le m+1$ if one point lies in B_3; and $c_\beta \le K$ for two points of one block. As $m+1 \le K$, we get $c(\beta) = K+1$.

The swaps. Let τ be the product of the following transpositions of two consecutive values:

(a) $(2K+m+q,\ 2K+m+q+1)$ for every odd $q \le K-1$; these are the values of $B_1(q)$ and $B_1(q+1)$;
(b) $(n-q,\ n+1-q)$ for every even $q \le K-1$; these are the values of $B_4(q+1)$ and $B_4(q)$;
(c) $(2K,\ 2K+1)$ if $m \ge 1$; these are the values of $B_5(K)$ and of the last point of B_3.

The transpositions are pairwise disjoint. Let $\pi = \tau\beta$, the permutation obtained from β by exchanging the two values of each transposition. Multiplying a permutation by a transposition of two elements of different cycles merges these two cycles. The transposition (a) or (b) with index q joins C_q and C_{q+1}, since $2K+m+q$ and $n+1-q$ belong to C_q and $2K+m+q+1$, $n-q$ belong to C_{q+1}; the transposition (c) joins C_1, which contains $2K$, with Z. The cycles are thus joined along the path $Z, C_1, C_2, \dots, C_K$, each transposition merges two cycles that are still distinct, and π is an n-cycle.

The effect on c. Call two points of the diagram *partners* if their values are exchanged by τ. Since partners have consecutive values, the relative order of the values of two points is the same in π as in β unless the two points are partners. Let x, y be two points. If they are partners, then $c_\pi = c_\beta = 2$ for this pair. Otherwise a third point z can lie in the rectangle of x, y for one of π, β and outside it for the other only if z is the partner of x or of y. Hence

$$c_\pi[x, y] \le c_\beta[x, y] + g,$$

where $g \le 2$ is the number of points among x, y that have a partner lying outside the rectangle of x, y in β.

Pairs with $c_\beta \le K$ therefore have $c_\pi \le K+2$. It remains to show $g \le 1$ for the pairs with $c_\beta = K+1$, which are the following. For $B_1(p), B_2(K)$: the point $B_2(K)$ has value 1 and no partner. For $B_1(1), B_4(t)$: the partner of $B_1(1)$ is $B_1(2)$, which lies in the rectangle. For $B_2(q), B_5(K)$ and, if $m = K-1$, for the pairs of type B_2B_4: the points of B_2 have values at most $K < 2K$ and no partner. For $B_4(1), B_5(u)$: the point $B_4(1)$ has value n, which no transposition of (b) moves. For the pairs $B_1(p), B_5(u)$ if $m = K-1$: the only point of B_5 with a partner is $B_5(K)$, and its partner is a point of B_3, which lies in the rectangle. Hence $c(\pi) \le K+2 = \lfloor n/5 \rfloor + 3$. □

For example, $n = 9$ gives $K = 2$, $m = 1$ and the 9-cycle $7\,6\,2\,1\,4\,9\,8\,3\,5$ with $c = 4$. The restriction to n-cycles therefore changes the answer (10.2) by at most one. Whether $\mathrm{cyc}(n) = \lfloor n/5 \rfloor + 2$ for every $n \not\equiv 4 \pmod 5$ remains open; by Theorem 10.5 this holds for all such $n \le 16$.

10.5. **Deformed Dumont sets.** We turn to [31]. For integers $n \ge 1$ and $0 \le j \le n$ let $\mathcal{D}_{2n-1,j}$ be the set of permutations $\sigma \in S_{2n-1}$ such that

$$\sigma(i) \le i \ \text{ for every even } i, \qquad \sigma(i) \ge i \ \text{ for every odd } i \ge 2(n-j)+1.$$

For $j = 0$ the second condition is empty, and for $j = n$ it concerns all odd positions. For $1 \le r \le 2n-1$ let $\mathcal{D}^{r}_{2n-1,j}$ be the subset defined by $\sigma(1) = r$. The paper [31] determines $|\mathcal{D}^{r}_{2n+1,0}| = (n!)^2 \lceil r/2 \rceil$ and treats the case $j = n$, and its Section 6 ends with the following question.

Question 10.6 (Poddar, Sawant and Shankar [31, Section 6])**.** *"Can we enumerate the cardinalities of the sets $\mathcal{D}_{2n-1,j}$ for other values of j? More specifically, can we give recurrences for the cardinalities of the sets $\mathcal{D}^{r}_{2n-1,j}$ for other values of j?"*

Write $x^{\underline{h}} = x(x-1)\cdots(x-h+1)$ for the falling factorial, with $x^{\underline{0}} = 1$, and put $a(n,j) = |\mathcal{D}_{2n-1,j}|$. Define polynomials $G_j(x)$, $j \ge 1$, by

$$G_1 = 1, \qquad G_{j+1}(x) = (x+1)\big[(x+2)\,G_j(x+1) - x\,G_j(x)\big]. \tag{10.5}$$

Theorem 10.7. *(i) $a(n,0) = (n!)^2$, and for $1 \le j \le n$*

$$a(n,j) = (n-j)!\,(n-j+1)!\,G_j(n-j).$$

(ii) For $0 \le j \le n$,

$$|\mathcal{D}_{2n+1,j+1}| = |\mathcal{D}_{2n+1,j}| - (n-j)(n-j+1)\,|\mathcal{D}_{2n-1,j}|.$$

(iii) $G_j(x) = \sum_{h=0}^{j-1} u_h^{(j)}\, x^{\underline{h}}$, where the nonnegative integers $u_h^{(j)}$ are given by $u_h^{(1)} = \delta_{h,0}$ and

$$u_h^{(j+1)} = (h+1)\Big[u_{h-1}^{(j)} + 2(h+1)\,u_h^{(j)} + (h+1)(h+2)\,u_{h+1}^{(j)}\Big]. \tag{10.6}$$

(iv) For $1 \le j \le n-1$ and $1 \le r \le 2(n-j)$,

$$|\mathcal{D}^{r}_{2n-1,j}| = \left\lceil \frac{r}{2} \right\rceil |\mathcal{D}_{2n-3,j}|.$$

The first polynomials are

$$G_2 = 2(x+1), \qquad G_3 = 2(x+1)(3x+4), \qquad G_4 = 8(x+1)(3x^2+9x+7),$$
$$G_5 = 8(x+1)(15x^3+75x^2+129x+76),$$
$$G_6 = 16(x+1)(45x^4+330x^3+933x^2+1200x+590).$$

Thus $a(n,1) = n!\,(n-1)!$, $a(n,2) = 2\,((n-1)!)^2$ and $a(n,3) = 2(3n-5)\,((n-2)!)^2$. The values $G_j(0) = a(j,j)$ are $1, 2, 8, 56, 608, 9440, 198272$ for $j \le 7$; this is the case $j = n$ of [31].[49] Table 9 lists $a(n,j)$ for $n \le 8$.

[49]The numbers $a(n,n)$ count the permutations of $[2n-1]$ with $\sigma(i) \le i$ for even i and $\sigma(i) \ge i$ for odd i. Part (iii) computes them as $u_0^{(n)}$ from a three-term recurrence of Jacobi type.

n	$a(n,0),\ a(n,1),\ \ldots,\ a(n,n)$
1	1, 1
2	4, 2, 2
3	36, 12, 8, 8
4	576, 144, 72, 56, 56
5	14400, 2880, 1152, 720, 608, 608
6	518400, 86400, 28800, 14976, 10656, 9440, 9440
7	25401600, 3628800, 1036800, 460800, 281088, 217152, 198272, 198272
8	1625702400, 203212800, 50803200, 19699200, 10483200, 7110144, 5807232, 5410688, 5410688

TABLE 9. The numbers $a(n,j) = |\mathcal{D}_{2n-1,j}|$, computed from Theorem 10.7.

Proof of Theorem 10.7. Let $1 \le j \le n$ and $f = n - j$. Call $[1, 2f]$ the *lower zone* and $[2f+1, 2n-1]$ the *upper zone*; both words apply to positions and to values. The upper zone contains j odd numbers and $j-1$ even numbers, and every odd position of the upper zone is subject to $\sigma(i) \ge i$.

The upper zone. For $\sigma \in \mathcal{D}_{2n-1,j}$ and $2f+1 \le t \le 2n$ let σ_t be the set of pairs $(i, \sigma(i))$ with $i \ge t$ and $\sigma(i) \ge t$. It is a partial injection of $[t, 2n-1]$ into itself. Let h_t be the number of positions $i \ge t$ with $\sigma(i) < t$; this is also the number of values $v \ge t$ with $\sigma^{-1}(v) < t$. We call these the *open* positions and the open values of σ_t. The partial injection σ_t is obtained from σ_{t+1} in one of five ways:

(1) $\sigma(t) = t$; then $h_t = h_{t+1}$;
(2) $\sigma(t) > t$ and $\sigma^{-1}(t) > t$: position t receives one of the h_{t+1} open values and the value t is given to one of the h_{t+1} open positions; then $h_t = h_{t+1} - 1$;
(3) $\sigma(t) > t$ and $\sigma^{-1}(t) < t$: position t receives one of the h_{t+1} open values and the value t becomes open; then $h_t = h_{t+1}$;
(4) $\sigma(t) < t$ and $\sigma^{-1}(t) > t$: the value t is given to one of the h_{t+1} open positions and position t becomes open; then $h_t = h_{t+1}$;
(5) $\sigma(t) < t$ and $\sigma^{-1}(t) < t$; then $h_t = h_{t+1} + 1$.

If t is even, the condition $\sigma(t) \le t$ allows (1), (4), (5). If t is odd, the condition $\sigma(t) \ge t$ allows (1), (2), (3). Conversely, every sequence of choices that respects these rules for $t = 2n-1, \ldots, 2f+1$ produces a partial injection σ_{2f+1} of the upper zone in which each odd position i has an image $\ge i$ and each even position either has an image $\le i$ or is open.

Let $U_t(h)$ be the number of partial injections σ_t that arise in this way and have h open positions, so that $U_{2n}(h) = \delta_{h,0}$. The five cases give

$$U_t(h) = (h+1)\,U_{t+1}(h) + U_{t+1}(h-1) \qquad (t \text{ even}), \tag{10.7}$$

$$U_t(h) = (h+1)\,U_{t+1}(h) + (h+1)^2\,U_{t+1}(h+1) \qquad (t \text{ odd}). \tag{10.8}$$

The first step $t = 2n-1$ is odd and gives $U_{2n-1}(h) = \delta_{h,0}$. Each further pair of steps, an even one followed by an odd one, replaces a vector u by

$$(h+1)\big[(h+1)u_h + u_{h-1}\big] + (h+1)^2\big[(h+2)u_{h+1} + u_h\big],$$

which is the right side of (10.6). The upper zone consists of the first odd step and $j-1$ such pairs. Therefore $U_{2f+1}(h) = u_h^{(j)}$, where $u^{(j)}$ is defined by (10.6); by induction $u_h^{(j)} = 0$ for $h \ge j$.

The lower zone. Fix a partial injection σ_{2f+1} with h open positions and h open values, and count its extensions to an element of $\mathcal{D}_{2n-1,j}$. The open positions are even positions of the

upper zone; they must receive lower values, and then $\sigma(i) \le i$ holds for them. The open values must be placed at lower positions; an even lower position i only accepts values $\le i \le 2f$, so the open values go to the f odd lower positions, in $f^{\underline{h}}$ ways. Next fill the even lower positions $2, 4, \dots, 2f$ in this order: position $2i$ may take any of the values $1, \dots, 2i$ not used by the $i-1$ earlier ones, which gives $i+1$ choices and $(f+1)!$ ways in all. There remain f lower values and f positions without a condition, namely h open positions and $f-h$ odd lower positions; they are matched in $f!$ ways. Hence

$$a(n,j) = f!\,(f+1)! \sum_{h\ge 0} u_h^{(j)}\, f^{\underline{h}} = f!\,(f+1)!\,\widetilde{G}_j(f), \qquad \widetilde{G}_j(x) := \sum_{h=0}^{j-1} u_h^{(j)}\, x^{\underline{h}}. \tag{10.9}$$

For $j = 0$ there is no upper zone: the $n-1$ even positions are filled in $n!$ ways as above and the n odd positions receive the remaining n values in $n!$ ways, so $a(n,0) = (n!)^2$.

The recurrence (10.5). We show $\widetilde{G}_j = G_j$. Let Φ be the linear operator $(\Phi G)(x) = (x+1)[(x+2)G(x+1) - xG(x)]$ on polynomials, and let Ψ be the linear operator that sends $\sum_h u_h x^{\underline{h}}$ to $\sum_h u'_h x^{\underline{h}}$ with u' given by the right side of (10.6). Then $\widetilde{G}_1 = 1 = G_1$ and $\widetilde{G}_{j+1} = \Psi \widetilde{G}_j$, so it suffices to prove $\Phi = \Psi$ on each $x^{\underline{h}}$. For $h = 0$ both give $2(x+1)$. Let $h \ge 1$ and $y = x - h + 1$. Collecting the contributions of u_h to u'_{h+1}, u'_h, u'_{h-1},

$$\Psi\, x^{\underline{h}} = (h+2)\, x^{\underline{h+1}} + 2(h+1)^2\, x^{\underline{h}} + h^2(h+1)\, x^{\underline{h-1}} = x^{\underline{h-1}}\big[(h+2)y(y-1) + 2(h+1)^2 y + h^2(h+1)\big].$$

On the other hand $(x+1)^{\underline{h}} = (x+1)x^{\underline{h-1}}$ and $x^{\underline{h}} = y\, x^{\underline{h-1}}$, so

$$\Phi\, x^{\underline{h}} = (x+1)\, x^{\underline{h-1}}\big[(x+2)(x+1) - xy\big] = x^{\underline{h-1}}\,(y+h)\big[(h+2)y + h^2 + h\big],$$

because $x + 1 = y + h$ and $(x+2)(x+1) - x(x-h+1) = (h+2)x + 2 = (h+2)y + h^2 + h$. Both brackets expand to $(h+2)y^2 + (2h^2+3h)y + h^2(h+1)$. This proves (iii), and with (10.9) it proves (i).

Part (ii). Let $f = n - j$. For $j \ge 1$, part (i) turns the claim into

$$f!\,(f+1)!\,G_{j+1}(f) = (f+1)!\,(f+2)!\,G_j(f+1) - f(f+1)\cdot f!\,(f+1)!\,G_j(f),$$

which is (10.5) at $x = f$ multiplied by $f!\,(f+1)!$. For $j = 0$ the claim is $n!\,(n+1)! = ((n+1)!)^2 - n(n+1)(n!)^2$, which is true.

Part (iv). Let $f = n - j \ge 1$, $r \le 2f$ and $\varrho = \lceil r/2 \rceil$. We repeat the count of the lower zone under the condition $\sigma(1) = r$; the upper zone is not affected, since 1 and r lie in the lower zone. The open values go to the odd lower positions other than 1, in $(f-1)^{\underline{h}}$ ways. The even position $2i$ has $i+1$ choices if $2i < r$ and i choices if $2i \ge r$, because the value r is taken. The first case occurs for $i \le \varrho - 1$, so the number of ways is $\varrho! \cdot f!/(\varrho-1)! = \varrho\, f!$. The remaining $f-1$ lower values are matched with the remaining $f-1$ positions in $(f-1)!$ ways. The total is

$$\varrho\, f!\,(f-1)! \sum_h u_h^{(j)}\,(f-1)^{\underline{h}} = \varrho\,(f-1)!\,f!\,G_j(f-1) = \varrho\, a(n-1,j),$$

by (i) applied to $n-1$, for which $n - 1 - j = f - 1$. □

Parts (i) and (ii) answer the first half of Question 10.6: the cardinalities are given in closed form for each fixed j, and for all j by a recurrence in which the factor $(n-j)(n-j+1)$ depends on the two indices only through their difference. Part (iv) answers the second half for first letters in the lower zone; for $j = 0$ the same count gives $|\mathcal{D}^r_{2n-1,0}| = \lceil r/2 \rceil\, ((n-1)!)^2$ for all r, which is the formula of [31].

Remark 10.8. Let $1 \le j \le n-1$. For a first letter $r > 2(n-j)$ the count is a linear recurrence of the same kind, and we have no closed form for it. The value r now lies in the upper zone and must be open from step $t = r$ on, until it is given to position 1. So at step $t = r$ only case (3) is allowed if r is odd, and only case (5) if r is even. At every later odd step one of the open values is reserved, which changes (10.8) to $U_t(h) = h\,U_{t+1}(h) + h(h+1)\,U_{t+1}(h+1)$; the even steps are unchanged. In the lower zone the other $h-1$ open values go to odd positions other than 1, the even positions are filled in $(f+1)!$ ways and the rest in $f!$ ways, so that $|\mathcal{D}^r_{2n-1,j}| = f!\,(f+1)!\sum_{h\ge1} U_{2f+1}(h)\,(f-1)^{\underline{h-1}}$ with $f = n-j$.[50]

11. The 2-Steiner distance matrix of a path

The last problem of this part concerns a matrix. Jana, Mahato and Sivasubramanian [25] study the characteristic polynomial of the 2-Steiner distance matrix of a tree, introduced by Azimi and Sivasubramanian [2], and end their treatment of the path with two conjectures. We prove one of them, for all $n \ge 3$, and show that the other fails for every n from 16 to 60.

11.1. **The matrix and the two conjectures.** Let T be a tree with vertex set $[n]$. The 2-*Steiner distance matrix* of T has its rows and columns indexed by the two-element subsets of $[n]$, and its entry at (p,q) is the least number of edges of a subtree of T that contains $p \cup q$. This matrix is singular in general, and [25] works with its restriction to a set of rows that is a basis of the row space.[51] For the path P_n with edges $\{i, i+1\}$ the basis used in [25] consists of the sets

$$e_i = \{i, i+1\} \quad (1 \le i \le n-1), \qquad f_j = \{j, j+2\} \quad (1 \le j \le n-2).$$

A subtree of a path is an interval, so the entry at (p,q) is $\max(p\cup q) - \min(p\cup q)$. We write $\mathfrak{D}_n$ for the resulting symmetric matrix of order $2n-3$; explicitly

$$\begin{aligned} &\mathfrak{D}_n(e_i,e_j) = |i-j|+1, && \mathfrak{D}_n(f_i,f_j) = |i-j|+2,\\ &\mathfrak{D}_n(e_i,f_j) = j-i+2 \quad (i \le j), && \mathfrak{D}_n(e_i,f_j) = i-j+1 \quad (i > j). \end{aligned} \tag{11.1}$$

Let

$$\chi_n(x) = \det(xI - \mathfrak{D}_n) = \sum_{i=0}^{2n-3} a_i\, x^i.$$

It is shown in [25] that $|a_0| = n-1$, $|a_1| = 4n^2 - 14n + 13$ and $|a_{2n-4}| = 3n-5$, and that an index ℓ at which $|a_0|, \dots, |a_{2n-4}|$ is largest satisfies $\ell \le \lfloor 7n/5 \rfloor$. The two conjectures are the following.

Conjecture 11.1 (Jana, Mahato and Sivasubramanian [25])**.** *"For a path P_n on $n > 5$ vertices, if $\chi_{\mathfrak{D}_{P_n}}(x) = \sum_{i=0}^{2n-3} a_i x^i$ and $|a_\ell| = \max\{|a_0|, |a_1|, \dots, |a_{2n-4}|\}$, then $\ell = n-1$."*

Conjecture 11.2 (Jana, Mahato and Sivasubramanian [25])**.** *"For a path P_n on $n > 5$ vertices, if $\chi_{\mathfrak{D}_{P_n}}(x) = \sum_{i=0}^{2n-3} a_i x^i$, then $a_{2n-5} = -\frac{1}{6}(n-1)(n-2)(2n^2+6n-15)$."*

Here $\mathfrak{D}_{P_n}$ is the matrix $\mathfrak{D}_n$. The authors of [25] report that the first statement was observed for $5 < n < 15$. Conjecture 11.2 is true, and the restriction $n > 5$ is not needed (Theorem 11.4). Conjecture 11.1 is true for $6 \le n \le 15$ and false for $16 \le n \le 60$ (Theorem 11.5).

[50] For $n = 3$, $j = 2$ the numbers $|\mathcal{D}^r_{5,2}|$, $r = 1, \dots, 5$, are $2, 2, 2, 2, 0$, with sum $a(3,2) = 8$. The first two are given by part (iv), and the last is 0 because $\sigma(5) \ge 5$ forces $\sigma(5) = 5$.

[51] The rank is $2n - \lambda - 1$ for a tree with λ leaves, by [2]; for the path this is $2n-3$. The restriction to a basis has one positive and $2n - \lambda - 2$ negative eigenvalues, and [25] deduces from this that the absolute values of the coefficients of its characteristic polynomial form a unimodal and log-concave sequence.

11.2. **The coefficient a_{2n-5}.**

Lemma 11.3. *For integers $N \geq 1$ and $r \geq 0$,*

$$\sum_{i,j=1}^{N} \big(|i-j|+r\big)^2 = \frac{N^2(N^2-1)}{6} + \frac{2r\,N(N^2-1)}{3} + r^2N^2.$$

Proof. There are $2(N-d)$ ordered pairs (i,j) with $|i-j| = d \geq 1$. Hence

$$\sum_{i,j} |i-j| = 2\sum_{d=1}^{N-1} d(N-d) = N^2(N-1) - \frac{(N-1)N(2N-1)}{3} = \frac{N(N^2-1)}{3}$$

and

$$\sum_{i,j} |i-j|^2 = 2\sum_{d=1}^{N-1} d^2(N-d) = \frac{N^2(N-1)(2N-1)}{3} - \frac{N^2(N-1)^2}{2} = \frac{N^2(N^2-1)}{6}.$$

Expanding the square gives the claim. □

Theorem 11.4. *Let $n \geq 3$, and let S_{ee}, S_{ff}, S_{ef} be the sums of the squares of the entries of $\mathfrak{D}_n$ in the blocks indexed by (e_i, e_j), by (f_i, f_j) and by (e_i, f_j). Then*

$$S_{ee} = \frac{(n-1)(n^3+n^2-6)}{6}, \qquad S_{ef} = \frac{(n-1)(n^3+n^2-12)}{6}, \tag{11.2}$$
$$S_{ff} = \frac{(n-2)(n^3+2n^2+3n-30)}{6},$$

the trace of $\mathfrak{D}_n^2$ is $\frac{1}{6}(4n^4-4n^2-66n+90)$, and

$$a_{2n-5} = -\frac{(n-1)(n-2)(2n^2+6n-15)}{6}. \tag{11.3}$$

Proof. For a symmetric matrix M of order m the coefficient of x^{m-2} in $\det(xI-M)$ is the sum of the principal minors of order 2, which is

$$\sum_{i<j}(M_{ii}M_{jj} - M_{ij}^2) = \frac{1}{2}\Big((\operatorname{tr} M)^2 - \sum_{i,j} M_{ij}^2\Big) = \frac{1}{2}\big((\operatorname{tr} M)^2 - \operatorname{tr} M^2\big). \tag{11.4}$$

For $M = \mathfrak{D}_n$ we have $m-2 = 2n-5$ and, by (11.1), $\operatorname{tr}\mathfrak{D}_n = (n-1)+2(n-2) = 3n-5$ and $\operatorname{tr}\mathfrak{D}_n^2 = S_{ee} + S_{ff} + 2S_{ef}$.

By Lemma 11.3 with $(N,r) = (n-1,1)$,

$$S_{ee} = \frac{N(N^3+4N^2+5N-4)}{6}\Big|_{N=n-1} = \frac{(n-1)(n^3+n^2-6)}{6},$$

and with $(N,r) = (n-2,2)$,

$$S_{ff} = \frac{N(N^3+8N^2+23N-8)}{6}\Big|_{N=n-2} = \frac{(n-2)(n^3+2n^2+3n-30)}{6}.$$

For the mixed block let $\sigma(M) = \sum_{s=1}^{M} s^2 = M(M+1)(2M+1)/6$. Fix j. The rows e_i with $i \leq j$ contribute $\sum_{s=2}^{j+1} s^2 = \sigma(j+1)-1$, and the rows with $i > j$ contribute $\sum_{s=2}^{n-j} s^2 = \sigma(n-j)-1$. As j runs from 1 to $n-2$, both $j+1$ and $n-j$ run over $2,\dots,n-1$. With the identity $\sum_{M=1}^{L}\sigma(M) = L(L+1)^2(L+2)/12$, which follows by induction on L, we get

$$S_{ef} = 2\sum_{M=2}^{n-1}\sigma(M) - 2(n-2) = \frac{(n-1)n^2(n+1)}{6} - 2(n-1) = \frac{(n-1)(n^3+n^2-12)}{6}.$$

This proves (11.2). Adding,

$$6\,\mathrm{tr}\,\mathfrak{D}_n^2 = (n-1)(3n^3+3n^2-30)+(n-2)(n^3+2n^2+3n-30) = 4n^4-4n^2-66n+90,$$

and (11.4) gives

$$a_{2n-5} = \frac{6(3n-5)^2-(4n^4-4n^2-66n+90)}{12} = \frac{-2n^4+29n^2-57n+30}{6}.$$

Finally $(n-1)(n-2)(2n^2+6n-15) = (n^2-3n+2)(2n^2+6n-15) = 2n^4-29n^2+57n-30$. □

For $n = 3,4,5,6$ the formula gives $a_{2n-5} = -7, -41, -130, -310$.[52] The three leading coefficients of χ_n are thus 1, $-(3n-5)$ and (11.3).

11.3. **The location of the largest coefficient.**

Theorem 11.5. *For $6 \le n \le 60$ the largest of the numbers $|a_0|, \dots, |a_{2n-4}|$ is attained at exactly one index $\ell(n)$, and*

$$\ell(n) = \begin{cases} n-1, & 6 \le n \le 15, \\ n, & 16 \le n \le 24, \\ n+1, & 25 \le n \le 34, \\ n+2, & 35 \le n \le 43, \\ n+3, & 44 \le n \le 52, \\ n+4, & 53 \le n \le 60. \end{cases}$$

In particular Conjecture 11.1 holds for $6 \le n \le 15$ and fails for every n with $16 \le n \le 60$. For $n = 16$,

$$|a_{14}| = 766504605632, \qquad |a_{15}| = 939602445008,$$
$$|a_{16}| = 956326515118, \qquad |a_{17}| = 802497798666.$$

Proof. The statement concerns the integer polynomials χ_n for $6 \le n \le 60$, which are determined by (11.1). They were computed exactly, by two independent methods for $n = 15$ and $n = 16$, and the computation is part of the verification archive [8]. For $n = 16$ the displayed values give $|a_{16}| > |a_{15}|$, so the maximum is not at $n-1 = 15$. □

The first failure is narrow: for $n = 15$ the two largest values are $|a_{14}| = 134221025536$ and $|a_{15}| = 133876572516$, which differ by less than three parts in a thousand, and at $n = 16$ the order is reversed.[53] Figure 15 shows $\ell(n)-(n-1)$. In the range of the theorem the index $\ell(n)$ stays far below the bound $\lfloor 7n/5 \rfloor$ of [25]; at $n = 60$ one has $\ell = 64$ against 84. The steps of $\ell(n)-n$ occur at $n = 16, 25, 35, 44, 53$, at distances 9, 10, 9, 9. We do not know the asymptotic behaviour of $\ell(n)$, and we state no conjecture for it.[54]

[52] For $n = 3$ the matrix is $\left(\begin{smallmatrix} 1 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 1 \end{smallmatrix}\right)$ in the order e_1, f_1, e_2, with characteristic polynomial $x^3 - 4x^2 - 7x - 2$; the constant term agrees with $|a_0| = n-1$.

[53] This explains why the pattern looked stable in the range $5 < n < 15$ examined in [25].

[54] The cases $n = 3,4,5$, excluded in Conjecture 11.1, have $\ell(n) = n-2$.

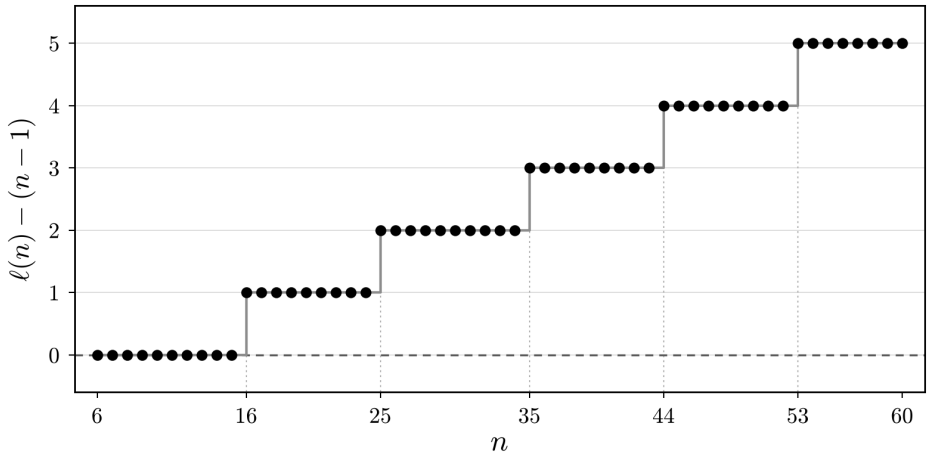


FIGURE 15. The excess $\ell(n) - (n-1)$ of the index of the largest coefficient of χ_n over the value predicted by Conjecture 11.1, for $6 \le n \le 60$.

12. Status of the conjectures and open problems

Table 10 lists the conjectures and questions from the literature that are treated in this paper, with the result obtained for each.

The following problems raised in this paper remain open.

(1) Odd families in rectangles of shape $(n+1) \times n$: the value 2^n and the Catalan classification (Conjecture 4.20), and the form with free endpoints (Conjecture 4.21).
(2) The exact values for two rows of cells (Conjecture 4.23) and the slope $\frac{3b-1}{4}$ for b rows (Conjecture 6.6).
(3) The constant $\lim \mathrm{M}_{\mathrm{od}}(n,n)/n^2$, which lies between $\frac{1}{2}$ and 1 if it exists (Question 6.7).
(4) Even families through a point next to the diagonal (Conjecture 5.17) and the least size of a set that determines an extremal family (Question 5.16).
(5) The exact values in higher dimensions (Conjectures 7.12 and 7.13) and the bound with free endpoints (Conjecture 7.14).
(6) Log-concavity of Q_k for all k (Conjecture 9.3).

Author contributions

Deep Bhattacharjee is the sole author. He posed the problems in the form treated here, proved the results, wrote the paper, drew the figures, and wrote the Lean 4 certificate and the Julia programs that check the finite statements.

Open Gluon and Anaxia

Two tools built by the author organised this project. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact integer and rational arithmetic. It was used here to keep the finite data of Sections 4–11 and their checks in one record. Every statement of this paper is proved in the text.

statement	result	reference
Shankar [35, Conj. 4.1]: the even families of 2^n paths are the C_n families $\mathcal{F}_M$	proved; also for all lengths and endpoints	Thms. 3.2, 3.3
Shankar [35]: bounds $\frac{1}{4}n^2 - O(1) \le \mathrm{M}_{\mathrm{od}}(n,n) \le n(n-1)+1$	improved to $\frac{1}{2}n^2 - 3n \le \mathrm{M}_{\mathrm{od}}(n,n)$; upper bound lowered unless $n \equiv 0,1$ (8)	Thms. 6.2, 6.5
Fulman, Kim, Lee, Petersen [23, Conj. 1.3]: $A_n^{\pm}$ real-rooted	proved	Thm. 8.4
Dey, Sivasubramanian [16]: $A_n^{\pm}$ real-rooted, $n \equiv 0,1$ (4)	proved	Thm. 8.4
Shankar [32]: $E_n^{\pm}$ real-rooted for $n \ge 5$	false for $n = 5$, true for $n \ge 6$	Thm. 8.5
Shankar [33, Conj. 9.1]: B_k real-rooted	proved, with interlacing	Thm. 9.10
Shankar [33, Conj. 9.2]: $Q(z)$ transcendental	proved	Thm. 9.14
Shankar [33, Conj. 9.3]: Q_k log-concave	open; true for $k \le 110$	Subsec. 9.5
Shankar [33, Conj. 9.5]: (B_k) and (Q_k) Stieltjes moment sequences	true for (B_k), with density; false for (Q_k)	Thms. 9.15, 9.16
Shankar [33, Qu. 9.4]: a finite model for Q_k	path model: yes; finite generating tree: no	Prop. 9.18, Cor. 9.19
Shankar [33, Qu. 9.6]: four letters	reduction and Catalan classes given; classification open	Props. 9.20, 9.21
Balachandran, Ramdas, Shankar [4]: extremal permutations	determined for $n = 5k+4$	Thm. 10.4
Balachandran, Ramdas, Shankar [4]: n-cycles	minimum raised by at most one; exact for $n = 5k+4$	Thm. 10.5
Poddar, Sawant, Shankar [31, Sec. 6]: $\lvert\mathcal{D}_{2n-1,j}\rvert$ and $\lvert\mathcal{D}^r_{2n-1,j}\rvert$	closed form for all j; $r \le 2(n-j)$	Thm. 10.7
Jana, Mahato, Sivasubramanian [25]: a_{2n-5}	proved for $n \ge 3$	Thm. 11.4
Jana, Mahato, Sivasubramanian [25]: peak at $n-1$	true for $6 \le n \le 15$, false for $16 \le n \le 60$	Thm. 11.5

TABLE 10. Conjectures and questions from the literature treated in this paper.

DECLARATIONS

Data and code availability. No data were generated. The Lean 4 certificate of the finite identities and the Julia programs that recompute every numerical statement of the paper are archived on Zenodo [8], doi:10.5281/zenodo.24169702. The Lean files use the core library only; every theorem is checked by the kernel, with no unproved statement and no axiom.

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